



Beyond Searching a Village: Learning to Recommend Diverse and Successful Collaborative Teams



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Under Review (1)

2027, arXiv, Graph Neural Team Recommendation: An Integrated Approach

Published (10)

2026, SIGIR, OpeNTF2: Fairness-Aware Graph Neural Team Recommendation

2026, COIN, A Probabilistic Greedy Attempt to be Fair in Neural Team Recommendation. Revision Requested

2025, SIGIR, Translative Neural Team Recommendation

2025, WSDM, Adaptive Loss-based Curricula for Neural Team Recommendation

2024, BIAS-ECIR, vivaFemme: Mitigating Gender Bias in Neural Team Recommendation via Female-Advocate Loss Regularization

2024, ECIR, A Streaming Approach to Neural Team Formation Training

2024, WISE, Skill Vector Representation Learning for Collaborative Team Recommendation: A Comparative Study

2023, BIAS-ECIR, Bootless Application of Greedy Re-ranking Algorithms in Fair Neural Team Formation

2022, CIKM, Effective Neural Team Formation via Negative Samples

2022, CIKM, OpeNTF: A Benchmark Library for Neural Team Formation

Tutorials (4)

2025, CIKM, Neural Shifts in Collaborative Team Recommendation

2025, WSDM, Bridging Subgraph Optimization and Graph Neural Network in Team Recommendations

2024, SIGIR-AP, Paradigm Shifts in Team Recommendation: From Subgraph Optimization to Graph Neural Network

2024, UMAP, Collaborative Team Recommendation for Skilled Users: Objectives, Techniques, and Perspectives





Beyond Searching a Village: Learning to Recommend Diverse and Successful Collaborative Teams



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recsys.acm.org/recsys26/

<https://fani-lab.github.io/OpenTF/tutorial/recsys26>

TIME AND LOCATION

ABSTRACT

TARGET AUDIENCE

OUTLINE

PIONEERING TECHNIQUES

NEURAL TEAM RECOMMENDATION

TIME AND LOCATION

Half-day, Afternoon, 1:30 PM - 5:30 PM (GMT/UTC -6), Monday, September 28, 2026

Elk/Deer Lake (4th floor), Marriott City Center, Downtown, Minneapolis, Minnesota, USA

Prev Tutorials at [UMAP24], [SIGIR-AP24], [WSDM25], [CIKM25] < [Full Outline] [Slides] [Recording]

Outline

I) Introduction and Background

II) Pioneering Techniques

III) Learning-based Heuristics

III) Team Refinement

V) Challenges and New Perspectives

VI) Applications

Hands-on: OpeNTF

Outline

I) Introduction and Background

II) Pioneering Techniques

III) Learning-based Heuristics

III) Team Refinement

V) Challenges and New Perspectives

VI) Applications

Hands-on: OpeNTF



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Where are Teams?

The Matrix, The Wachowskis, 1999

"Morpheus: The Matrix is [Teams are] everywhere. It is all around us. Even now, in this very room! ..."











COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. The Imitation Game

Why Teams? (1 vs. 442)

Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models

Transactions on Machine Learning Research, Published: 11 May 2023

Alphabetic author list:*

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R. Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, Agnieszka Kluska, Aitor Lewkowycz, Akshat Agarwal, Alethea Power, Alex Ray, Alex Warstadt, Alexander W. Kocurek, Ali Safaya, Ali Tazarv, Alice Xiang, Alicia Parrish, Allen Nie, Aman Hussain, Amanda Askill, Amanda Souza, Ambrose Slone, Ameet Rahane, Anantharaman S. Iyer, Anders Andreassen, Andrea Madotto, Andrea Santilli, Andreas Stuhlmüller, Andrew Dai, Andrew La, Andrew Lampinen, Andy Zou, Angela Jiang, Angelica Chen, Anh Vuong, Animesh Gupta, Anna Gottardi, Antonio Norelli, Anu Venkatesh, Arash Gholamidavoodi, Arfa Tabassum, Arul Menezes, Arun Kirubakaran, Asher Mullokandov, Ashish Sabharwal, Austin Herrick, Avia Efrat, Aykut Erdem, Ayla Karakas, B. 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What Is a Team?

A group of users who **collaborate** together with a **common purpose** in order to accomplish the requirements of a task.

[Brannik et al., Psychology Press, 1997]



What Is a Team?

A group of users who **independently** endeavor to accomplish their individual tasks to reach a **shared goal** or value, while **actively interacting and adapting**.

[Zzkarian et. Al., IIE transactions, 1999]



Team vs. Group

Dead Poet Society, 1998, Peter Weir

Robin Williams: *"we don't read and write poetry because it's cute ..."*



Group Learning Teams

FORMING

A short-lived phase; the Team gets acquainted, learns roles and responsibilities.

STORMING

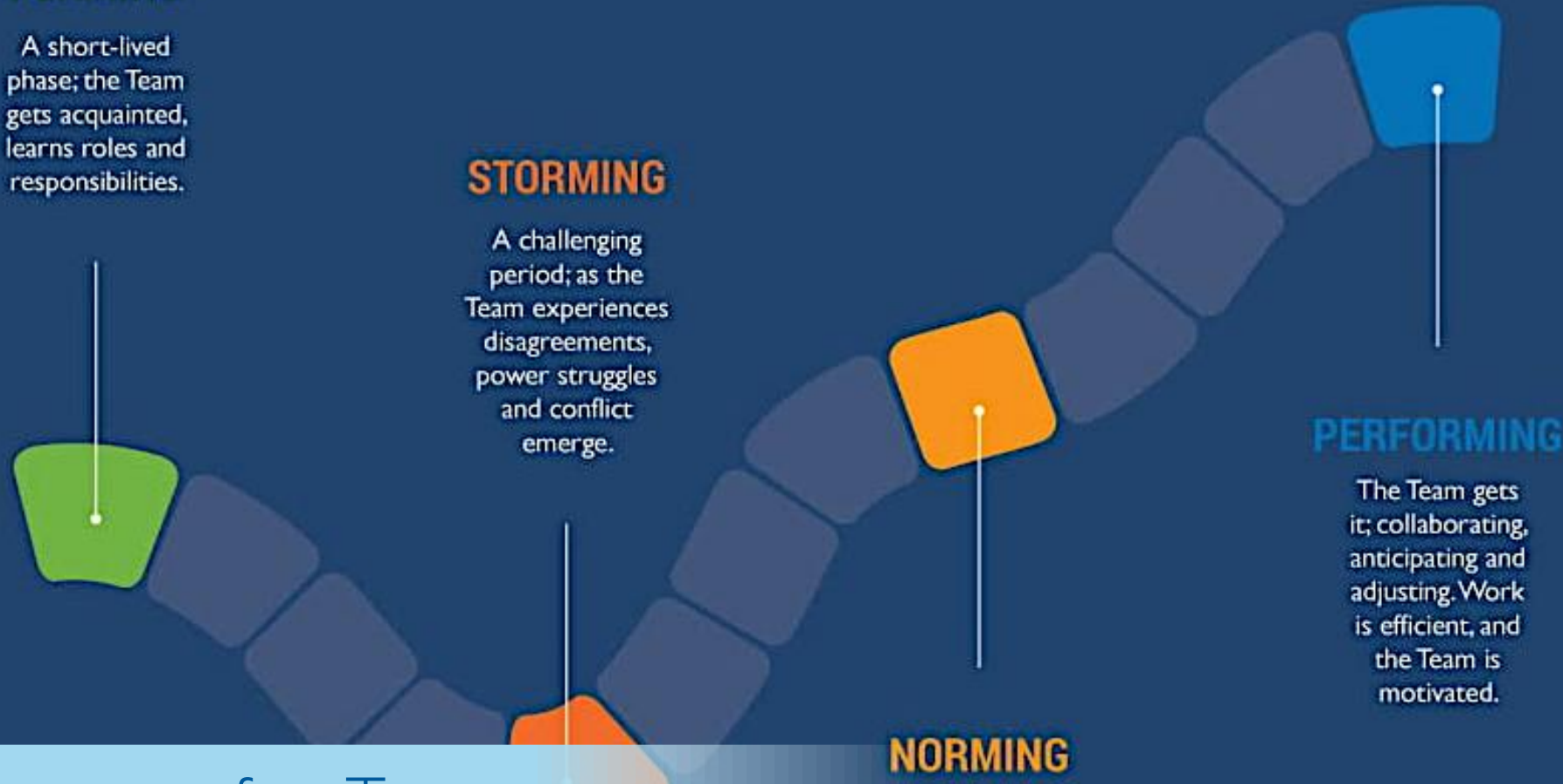
A challenging period; as the Team experiences disagreements, power struggles and conflict emerge.

PERFORMING

The Team gets it; collaborating, anticipating and adjusting. Work is efficient, and the Team is motivated.

NORMING

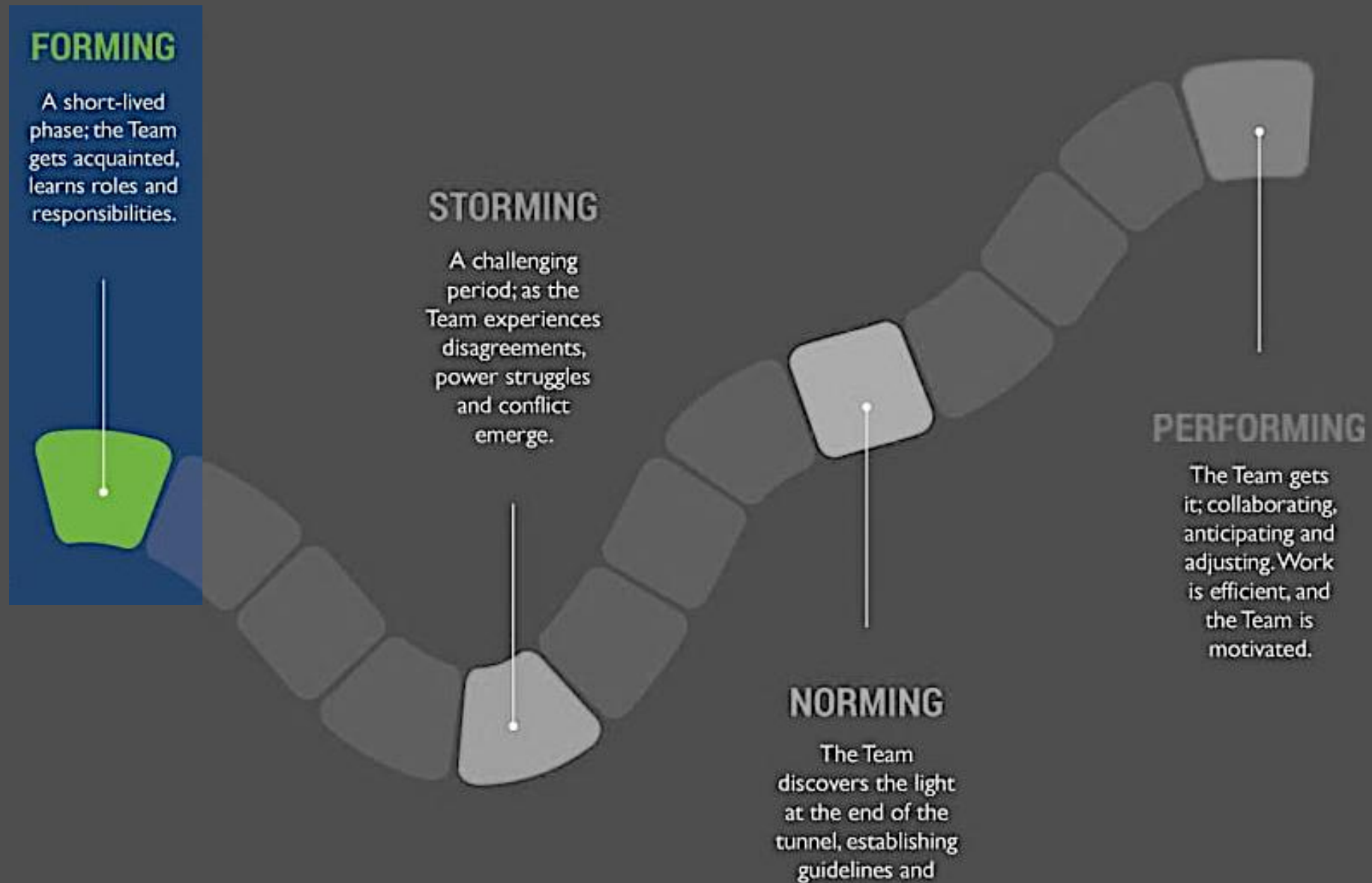
The Team discovers the light at the end of the tunnel, establishing guidelines and understanding process.



Development of a Team

Four distinct phases of development for a team

[Tuckman et al., Psychological bulletin ,1965]



FORMING

A short-lived phase; the Team gets acquainted, learns roles and responsibilities.

STORMING

A challenging period; as the Team experiences disagreements, power struggles and conflict emerge.

NORMING

The Team discovers the light at the end of the tunnel, establishing guidelines and

PERFORMING

The Team gets it; collaborating, anticipating and adjusting. Work is efficient, and the Team is motivated.

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A short-lived phase; the Team gets acquainted, learns roles and responsibilities.

STORMING

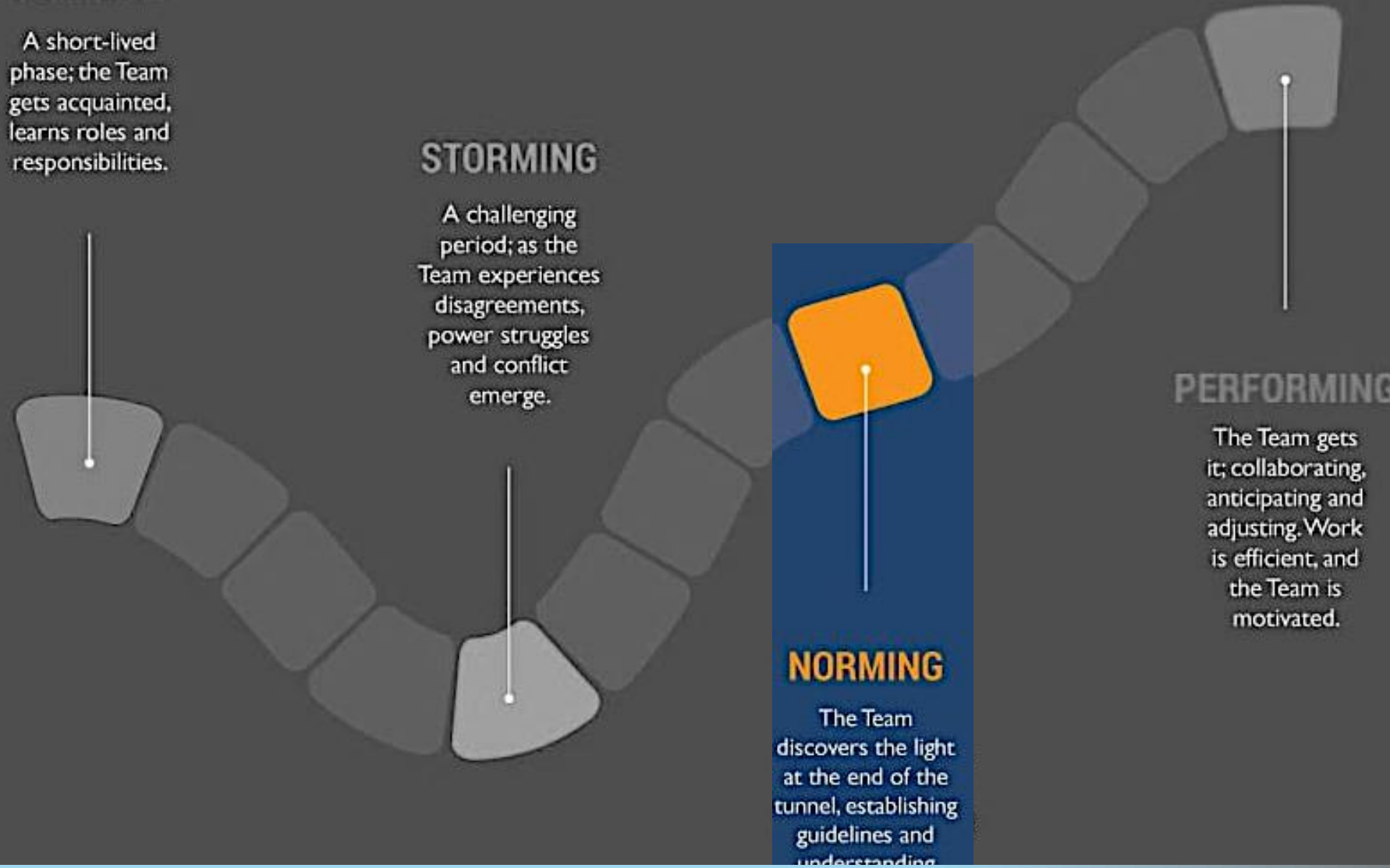
A challenging period; as the Team experiences disagreements, power struggles and conflict emerge.

PERFORMING

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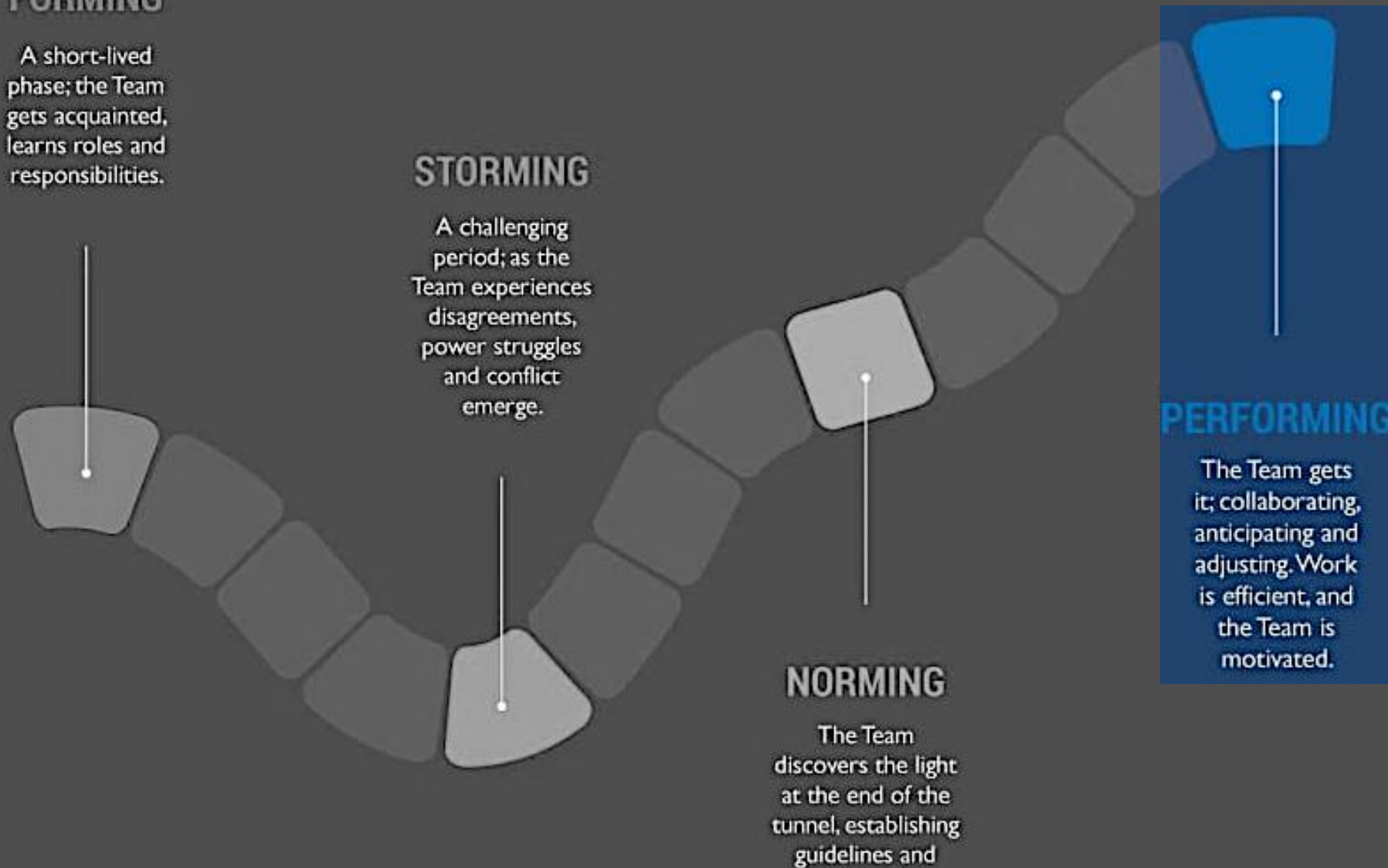
A challenging period; as the Team experiences disagreements, power struggles and conflict emerge.

NORMING

The Team discovers the light at the end of the tunnel, establishing guidelines and

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23

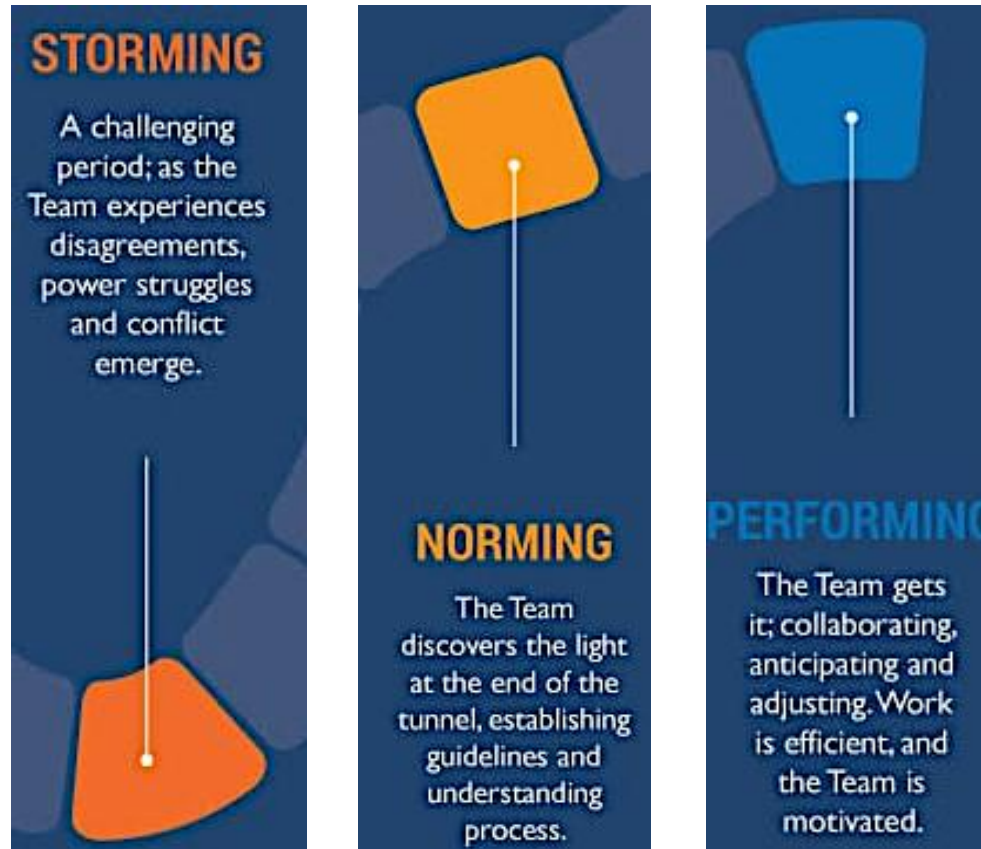
Team Recommendation → Team Formation

FORMING

A short-lived phase; the Team gets acquainted, learns roles and responsibilities.



Team Recommendation → Team Refinement





Eternal Sunshine of the Spotless Mind, Michel Gondry, 2004



Eternal Sunshine of the Spotless Mind, Michel Gondry, 2004

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Information Retrieval in a Village: Social Search Social Information Retrieval (Social IR)

30

Traditional Approach

Manual, based on human experience, instinct; error-prone, and suboptimal

- Large number of candidates
 - Different knowledge
 - Different culture
 - Different characteristic
- Hidden personal and societal biases
 - Race
 - Gender
 - Popularity
- Multitude of criteria to optimize
 - Communication cost
 - Budget
 - Time

31

Traditional Approach

Manual, based on human experience, instinct; error-prone, and suboptimal

- Large number of candidates
 - Different knowledge
 - Different culture
 - Different characteristic

Manual team recommendation on a large scale is almost impossible!

× Not all teams with best experts are necessarily successful → Synergistic Balance → Unknown!

- Multitude of criteria to optimize
 - Communication cost
 - Budget
 - Time

Team Recommendation ↔ Information Retrieval



Available in English

Owner Google

Commercial Yes

Launched Early 2008

Current status Abandoned, shut down by Google

The anatomy of a large-scale social search engine

Authors: [Damon Horowitz](#), [Sepandar D. Kamvar](#) | [Authors Info & Claims](#)

WWW '10: Proceedings of the 19th international conference on World wide web • Pages 431 - 440
<https://doi.org/10.1145/1772690.1772735>

Published: 26 April 2010 [Publication History](#)



297 3,631



Abstract

We present Aardvark, a web input, text message network most likely to challenge lies in finding engine like Aardvark lie

Searching the village: models and methods for social search

Authors: [Damon Horowitz](#), [Sepandar D. Kamvar](#) | [Authors Info & Claims](#)

Communications of the ACM, Volume 55, Issue 4 • Pages 111 - 118 • <https://doi.org/10.1145/2133806.2133830>

Published: 01 April 2012 [Publication History](#)

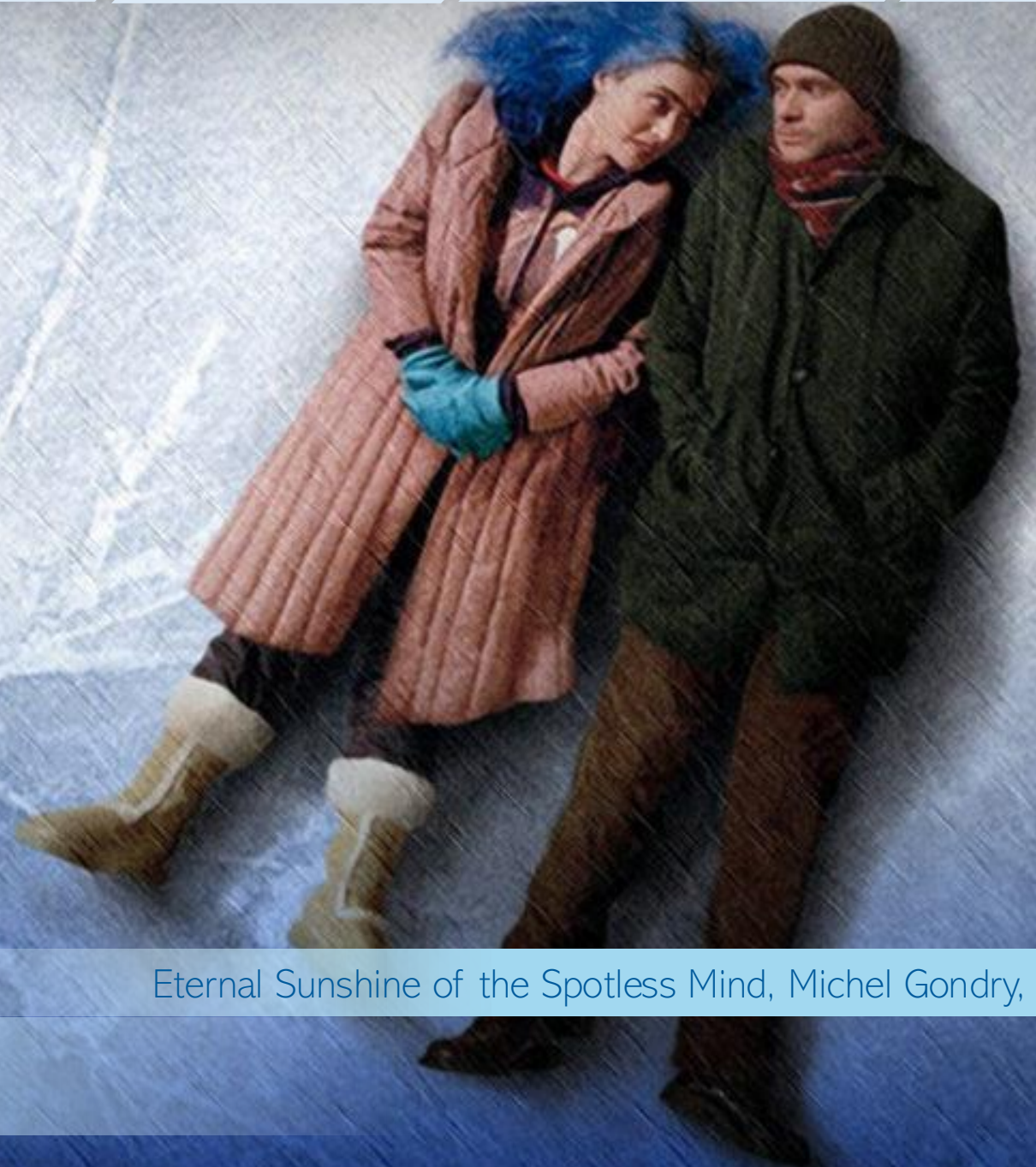


Feb. 2010: Google acquired Aardvark for \$50M.

Sept 2011: Google discontinued Aardvark.

× Talent search, hiring search, search committees, ..., fall short → Collaboration is missing!

ask a question, either by instant message, e-mail, Web input, text message, or voice. Aardvark then routes the question to the person in the user's extended social network most likely to be able to answer that question. As compared to a traditional Web search engine, where the challenge lies in finding the right *document* to satisfy a user's information need, the challenge in a social search engine like Aardvark lies in finding the right *person* to satisfy a user's information need. Further, while trust in a traditional search engine is based on authority, in a social search engine like Aardvark, trust is based on intimacy. We describe how these considerations inform the architecture, algorithms, and user interface of Aardvark, and how they are reflected in the behavior of Aardvark users.



Eternal Sunshine of the Spotless Mind, Michel Gondry, 2004



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What is Success?

Kilfoil wins gold as Canada defeats Czechia at the 2024 Hlinka Gretzky Cup



Tomas Mikolov

Efficient estimation of word representations in vector space

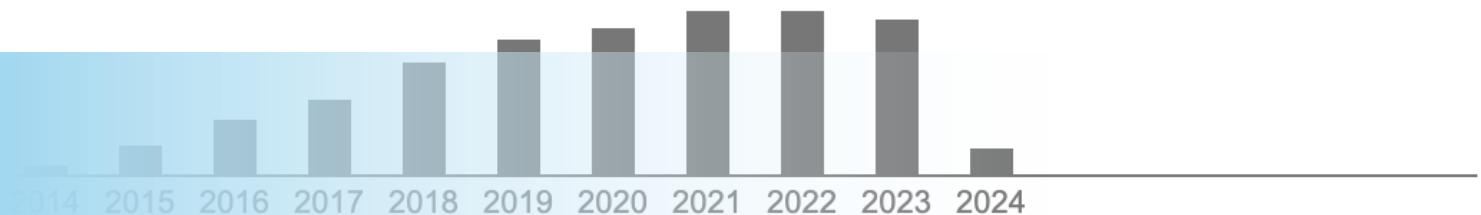
Authors Tomas Mikolov, Kai Chen, Greg Corrado, Jeffrey Dean

Publication date 2013/1/16

Journal arXiv preprint arXiv:1301.3781

Description We propose two novel model architectures for computing continuous vector representations of words from very large data sets. The quality of these representations is measured in a word similarity task, and the results are compared to the previously best performing techniques based on different types of neural networks. We observe large improvements in accuracy at much lower computational cost, i.e. it takes less than a day to learn high quality word vectors from a 1.6 billion words data set. Furthermore, we show that these vectors provide state-of-the-art performance on our test set for measuring syntactic and semantic word similarities.

Total citations Cited by 40332



**Tomas Mikolov**

December 13, 2023 · 🌐

<https://openreview.net/forum?id=idpCdOWtqXd60>

Yesterday we received a **Test of Time Award at NeurIPS** for the word2vec paper from ten years ago. I'm really happy about it! I think it's the first "best paper" type of award I ever received. In fact, the original word2vec paper was **rejected at the first ICLR conference in 2013** (despite the **acceptance rate of around 70%**), so it made me think how difficult it is for reviewers to predict future impact of research papers.

<https://www.facebook.com/share/p/kXYaYaRvRCr5K2Ze>

What is Success?



Produced

Releases50

PyTorch 2.2.1 Release, bug fi...

last month

Latest

+ 49 releases

Contributors3,202



+ 3,188 contributors

Languages

Python50.0%

C++40.6%

Cuda3.7%

C2.0%

Objective-C++1.3%

CMake0.8%

Other1.6%

Released

US1781541A

United States

Download PDF

Find Prior Art

Similar

Inventor: Einstein Albert, Szilard Leo

Current Assignee : Electrolux Servel Corp

Worldwide applications

1927 · US

Application US240566A events ⓘ

1927-12-16 · Application filed by Electrolux Servel Corp

1930-11-11 · Application granted

1930-11-11 · Publication of US1781541A

1947-11-11 · Anticipated expiration

Status

Expired - Lifetime

Info: Cited by (20), Similar documents, Priority and Related Applications

Issued

A Streaming Approach to Neural Team Formation Training

Hossein Fani^[0000–0002–6033–6564], Reza Barzegar^[0009–0002–2831–4143], Arman Dashti^[0000–0001–9022–5403], and Mahdis Saeedi^[0000–0002–6297–3794]

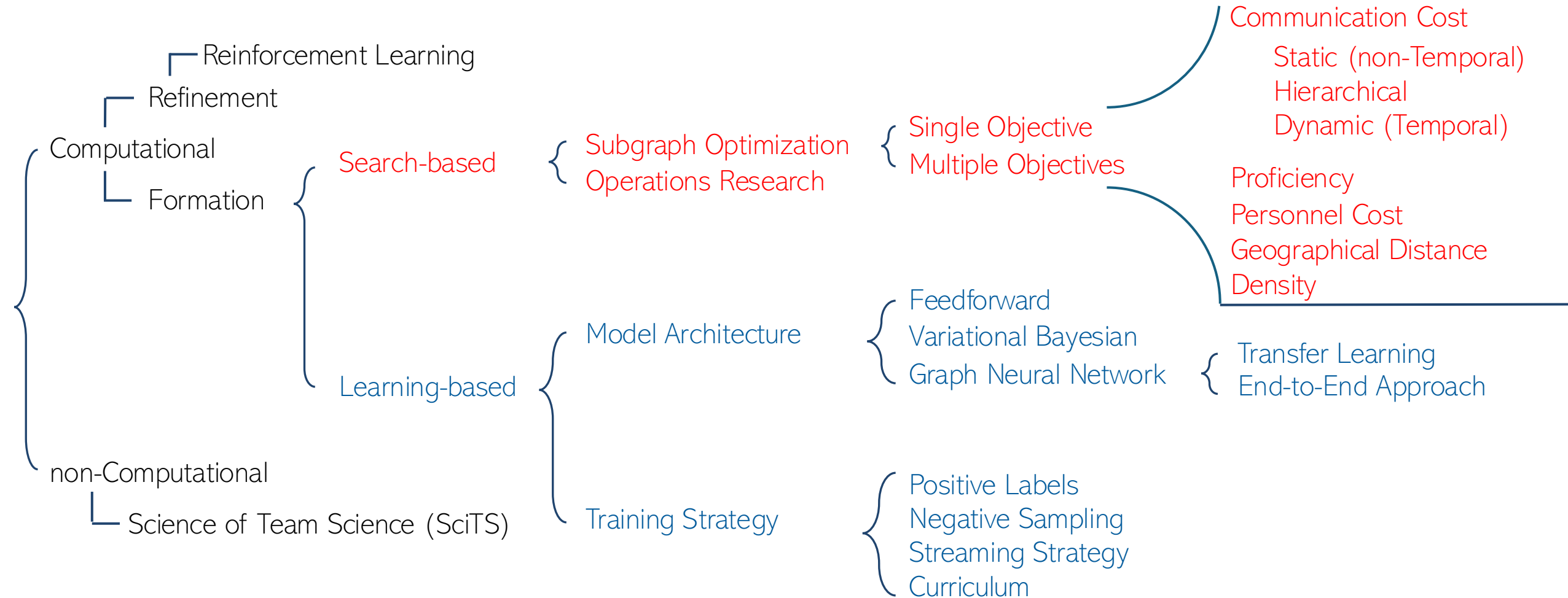
University of Windsor, Windsor, ON., Canada
{hfani, barzegar, vaghehd, msaeedi}@uwindsor.ca

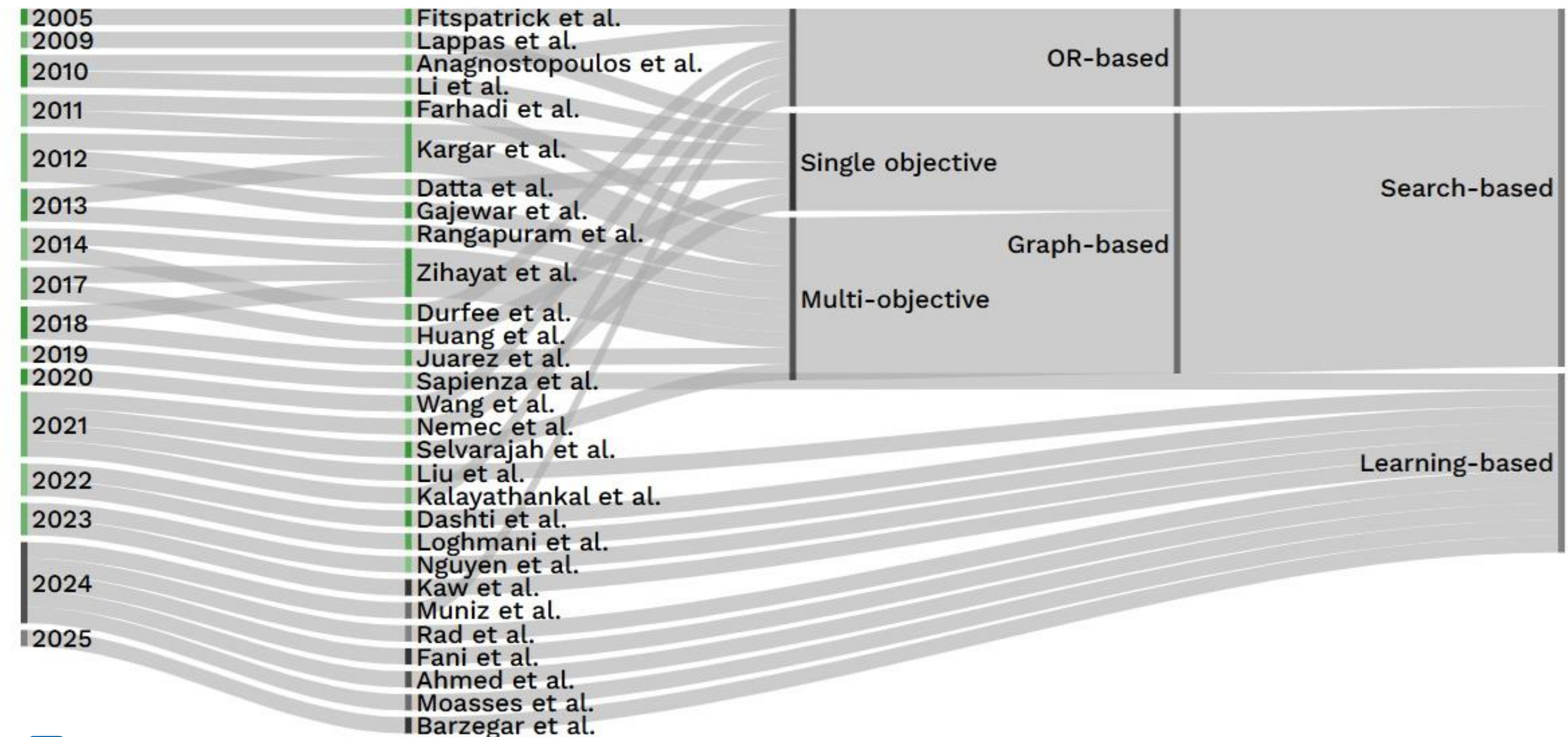
Abstract. Predicting *future* successful teams of experts who can effectively collaborate is challenging due to the experts' temporality of skill sets, levels of expertise, and collaboration ties, which is overlooked by prior work. Specifically, state-of-the-art neural-based methods learn vector representations of experts and skills in a *static* latent space, falling short of incorporating the possible drift and variability of experts' skills and collaboration ties in time. In this paper, we propose (1) a streaming-based training strategy for neural models to capture the evolution of experts' skills and collaboration ties over time and (2) to consume time information as an additional signal to the model for predicting future successful teams. We empirically benchmark our proposed method against state-of-the-art neural team formation methods and a strong temporal recommender system on datasets from varying domains with distinct distributions of skills and experts in teams. The results demonstrate neural models that utilize our proposed training strategy excel at efficacy in terms of classification and information retrieval metrics. The codebase is available at <https://github.com/fani-lab/OpENTF/tree/ecir24>.

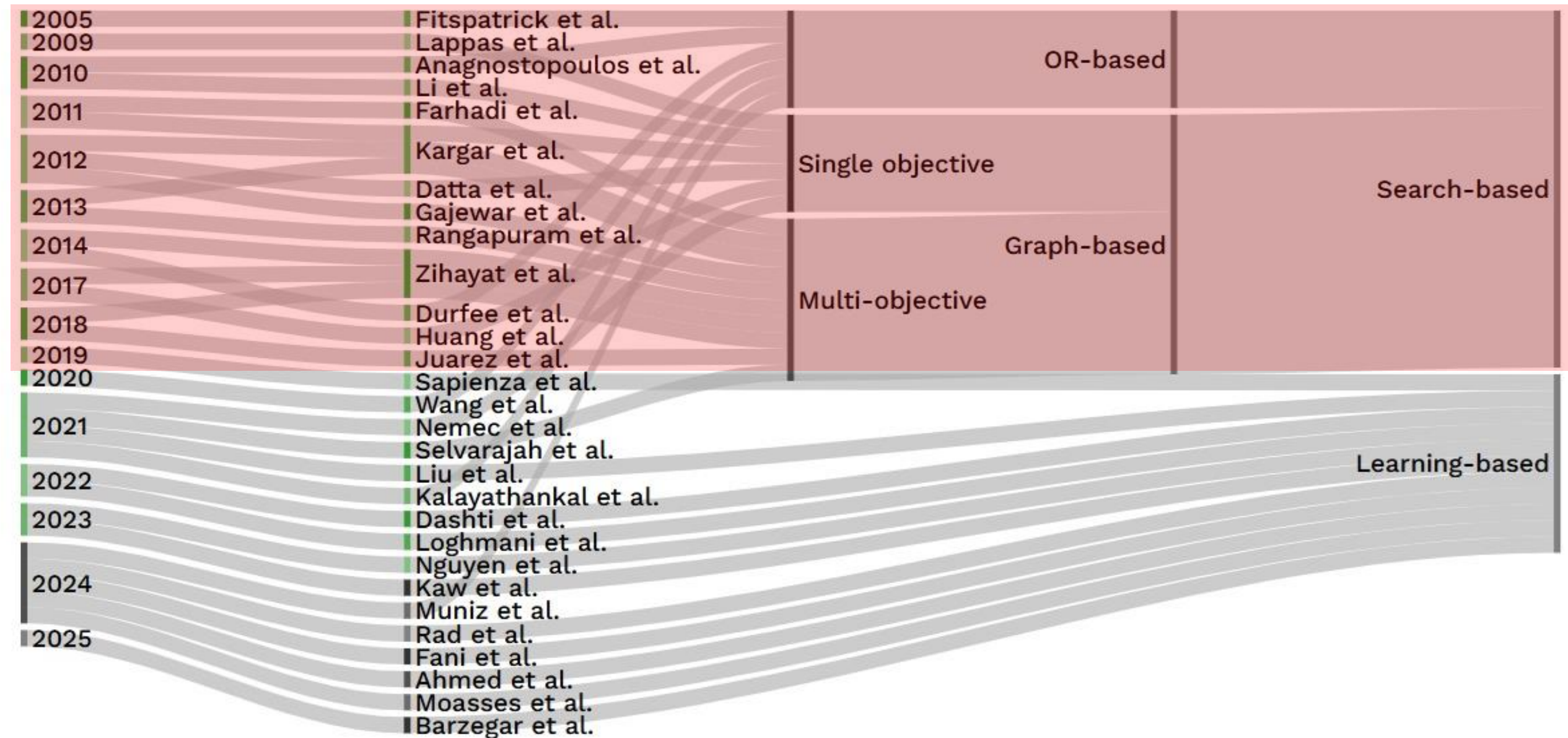
Keywords: Neural Team Formation · Training Strategy · OpENTF.

Published

39 Computational Approach

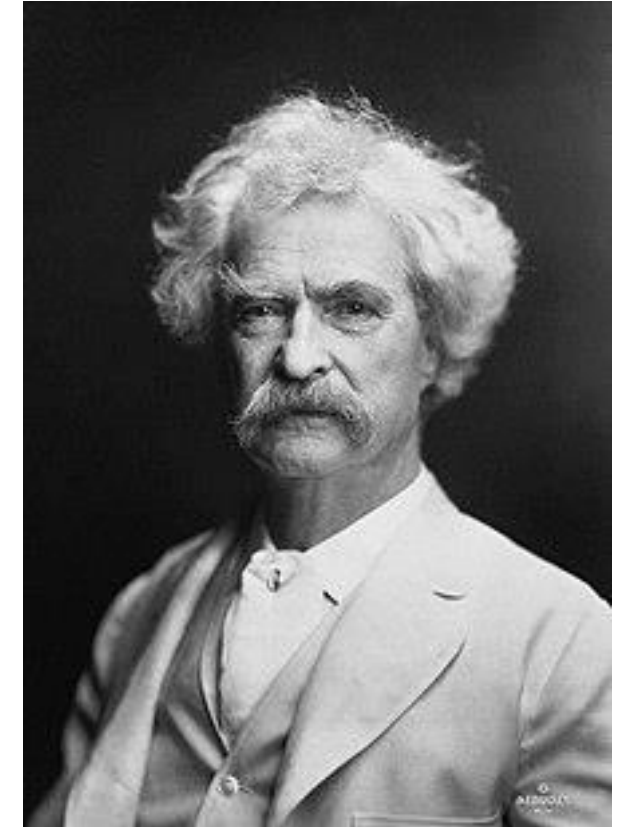




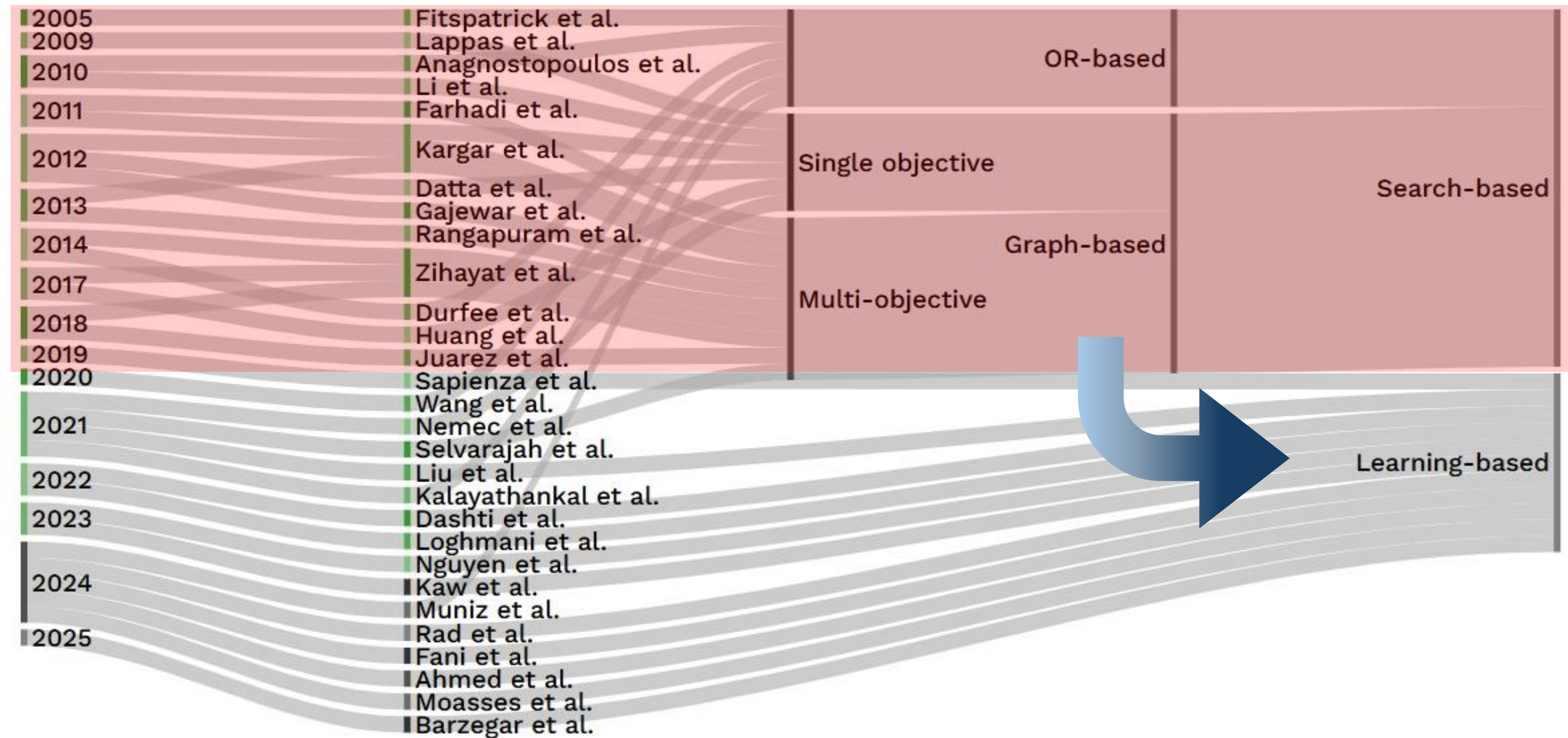


Core Objective

"There is no such thing as a new idea. It is impossible. We simply take a lot of old ideas and put them into a sort of mental kaleidoscope. We give them a turn and they make new and curious combinations. We keep on turning and making new combinations indefinitely; but they are the same old pieces of colored glass that have been in use through all the ages."



Samuel Langhorne Clemens (1835 – 1910)
Mark Twain, American Writer



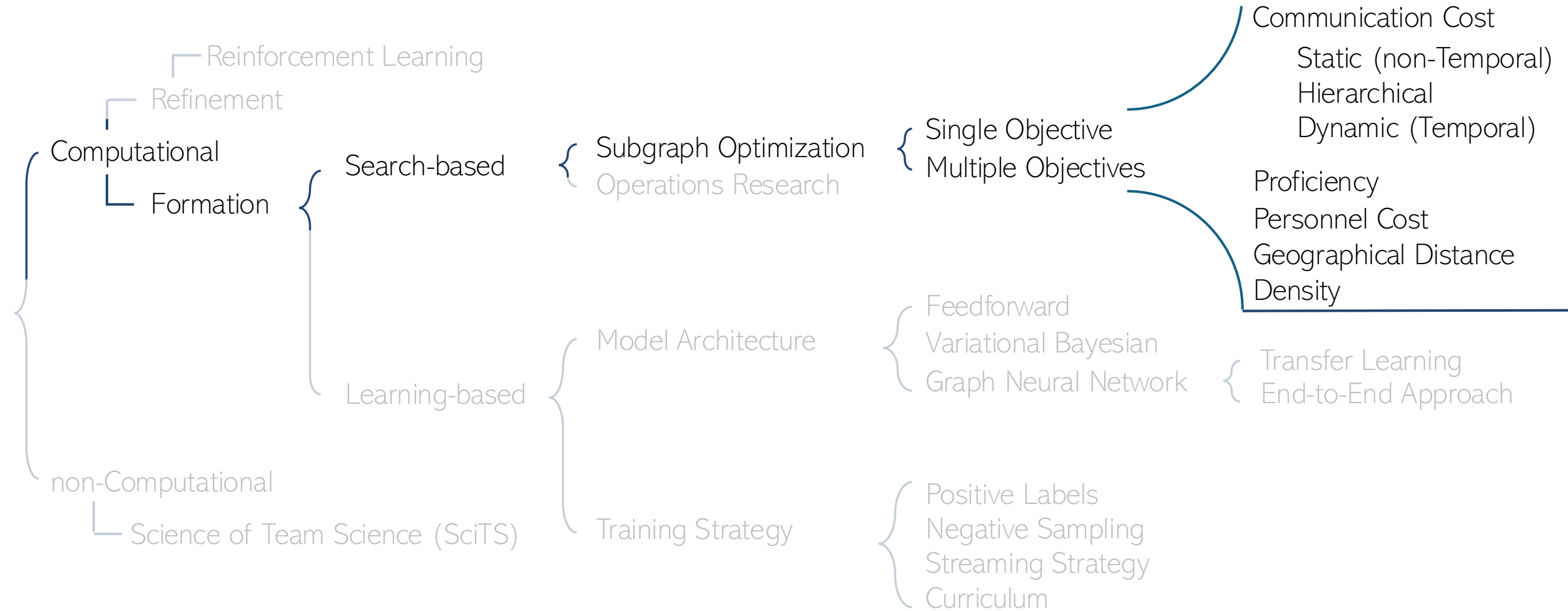
Distill the Knowledge for Neural Team Recommendation

Outline

- I) Introduction and Background
- II) Pioneering Techniques
- III) Learning-based Heuristics
- III) Team Refinement
- V) Challenges and New Perspectives
- VI) Applications

Hands-on: OpeNTF

45 Computational Approach



Team Allocation, Team Selection, Team Composition, Team Configuration, Team Recommendation, Team Formation

Graph-based Approach



Underlying **network structure** is a key for Effective Team

- Organizations' inherent hierarchical structure.
- Experts' social and collaborative ties.

Team Recommendation

→ Social Network Analysis

→ Graph Theory

[Miller, Computational Modeling and Organization Theories. AAAI Press, 2001]

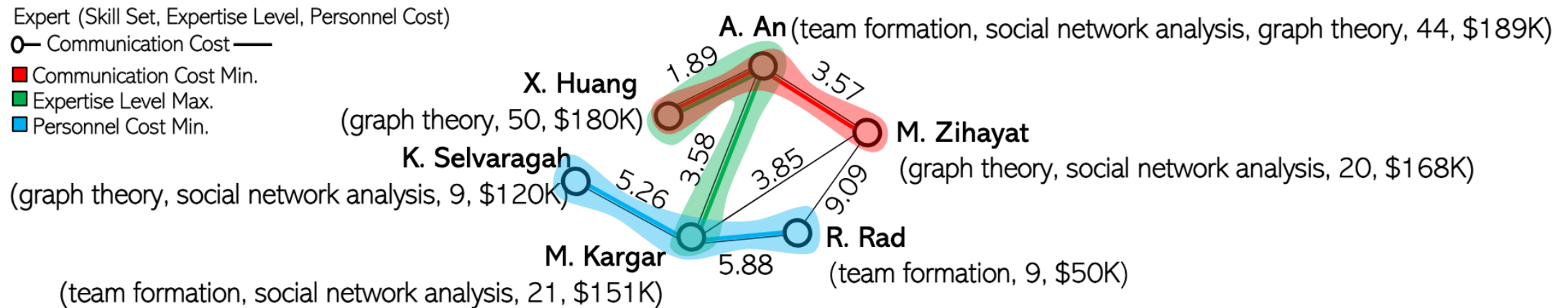
[Gaston et al., Proceedings of the 1st NAACSOS Conference, 2003]

[Gaston et al., AAAI Technical Report, 2004]

[Chen, INCoS, 2010]

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Subgraph Optimization on Expert Network



Undirected Homogeneous Weighted Attributed Graph:

Nodes	→	Experts
Node Attributes	→	Expert Skills, Salary, Expertise Level, ...
Weighted Edges	→	Depends on the Optimization Objective

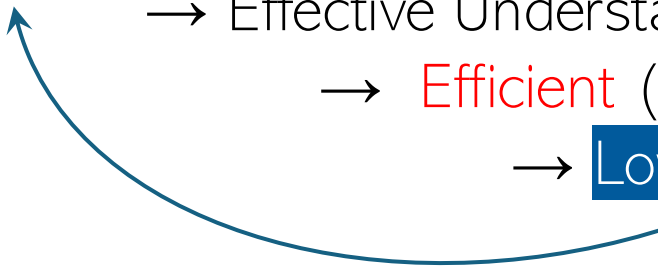
	Year	Hybrid	Communication Cost					Proficiency				Personnel Cost	Geographical Distance	Density	
			Diameter	MST	Leader Distance	Sum of Edge Weights	Random Walk	Sum of Distance	Trust Score	Expertise Level	Connector Authority				Skill Holder Authority
Lappas et al. [43]	2009		✓	✓											
Li et al. [92]	2010			✓											
Farhadi et al. [99] (2011)	2011		✓												
Kargar et al. [44]	2011				✓										
Datta et al. [93] (2012)	2012		✓	✓											
Kargar et al. [98]	2012												✓		
Gajewar et al. [105]	2012														✓
Kargar et al. [97]	2013		✓										✓		
Kargar et al. [96]	2013		✓										✓		
Rangapuram et al. [106]	2013													✓	✓
Zihayat et al. [94]	2014									✓			✓		
Huang et al. [107]	2016				✓										
Zihayat et al. [95]	2017					✓					✓	✓			
Zihayat et al. [84]	2018					✓					✓	✓	✓		
Juarez et al. [78]	2018									✓					✓
Nemec et al. [103]	2021						✓								
Selvarajah et al. [101]	2021							✓	✓					✓	

	Year	Hybrid	Communication Cost					Proficiency				Personnel Cost	Geographical Distance	Density	
			Diameter	MST	Leader Distance	Sum of Edge Weights	Random Walk	Sum of Distance	Trust Score	Expertise Level	Connector Authority				Skill Holder Authority
Lappas et al. [43]	2009		✓	✓											
Li et al. [92]	2010			✓											
Farhadi et al. [99] (2011)	2011		✓												
Kargar et al. [44]	2011				✓										
Datta et al. [93] (2012)	2012		✓	✓											
Kargar et al. [98]	2012												✓		
Gajewar et al. [105]	2012													✓	
Kargar et al. [97]	2013		✓										✓		
Kargar et al. [96]	2013		✓										✓		
Rangapuram et al. [106]	2013													✓	✓
Zihayat et al. [94]	2014								✓				✓		
Huang et al. [107]	2016				✓										
Zihayat et al. [95]	2017					✓					✓	✓			
Zihayat et al. [84]	2018					✓					✓	✓	✓		
Juarez et al. [78]	2018								✓						✓
Nemec et al. [103]	2021						✓								
Selvarajah et al. [101]	2021							✓	✓					✓	

Communication Cost $\leftrightarrow$ Team Performance

Effective Team

- Effective Understanding among Team Members
 - **Efficient** (Easy, Fast, ...) Communication
 - **Low Communication**



Communication Cost $\leftrightarrow$ Team Performance

More Joint Teamwork

→ Effective Understanding among Team Members

→ **Efficient** (Easy, Fast, ...) Communication

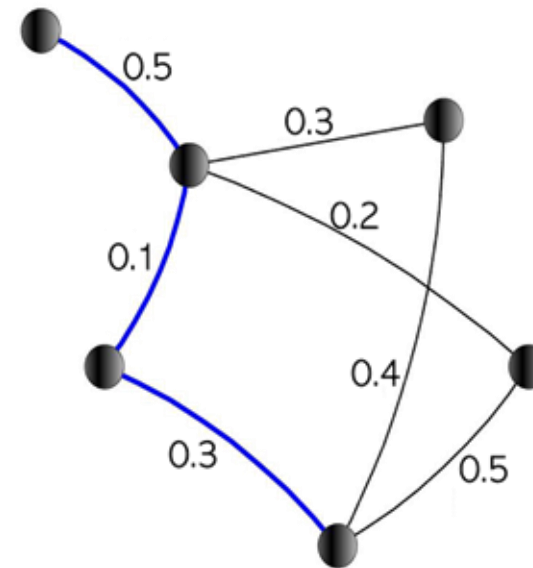
→ Low Communication

$$\omega_{\downarrow}(u, v) = 1 - \left| \frac{P_u \cap P_v}{P_u \cup P_v} \right| \in \mathfrak{R}^{[0,1]}$$

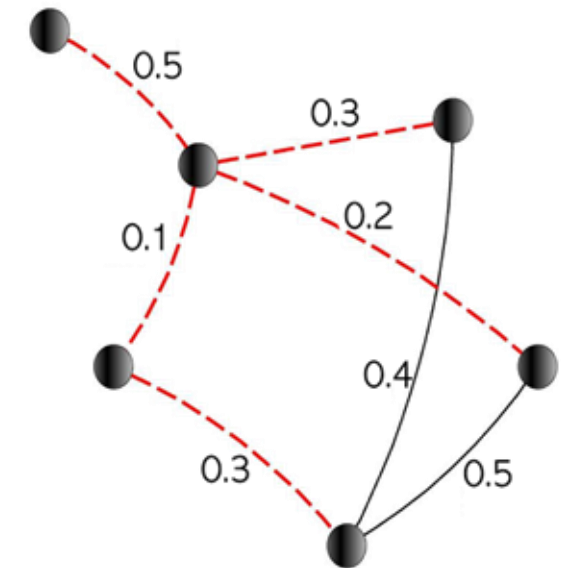
$$\omega_{\downarrow}(u, v) = \left| \frac{1}{P_u \cap P_v} \right| \in \mathfrak{R}^{(0,1] \cup \{\infty\}}$$

Communication Cost $\leftrightarrow$ Team Performance

Subgraph w/ Minimum Sum of Distances
Subgraph w/ Minimum Sum of Edge Weights
Subgraph w/ Minimum Diameter
Subgraph w/ Minimum Spanning Tree



Minimum Diameter



Minimum Spanning Tree

Communication Cost $\leftrightarrow$ Team Performance

More Joint Teamwork **Recently**

- Effective Understanding among Team Members
 - **Efficient** (Easy, Fast, ...) Communication
 - Low Communication

$$\min_{V_P \in \mathcal{P}(G)} \varphi_{DT}(G[V_P]) = \sum_{v, u \in V_P} d(v, u) + \alpha(t - t')$$

	Year	Hybrid	Communication Cost					Proficiency				Personnel Cost	Geographical Distance	Density	
			Diameter	MST	Leader Distance	Sum of Edge Weights	Random Walk	Sum of Distance	Trust Score	Expertise Level	Connector Authority				Skill Holder Authority
Lappas et al. [43]	2009		✓	✓											
Li et al. [92]	2010			✓											
Farhadi et al. [99] (2011)	2011		✓												
Kargar et al. [44]	2011				✓										
Datta et al. [93] (2012)	2012		✓	✓											
Kargar et al. [98]	2012												✓		
Gajewar et al. [105]	2012														✓
Kargar et al. [97]	2013		✓										✓		
Kargar et al. [96]	2013		✓										✓		
Rangapuram et al. [106]	2013													✓	✓
Zihayat et al. [94]	2014								✓				✓		
Huang et al. [107]	2016				✓										
Zihayat et al. [95]	2017					✓					✓	✓			
Zihayat et al. [84]	2018					✓					✓	✓	✓		
Juarez et al. [78]	2018								✓						✓
Nemec et al. [103]	2021						✓								
Selvarajah et al. [101]	2021							✓	✓					✓	

55

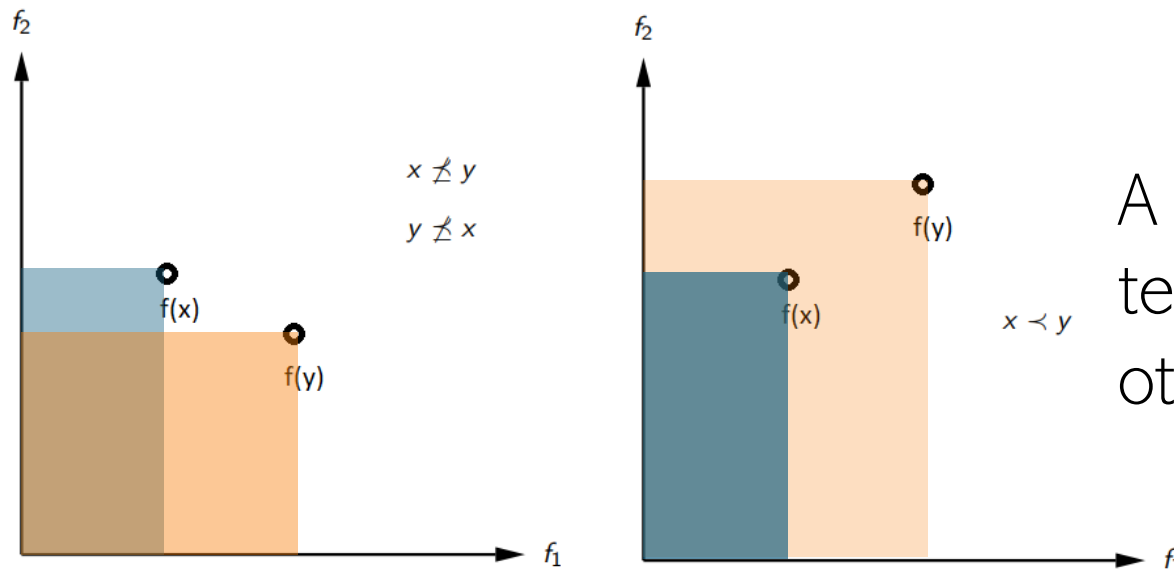
Multi-Objective → Optimizer

Interpolate (mainly linear) to a single edge weights

$$\omega_{\downarrow}(u, v) = \alpha f_1(u, v) + (1 - \alpha)f_2(u, v)$$

Pareto Optimality in Team Recommendation

[An et al. Finding affordable and collaborative teams from a network of experts. SIAM 2013.]



A subgraph is a Pareto Optimum team if it is not dominated by any other subgraphs.

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Multi-Objective Optimization

Communication Cost + Proficiency (e.g., h-index, #citations)

[Zihayat et. al., AMW, 2018], [Zihayat et. al., EDBT, 2017]

Communication Cost + Personnel Cost

[Aijun et. al., SDM, 2013], [Kargar et. al., CSE, 2013], [Zihayat et. al., AMW, 2018]

Communication Cost + Personnel Cost + Proficiency

[Zihayat et. al., WI-IAT, 2014]

Communication Cost + Geographical Proximity + Proficiency

[Selvarajah et. al., Expert Syst. Appl., 2021]

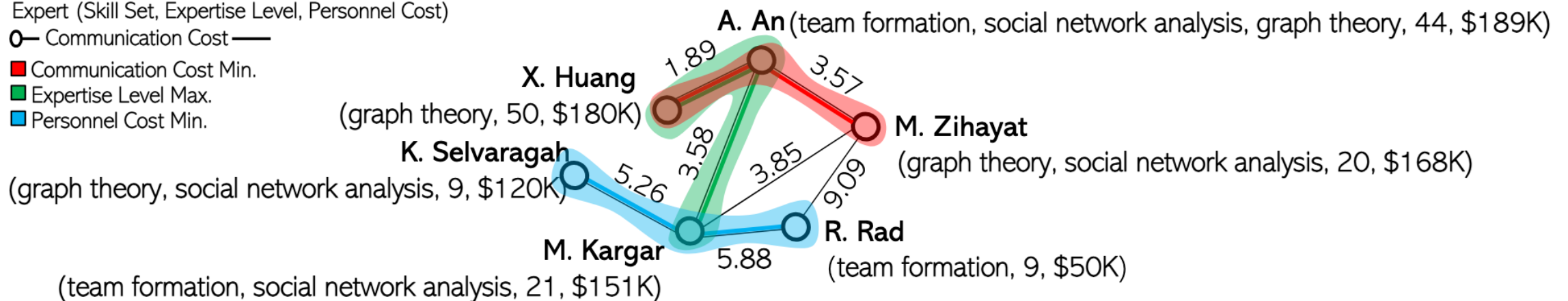
Expert (Skill Set, Expertise Level, Personnel Cost)

○ — Communication Cost —

■ Communication Cost Min.

■ Expertise Level Max.

■ Personnel Cost Min.



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Optimization Objective $\rightarrow$ Optimizer

Subgraph Optimization is **NP-hard**

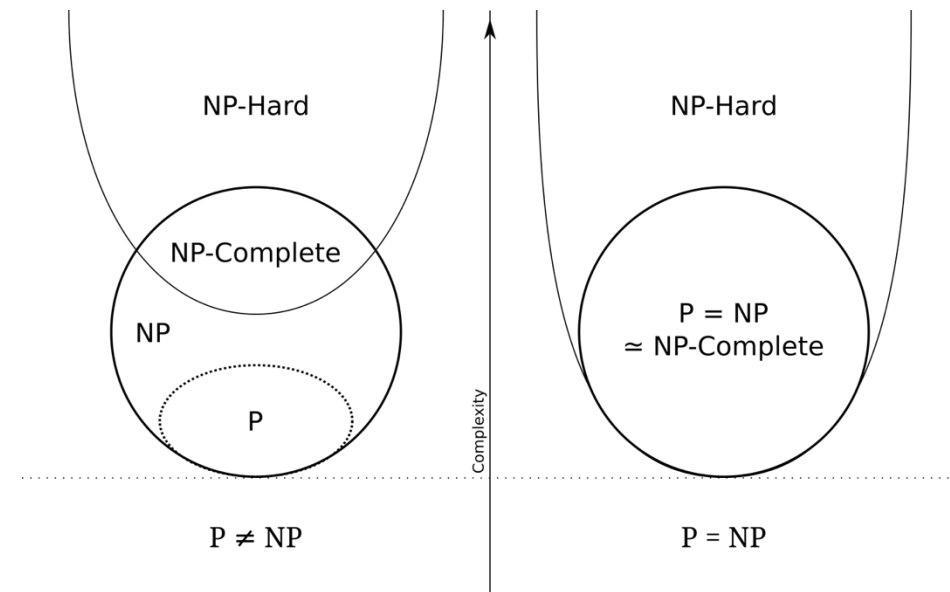
[Karp, Complexity of computer computations, Springer, 2010]



Heuristics for Polynomial Time

Greedy

Approximation



Optimization Objective $\rightarrow$ Optimizer

Lappas et al. Finding a team of experts in social networks, KDD 2009.

Algorithm 1 The **RarestFirst** algorithm for the DIAMETER-TF problem.

Input: Graph $G(\mathcal{X}, E)$; individuals' skill vectors $\{X_1, \dots, X_n\}$ and task T .

Output: Team $\mathcal{X}' \subseteq \mathcal{X}$ and subgraph $G[\mathcal{X}']$.

```

1: for every  $a \in T$  do
2:    $S(a) = \{i \mid a \in X_i\}$ 
3:  $a_{rare} \leftarrow \arg \min_{a \in T} |S(a)|$ 
4: for every  $i \in S(a_{rare})$  do
5:   for  $a \in T$  and  $a \neq a_{rare}$  do
6:      $R_{ia} \leftarrow d(i, S(a))$ 
7:    $R_i \leftarrow \max_a R_{ia}$ 
8:  $i^* \leftarrow \arg \min R_i$ 
9:  $\mathcal{X}' = i^* \cup \{Path(i^*, S(a)) \mid a \in T\}$ 

```

$$\mathcal{O}(|S(a_{rare})| \times n) \xrightarrow{|S(a_{rare})| = \mathcal{O}(n)} \mathcal{O}(n^2)$$

Required Skills: $T = \{gn, nn, ml\}$

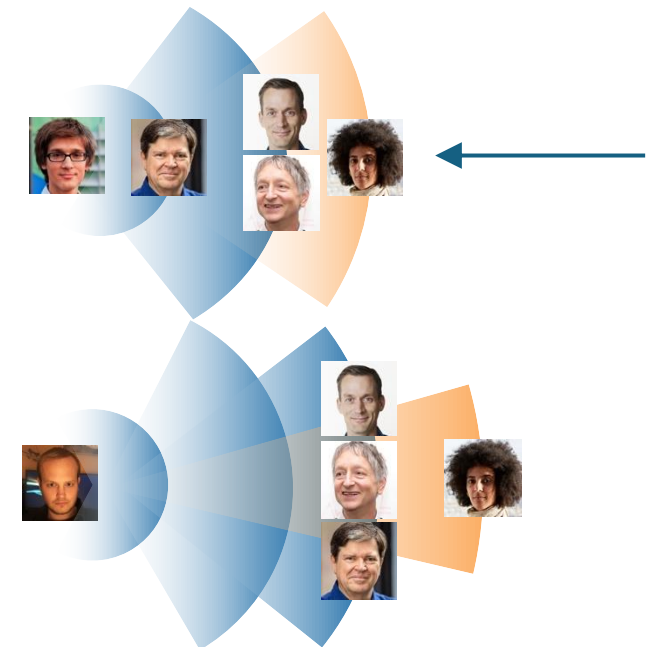
Support (gn) = 

Support (nn) = 

Support (ml) = 

Subgraph of Experts that Covers Required Skills

Start with Experts with the **Rarest Skills** as **Seed Support Group**



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A Survey of Subgraph Optimization for Expert Team Formation

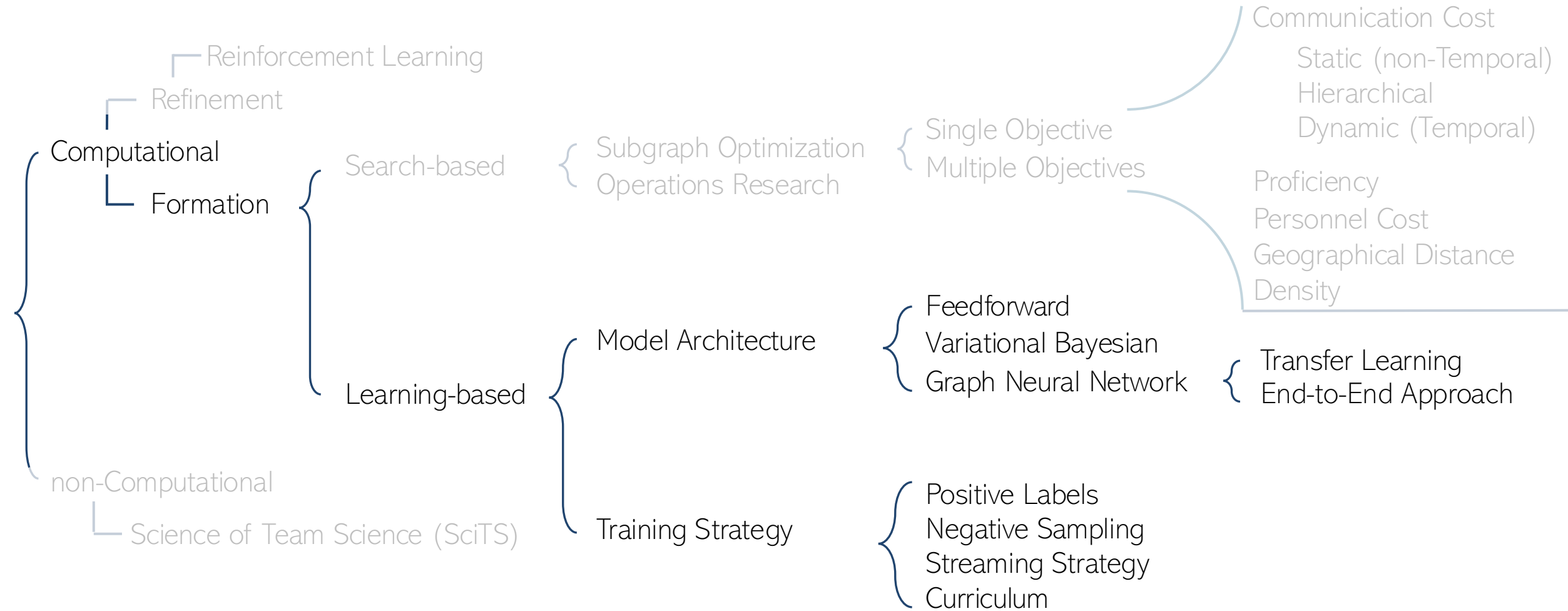
ACM Computing Surveys, 2025.

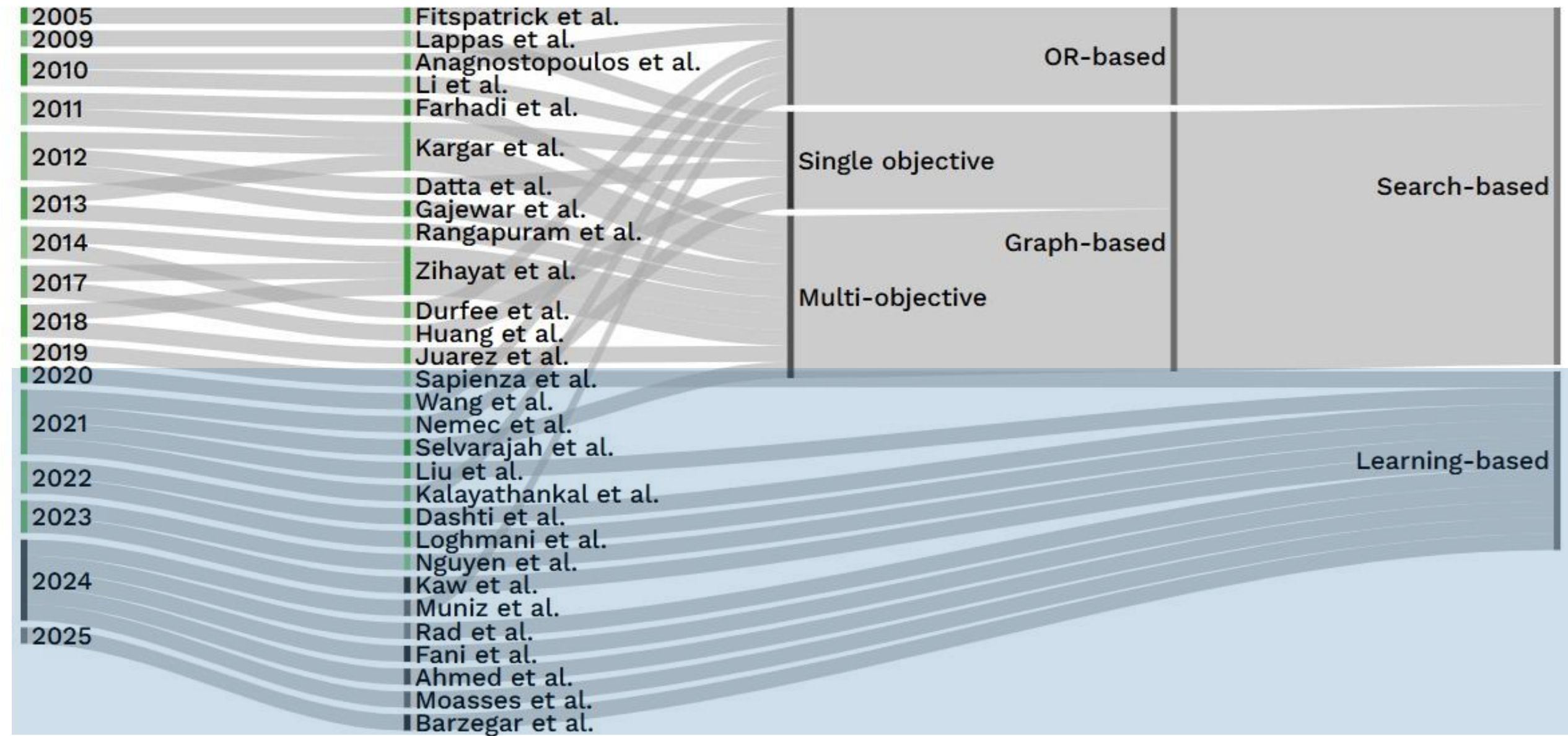
Outline

- I) Introduction and Background
- II) Pioneering Techniques
- III) Learning-based Heuristics
- III) Team Refinement
- V) Challenges and New Perspectives
- VI) Applications

Hands-on: OpeNTF

Computational Approach





Neural Team Recommendation

The Seventh Seal, Ingmar Bergman, 1957

1957: The Seventh Seal, Ingmar Bergman, Gunnar Fischer, Max von Sydow, Gunnar Björnstrand, Bengt Ekerot, Bibi Andersson, Nils Poppe
1957: Wild Strawberries, Ingmar Bergman, Gunnar Fischer, Sjöström, Bibi Andersson, Ingrid Thulin, Gunnar Björnstrand, Jullan Kindahl
1966: Persona, Ingmar Bergman, Sven Nykvist, Bibi Andersson, Liv Ullmann, Margaretha Krook, Gunnar Björnstrand, Jörgen Lindström
1972: Cries and Whispers, Ingmar Bergman, Sven Nykvist, Harriet Andersson, Ingrid Thulin, Liv Ullmann, Kari Sylwan, Erland Josephson
1982: Fanny and Alexander, Ingmar Bergman, Sven Nykvist, Bertil Guve, Pernilla Allwin, Ewa Fröling, Jan Malmström, Gunn Wållgren



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Neural Team Recommendation

The Seventh Seal, Ingmar Bergman, 1957

1984: After the Rehearsal, Ingmar Bergman, [...],



Neural Team Recommendation

Utilize past (un)successful teams as training samples.
Predict future teams whose success is **almost surely** guaranteed!

Efficiency: iterative and online learning procedures

Efficacy: improved prediction and team performance

Scalability: can handle larger pool of experts

Dynamic Adaptation: better suited for temporal studies

Neural Team Recommendation → Team

Experts Pool

Skill Pool

Definition 1 (Team). Given a set of skills $\mathcal{S} = \{i\}$ and a set of experts $\mathcal{E} = \{j\}$, a team of experts $\mathbf{e} \subseteq \mathcal{E}$; $\mathbf{e} \neq \emptyset$ that collectively cover the skill set $\mathbf{s} \subseteq \mathcal{S}$; $\mathbf{s} \neq \emptyset$ is shown by $(\mathbf{s}, \mathbf{e})$ along with its success status $y \in \{0, 1\}$. Further, $\mathcal{T} = \{(\mathbf{s}, \mathbf{e})_y : y \in \{0, 1\}\}$ indexes all previous teams.

Team is a subset of experts for a subset of required skills

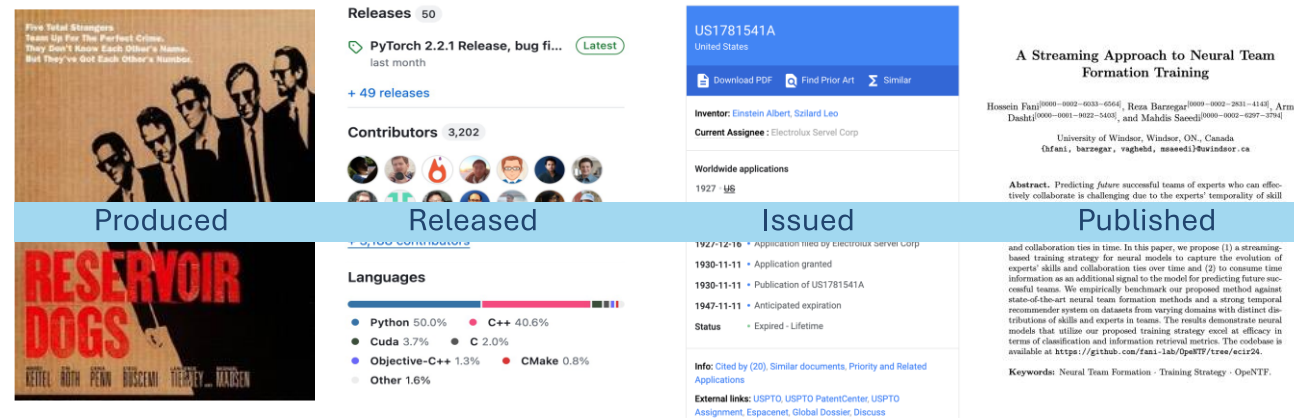
Label of Success

Neural Team Recommendation $\rightarrow$ Problem

The Seventh Seal, Ingmar Bergman, 1957

Learn a function f of parameters θ from the powerset of skills S to the powerset of E

$$f_{\theta}(s) = e^* \rightarrow (s, e^*)_{v=1} ; f_{\theta}(s) \neq e' \rightarrow (s, e')_{v=0}$$



Teams	Movies	Software Repos	Patents	Papers
Skills	Genres	Programing Languages	Subclasses	Keywords
Experts	Cast'nCrew	Programmers	Inventors	Authors

Neural Team Recommendation → Solutions

The Seventh Seal, Ingmar Bergman, 1957

Maximizing A Posterior (MAP) of θ in $\mathbf{f}_\theta$ over training set $\mathcal{T}$ in a multilayer neural network

$$\operatorname{argmax}_{\theta} p(\theta|\mathcal{T}) \propto p(\mathcal{T}|\theta)p(\theta) = p(\theta) \prod_{(\mathbf{s}, \mathbf{e}) \in \mathcal{T}} p(\mathbf{e}|\mathbf{s}, \theta)$$



Produced

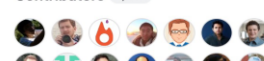


Releases 50

PyTorch 2.2.1 Release, bug fi...
last month

+ 49 releases

Contributors 3,202



Released

Languages



Inventor: Einstein Albert, Szilard Leo

Current Assignee: Electrolux Servel Corp

Worldwide applications

1927 - US

Issued

1927-12-18 • Application filed by Electrolux Servel Corp
1930-11-11 • Application granted
1930-11-11 • Publication of US1781541A
1947-11-11 • Anticipated expiration
Status • Expired - Lifetime
Info: Cited by (20), Similar documents, Priority and Related Applications
External links: USPTO, USPTO PatentCenter, USPTO Assignment, Espacenet, Global Dossier, Discuss

A Streaming Approach to Neural Team Formation Training

Hossein Fanj^[0000-0002-0031-0506], Reza Barzegar^[0009-0002-2831-4140], Arman Dashti^[0000-0001-0022-5400], and Mahdi Saadat^[0000-0002-0297-3704]
University of Windsor, Windsor, ON, Canada
{rfan1, barzegar, vaghebi, msaadi}@uwindsor.ca

Abstract. Predicting future successful teams of experts who can effectively collaborate is challenging due to the experts' temporality of skill

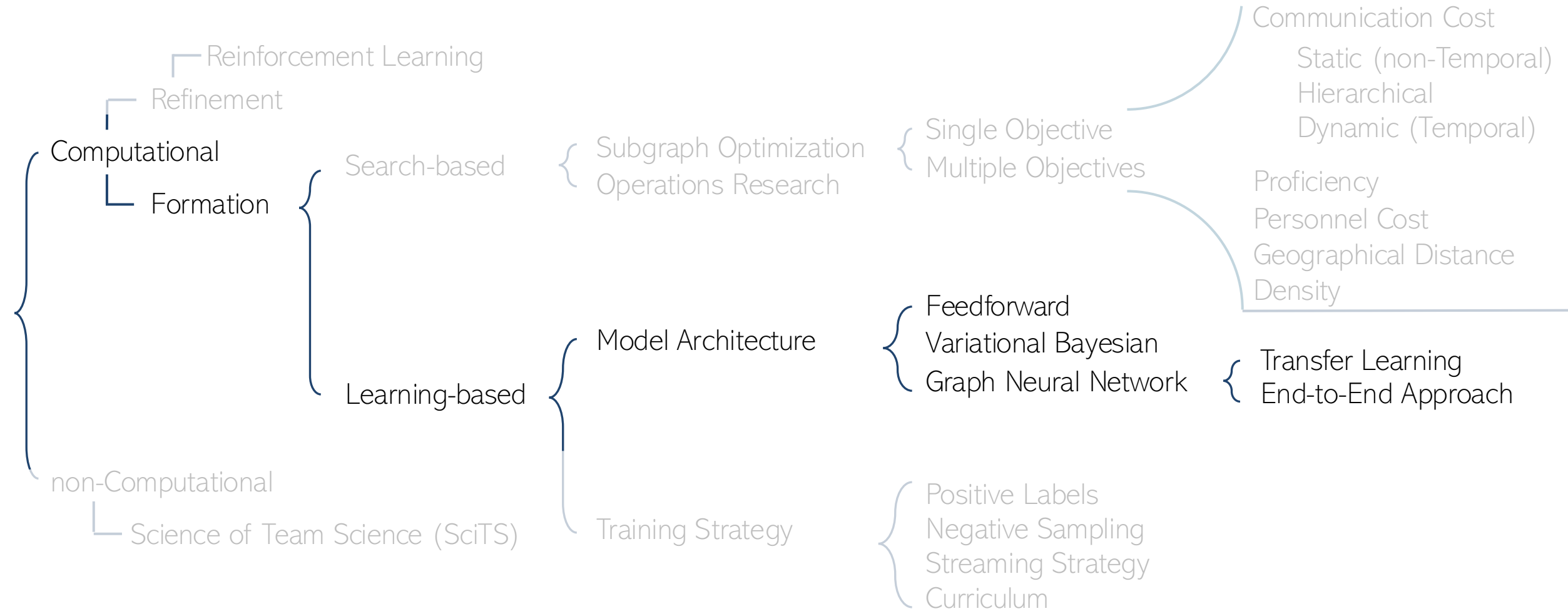
and collaboration ties in time. In this paper, we propose (1) a streaming-based training strategy for neural models to capture the evolution of experts' skills and collaboration ties over time and (2) to consume time information as an additional signal to the model for predicting future successful teams. We empirically benchmark our proposed method against state-of-the-art neural team formation methods and a strong temporal recommender system on datasets from varying domains with distinct distributions of skills and experts in teams. The results demonstrate neural models that utilize our proposed training strategy excel at efficacy in terms of classification and information retrieval metrics. The codebase is available at <https://github.com/fan1-lab/Opentf/tree/6c1c296>.

Keywords: Neural Team Formation · Training Strategy · Opentf.

Published

Teams	Movies	Software Repos	Patents	Papers
Skills	Genres	Programing Languages	Subclasses	Keywords
Experts	Cast'nCrew	Programmers	Inventors	Authors

70 Computational Approach

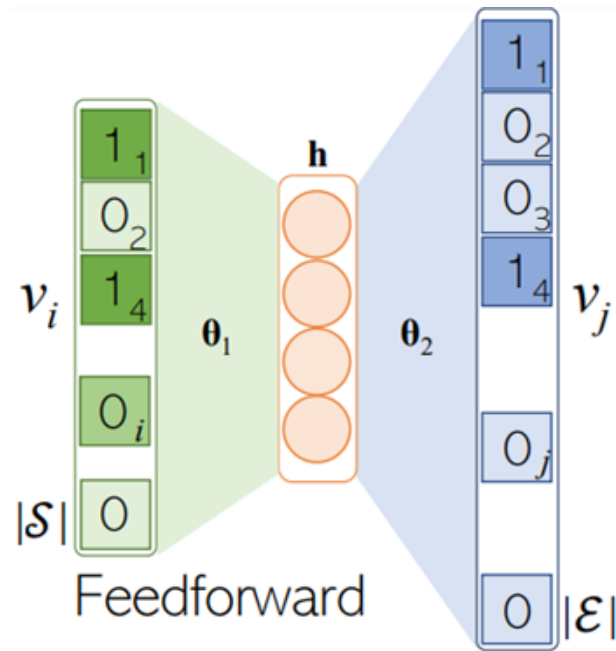


Rad et al. CIKM2020, CIKM2021, TOIS2023

Learning to Form Skill-based Teams of Experts. CIKM 2020.

PyTFL: A Python-based Neural Team Formation Toolkit. CIKM 2021.

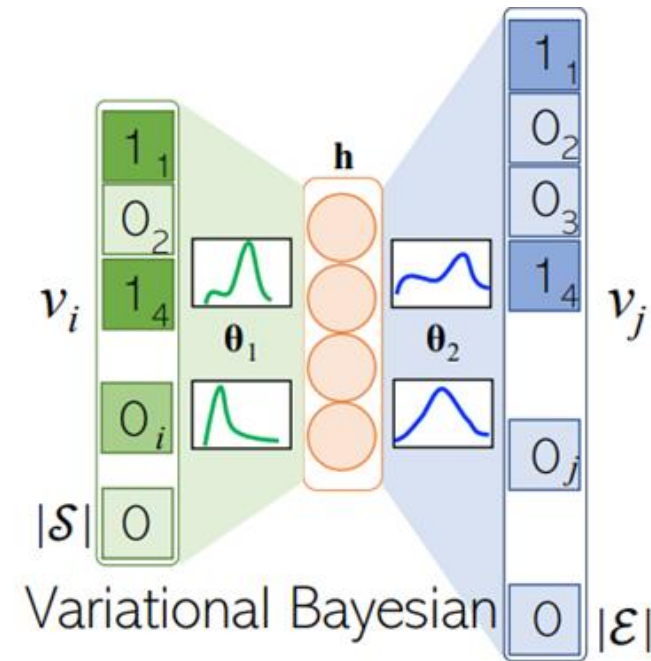
A Variational Neural Architecture for Skill-based Team Formation. TOIS, 42(1): 1-28, 2023.



$$\mathbf{h} = \pi(\boldsymbol{\theta}_1 \mathbf{v}_s + \mathbf{b}_1)$$

$$\text{logits} \rightarrow \mathbf{z} = \boldsymbol{\theta}_2 \mathbf{h} + \mathbf{b}_2$$

$$\mathbf{v}_e = \sigma(\mathbf{z})$$



$$\boldsymbol{\theta} \sim (\boldsymbol{\mu}, \boldsymbol{\sigma}); q(\boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}) \rightarrow \min \text{KL}(q \parallel p)$$

Minimizing the **Kullback-Leibler Divergence** between $q(\boldsymbol{\theta})$ and $p(\boldsymbol{\theta})$

Challenges

Learning to Form Skill-based Teams of Experts. CIKM 2020.

PyTFL: A Python-based Neural Team Formation Toolkit. CIKM 2021.

A Variational Neural Architecture for Skill-based Team Formation. TOIS, 42(1): 1-28, 2023.

Large set
- End-to-End GNN

Small vs. Large set

- o Dense Representation Learning
- GNN-based (Rad et al. SIGIR 2021)

Definition 1 (Team). Given a set of skills $S = \{i\}$ and a set of experts $\mathcal{E} = \{j\}$, a team of experts $e \subseteq \mathcal{E}$; $e \neq \emptyset$ that collectively cover the skill set $s \subseteq S$; $s \neq \emptyset$ is shown by (s, e) along with its success status $y \in \{0, 1\}$. Further, $\mathcal{T} = \{(s, e)_y : y \in \{0, 1\}\}$ indexes all previous teams.

Lack of negative samples (unsuccessful teams)

Challenges

Learning to Form Skill-based Teams of Experts. CIKM 2020.

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A Variational Neural Architecture for Skill-based Team Formation. TOIS, 42(1): 1-28, 2023.

Large set
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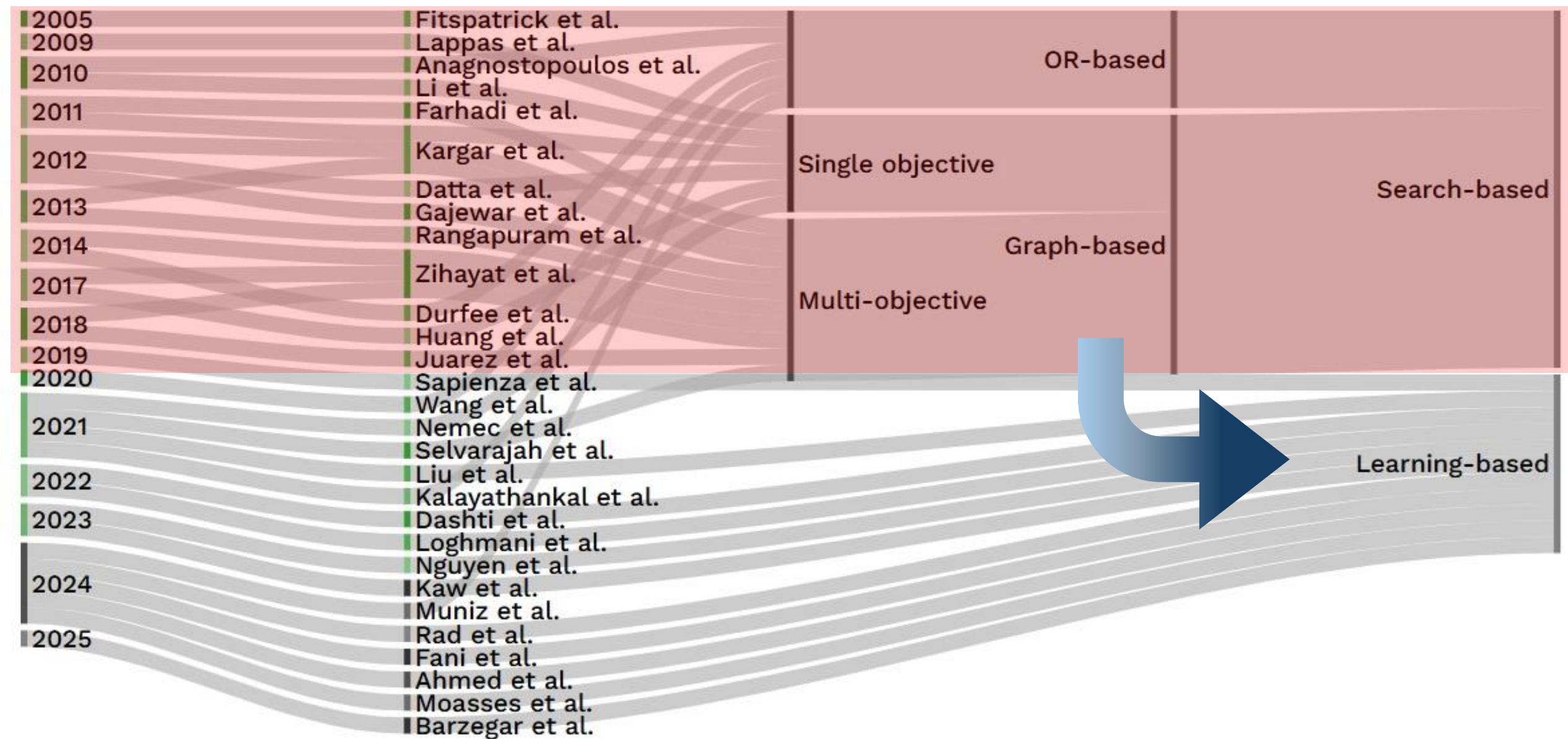
- o Dense Representation Learning
 - GNN-based (Rad et al. SIGIR 2021)

Biases

Popularity Bias

Gender Bias (Male Dominant)

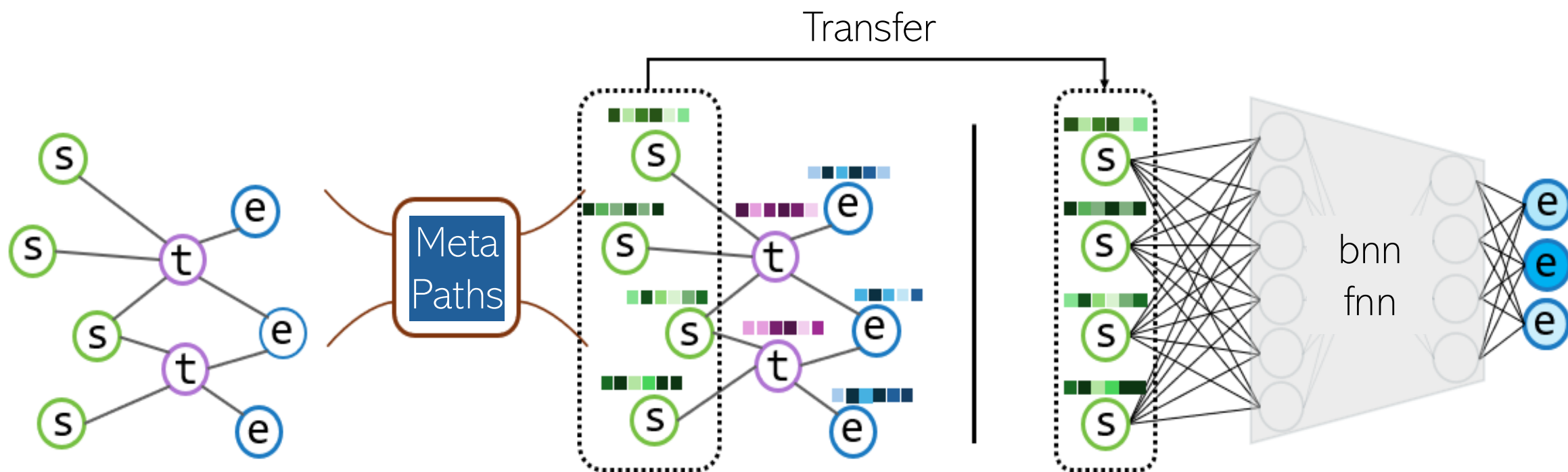
Lack of negative samples (unsuccessful)



75

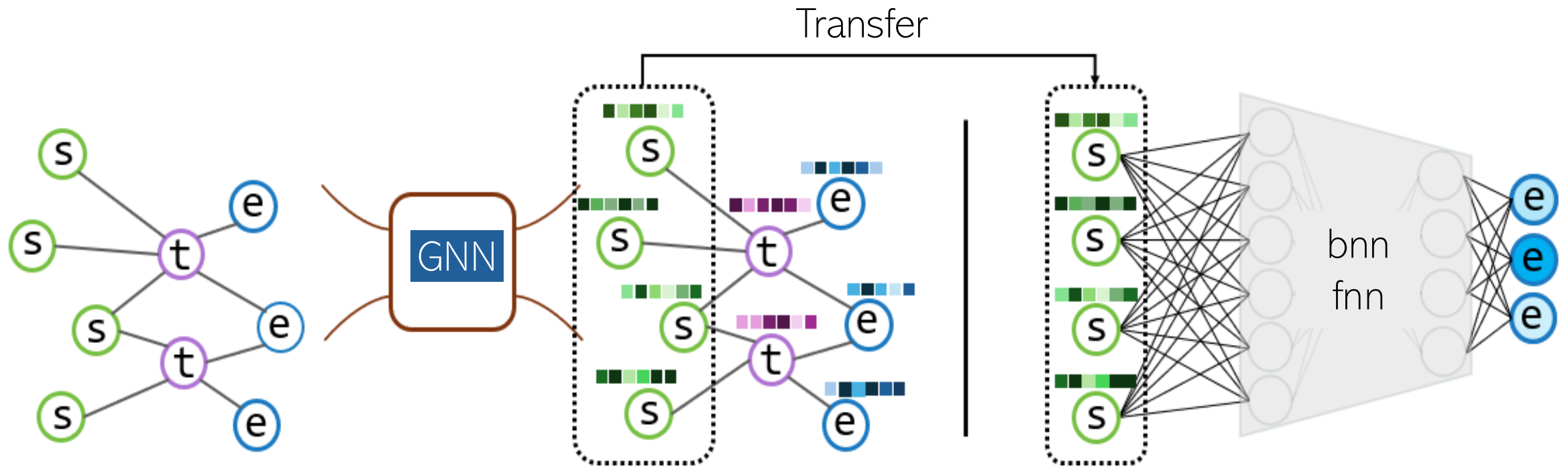
Graph Representation Learning → Skills

Rad et. al., Retrieving skill-based teams from collaboration networks. SIGIR 2021.



Graph Representation Learning → Skills

Ahmed et. al., Skill Vector Representation Learning for Collaborative Team Recommendation: A Comparative Study. WISE 2024.

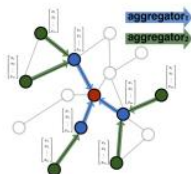
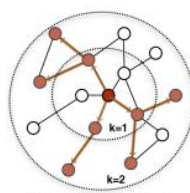


Message Passing-based GNN

$$v_{s_i}^{k+1} = \text{comb}(v_{s_i}^k, \text{agg}(v_x^k : x \in \mathcal{N}_{s_i}^k))$$

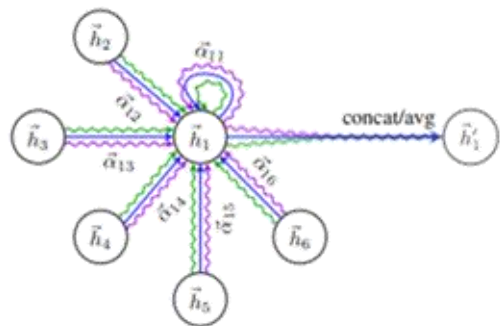
Graph Neural Network → Skills

Graph SAmple-aggreGatE (GraphSAGE) [Hamilton et. al., NIPS 2017]



$$\begin{aligned} \mathbf{h}_{\mathcal{N}(v)}^k &\leftarrow \text{AGGREGATE}_k(\{\mathbf{h}_u^{k-1}, \forall u \in \mathcal{N}(v)\}); \\ \mathbf{h}_v^k &\leftarrow \sigma \left(\mathbf{W}^k \cdot \text{CONCAT}(\mathbf{h}_v^{k-1}, \mathbf{h}_{\mathcal{N}(v)}^k) \right) \end{aligned} \quad \left\{ \begin{array}{l} \text{LSTM} \\ \max(\{\sigma(\mathbf{W}_{\text{pool}} \mathbf{h}_{u_i}^k + \mathbf{b}), \forall u_i \in \mathcal{N}(v)\}) \\ \text{MEAN}(\{\mathbf{h}_v^{k-1}\} \cup \{\mathbf{h}_u^{k-1}, \forall u \in \mathcal{N}(v)\}) \end{array} \right.$$

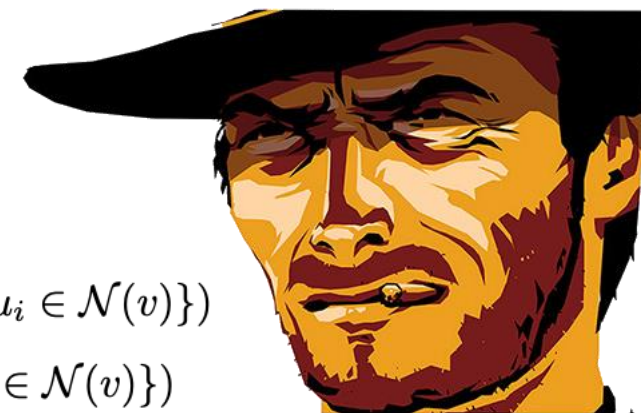
Graph Attention Network (GAT) [Velickovic et. al., ICLR 2018]



$$\vec{h}'_i = \sigma \left(\frac{1}{K} \sum_{k=1}^K \sum_{j \in \mathcal{N}_i} \alpha_{ij}^k \mathbf{W}^k \vec{h}_j \right) \quad \alpha_{ij} = \frac{\exp \left(\text{LeakyReLU} \left(\vec{\mathbf{a}}^T [\mathbf{W} \vec{h}_i \| \mathbf{W} \vec{h}_j] \right) \right)}{\sum_{k \in \mathcal{N}_i} \exp \left(\text{LeakyReLU} \left(\vec{\mathbf{a}}^T [\mathbf{W} \vec{h}_i \| \mathbf{W} \vec{h}_k] \right) \right)}$$

Graph Isomorphism Network (GIN) [Xu et al. ICLR 2019]

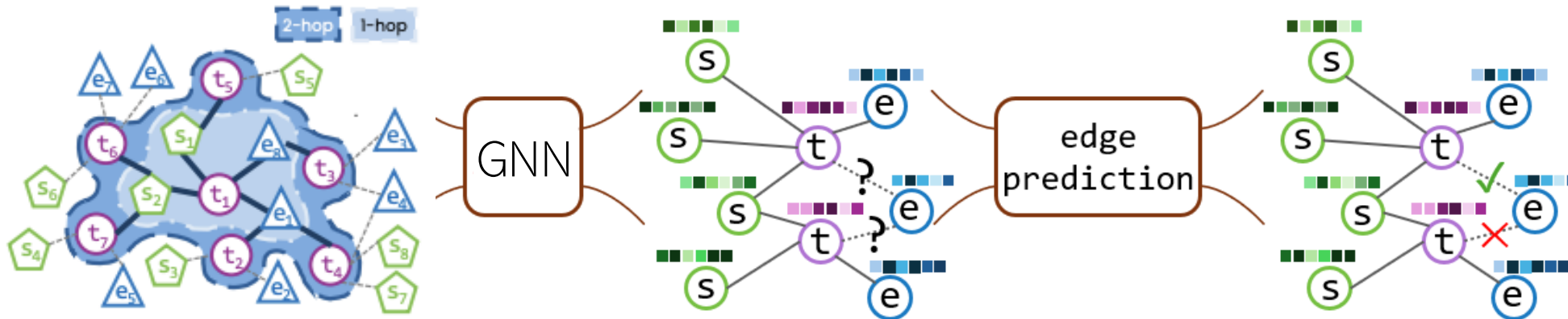
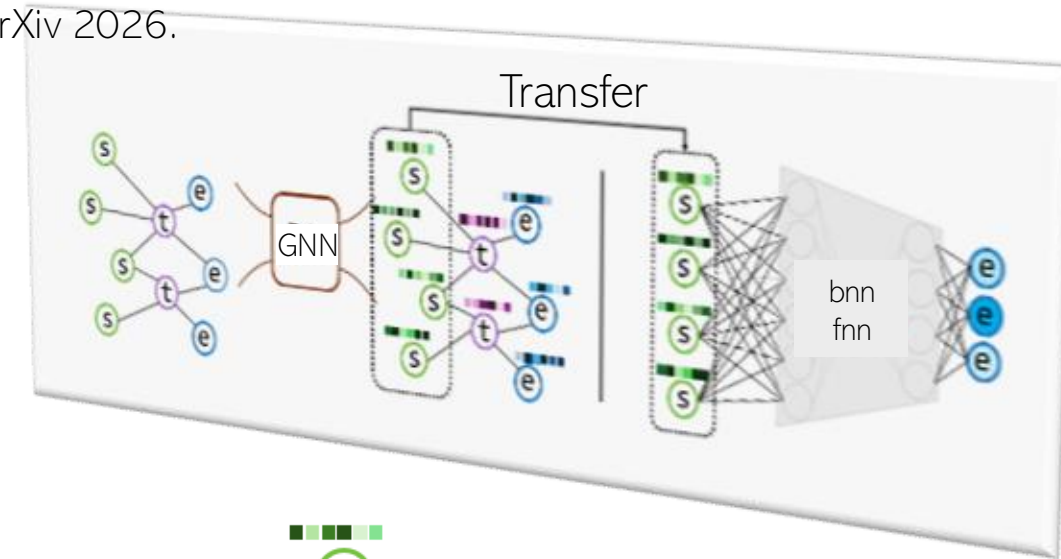
$$h_v^{(k)} = \phi \left(h_v^{(k-1)}, f \left(\left\{ h_u^{(k-1)} : u \in \mathcal{N}(v) \right\} \right) \right)$$



Graph Neural Team Recommendation → End-to-End

Ahmed et. al., Graph Neural Team Recommendation: An Integrated Approach. arXiv 2026.

- + Captures Cross-Team Collaborations (Multi-Hup)
- + One-Step Training Phase
- + No Curse of Sparsity in the Last Layer



Graph Neural Team Recommendation → End-to-End

Efficacy of end-to-end vs. transfer-based approaches in dblp and imdb using skill-team-expert graph structure.

		dblp								imdb							
		%pr		%rec		%ndcg		%map		%pr		%rec		%ndcg		%map	
		@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10
e2e	gs	14.37	09.24	54.00	67.79	33.21	38.39	25.18	28.10	15.39	09.16	49.73	57.44	33.70	36.78	27.00	29.03
	gin	14.47	09.07	54.68	66.95	31.50	36.12	22.74	25.36	14.08	08.91	45.34	56.49	28.49	32.79	21.76	24.30
	gine	07.66	07.65	21.56	44.17	16.50	24.74	12.67	16.18	19.33	12.21	53.14	66.25	38.80	43.86	31.13	34.07
	gat	14.33	09.46	53.81	68.86	38.11	43.78	31.44	34.65	14.66	09.17	47.44	57.47	33.45	37.40	27.17	29.68
	gatv2	14.03	09.48	52.73	69.01	37.87	43.98	31.43	34.86	14.49	09.09	46.72	57.18	33.81	37.88	27.86	30.39
	han	28.97	15.17	83.19	86.38	70.42	71.74	64.28	65.27	23.37	15.42	57.82	74.15	45.58	52.38	37.42	42.21
t-fnn	m2v	00.82	00.70	01.24	02.13	01.13	01.55	00.66	00.82	00.84	00.77	00.95	01.76	00.95	01.33	00.46	00.57
	lant	00.90	00.83	01.35	02.48	01.17	01.69	00.67	00.84	00.74	00.68	00.81	01.52	00.85	01.18	00.40	00.50
	gs	00.87	00.74	01.30	02.24	01.26	01.63	00.60	00.76	00.83	00.73	00.91	01.64	00.95	01.27	00.44	00.55
	gin	00.95	00.78	01.43	02.35	01.37	01.57	00.59	00.76	00.76	00.68	00.84	01.56	00.87	01.19	00.40	00.51
	gine	01.02	00.83	01.52	02.49	01.35	01.80	00.76	00.92	00.73	00.64	00.82	01.45	00.85	01.14	00.41	00.50
	gat	01.19	00.92	01.78	02.75	01.62	02.08	00.92	01.09	00.92	00.82	01.00	01.83	01.06	01.43	00.50	00.62
	gatv2	01.01	00.83	01.51	02.51	01.41	01.88	00.80	00.96	00.87	00.80	00.96	01.80	00.98	01.37	00.47	00.59
	han	00.93	00.82	01.42	02.49	01.33	01.82	00.80	00.97	00.77	00.72	00.86	01.63	00.85	01.21	00.39	00.50
t-bnn	m2v	00.31	00.32	00.48	00.98	00.42	00.65	00.23	00.30	00.76	00.71	00.83	01.58	00.85	01.20	00.41	00.52
	lant	00.66	00.61	01.01	01.87	00.88	01.28	00.49	00.61	01.01	00.96	01.14	02.18	01.18	01.66	00.58	00.73
	gs	00.71	00.66	01.08	02.00	00.94	01.36	00.52	00.65	00.99	00.90	01.12	02.03	01.12	01.55	00.54	00.67
	gin	00.60	00.52	00.91	01.59	00.84	01.15	00.49	00.60	00.90	00.83	01.04	01.92	01.07	01.47	00.52	00.65
	gine	00.75	00.63	01.12	01.90	01.08	01.44	00.66	00.79	01.01	00.93	01.15	02.14	01.16	01.62	00.58	00.71
	gat	00.66	00.62	00.99	01.86	00.87	01.28	00.49	00.62	00.96	00.86	01.07	01.94	01.10	01.50	00.53	00.66
	gatv2	00.64	00.54	00.96	01.64	00.87	01.19	00.49	00.58	00.93	00.80	01.05	01.84	01.05	01.41	00.51	00.62
	han	00.80	00.71	01.20	02.15	01.09	01.53	00.61	00.76	01.06	00.99	01.19	02.25	01.21	01.70	00.59	00.74

Neural Team Recommendation → Pioneer

Paradigm Shift to Learning-based Methods

- Teams for Online Games



ORIGINAL RESEARCH
published: 13 June 2019
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Deep Neural Networks for Optimal Team Composition

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USC Information Sciences Institute, Los Angeles, CA, United States

Cooperation is a fundamental social mechanism, whose effects on human performance have been investigated in several environments. Online games are modern-days natural settings in which cooperation strongly affects human behavior. Every day, millions of players connect and play together in team-based games: the patterns of cooperation can either foster or hinder individual skill learning and performance. This work has three goals: (i) identifying teammates' influence on players' performance in the short and long term, (ii) designing a computational framework to recommend teammates to improve players' performance, and (iii) setting to demonstrate that such improvements can be predicted via deep learning. We leverage a large dataset from Dota 2, a popular Multiplayer Online Battle Arena game. We generate a directed co-play network, whose links' weights depict the effect of teammates on players' performance. Specifically, we propose a measure of network influence that captures skill transfer from player to player over time. We then use such framing to design a recommendation system to suggest new teammates based on a modified deep neural autoencoder and we demonstrate its state-of-the-art recommendation performance. We finally provide insights into skill transfer effects: our experimental results demonstrate that such dynamics can be predicted using deep neural networks.

Keywords: recommendation system, link prediction, deep neural network, graph factorization, multiplayer online games

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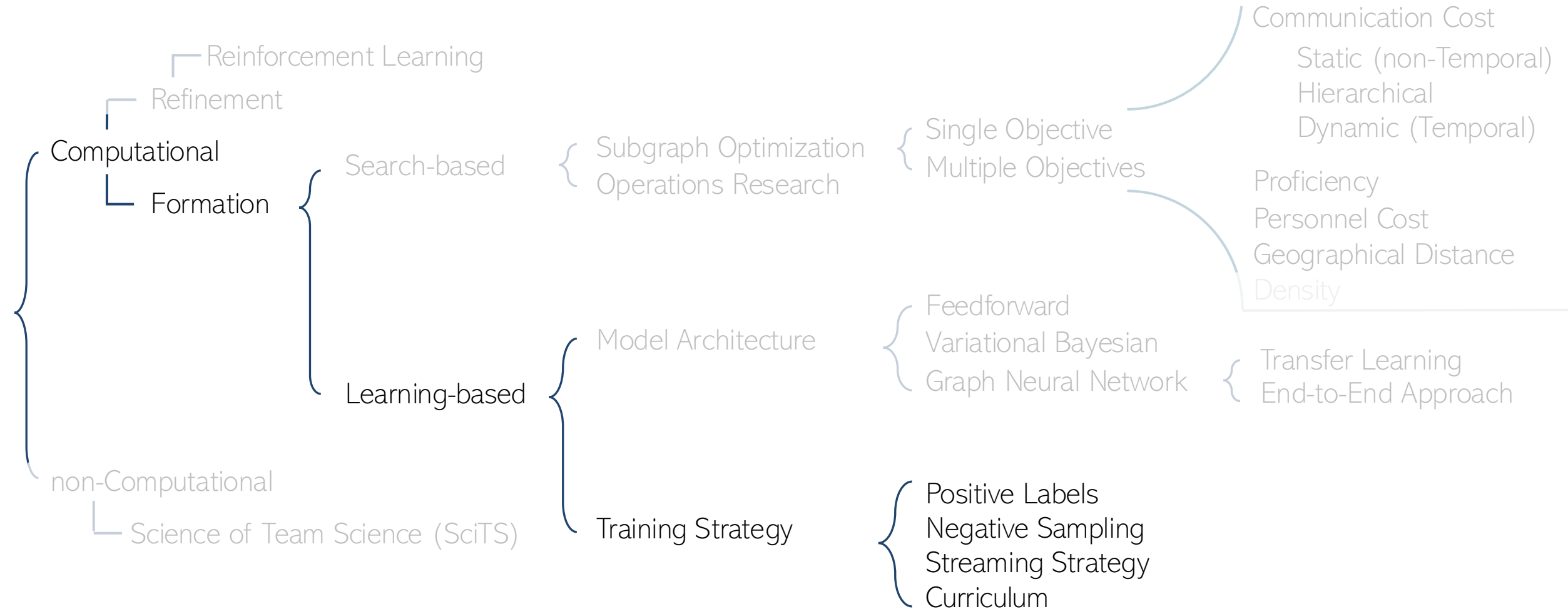
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Emilio Ferrara

86 Computational Approach



87

Negative Sampling

Most training sets only consists of successful teams!



Hamid Sourian, 59KG Greco-Roman Wrestler, Iran, Rio Olympic 2016
7 - 0 → Lost to a Fall (Pin)

88

Negative Sampling → Closed-World Assumption

No currently known (successful) team



Unsuccessful



Medal record [hide]			
Representing  Iran			
Men's Greco-Roman Wrestling			
Event	1st	2nd	3rd
Olympic Games	1	0	0
World Championships	6	0	0
World Junior Championship	1	0	0
World Cup	1	1	0
Other	6	1	2
Total	15	2	2

Negative Sampling → Inspired by ?

Dashti et. al., Effective Neural Team Formation via Negative Samples. CIKM 2022.

$$\sum_{t_{se} \in \mathcal{T}} [\log \sigma(v_e^\top \cdot v_s) + \sum_{t_{se'} \sim \mathbb{P}: t_{se'} \notin \mathcal{T}}^k \log \sigma(-v_{e'}^\top \cdot v_s)]$$

Positive Samples

Probability Distribution

Negative Samples

2.2 Negative Sampling

An alternative to the hierarchical softmax is Noise Contrastive Estimation (NCE), which was introduced by Gutmann and Hyvarinen [4] and applied to language modeling by Mnih and Teh [11]. NCE posits that a good model should be able to differentiate data from noise by means of logistic regression. This is similar to hinge loss used by Collobert and Weston [2] who trained the models by ranking the data above noise.

While NCE can be shown to approximately maximize the log probability of the softmax, the Skip-gram model is only concerned with learning high-quality vector representations, so we are free to simplify NCE as long as the vector representations retain their quality. We define Negative sampling (NEG) by the objective

$$\log \sigma(v_{w_o}^\top v_{w_i}) + \sum_{i=1}^k \mathbb{E}_{w_i \sim P_n(w)} [\log \sigma(-v_{w_i}^\top v_{w_i})]$$

(4)

Table 1: Average performance of 5-fold neural models on test set in dblp.v12.

	%map2	%map5	%map10	%ndcg2	%ndcg5	%ndcg10	%pr2	%pr5	%pr10	%rec2	%rec5	%rec10	rocauc
random	0.0001	0.0001	0.0002	0.0002	0.0002	0.0004	0.0001	0.0002	0.0002	0.0001	0.0003	0.0006	0.4992
fnn	0.0045	0.0307	0.0612	0.0082	0.0227	0.0369	0.0045	0.0113	0.0155	0.0067	0.0188	0.0188	0.5000
fnn-uniform	0.0487	0.0741	0.0943	0.1074	0.1350	0.1993	0.1020	0.1030	0.0986	0.0597	0.1522	0.2913	0.6512
fnn-unigram	0.0437	0.0677	0.0880	0.0952	0.1249	0.1907	0.0932	0.0985	0.0971	0.0552	0.1447	0.2854	0.6505
fnn-unigram-b	0.0436	0.0665	0.0847	0.1005	0.1249	0.1846	0.0993	0.0979	0.0932	0.0569	0.1429	0.2702	0.6500
fnn-emb	0.0084	0.0402	0.0688	0.0134	0.0302	0.0428	0.0073	0.0174	0.0209	0.0134	0.0255	0.0215	0.4999
fnn-emb-uniform	0.0668	0.1043	0.1305	0.1537	0.1901	0.2716	0.1543	0.1505	0.1346	0.0870	0.2179	0.3925	0.6313
fnn-emb-unigram	0.0700	0.1084	0.1356	0.1564	0.1942	0.2803	0.1523	0.1500	0.1378	0.0884	0.2194	0.4038	0.6331
fnn-emb-unigram-b	0.0656	0.1015	0.1277	0.1444	0.1782	0.2607	0.1415	0.1374	0.1291	0.0830	0.2011	0.3770	0.6322
bnn	0.0061	0.0254	0.0569	0.0123	0.0204	0.0365	0.0061	0.0107	0.0155	0.0101	0.0161	0.0195	0.5000
bnn-uniform	0.0487	0.0741	0.0943	0.1074	0.1350	0.1993	0.4005	0.3555	0.3102	0.2297	0.5124	0.8974	0.7150
bnn-unigram	0.0437	0.0677	0.0880	0.0952	0.1249	0.1907	0.3388	0.2885	0.2670	0.1944	0.4175	0.7737	0.7132
bnn-unigram-b	0.1757	0.2499	0.3039	0.3983	0.4505	0.6312	0.3904	0.3402	0.3017	0.2256	0.4929	0.8774	0.7168
bnn-emb	0.0112	0.0326	0.0634	0.0175	0.0267	0.0413	0.0089	0.0145	0.0186	0.0168	0.0201	0.0201	0.5000
bnn-emb-uniform	0.1620	0.2296	0.2817	0.3663	0.4176	0.5895	0.3656	0.3405	0.2909	0.2121	0.4934	0.8463	0.7123
bnn-emb-unigram	0.1792	0.2569	0.3060	0.4022	0.4623	0.6205	0.3783	0.3298	0.2858	0.2213	0.4802	0.8313	0.7077
bnn-emb-unigram-b	0.1752	0.2537	0.3094	0.4069	0.4728	0.6515	0.3938	0.3518	0.3033	0.2252	0.5065	0.8775	0.7090

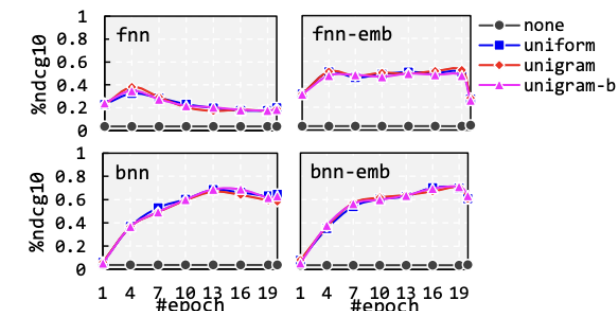
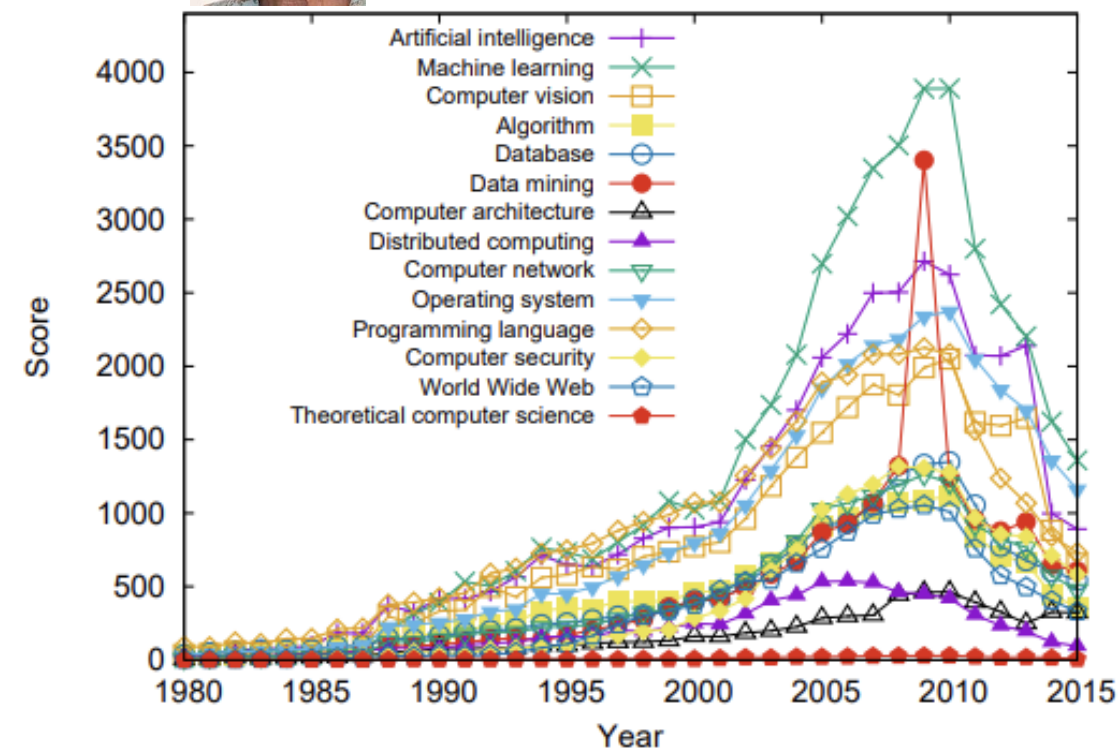
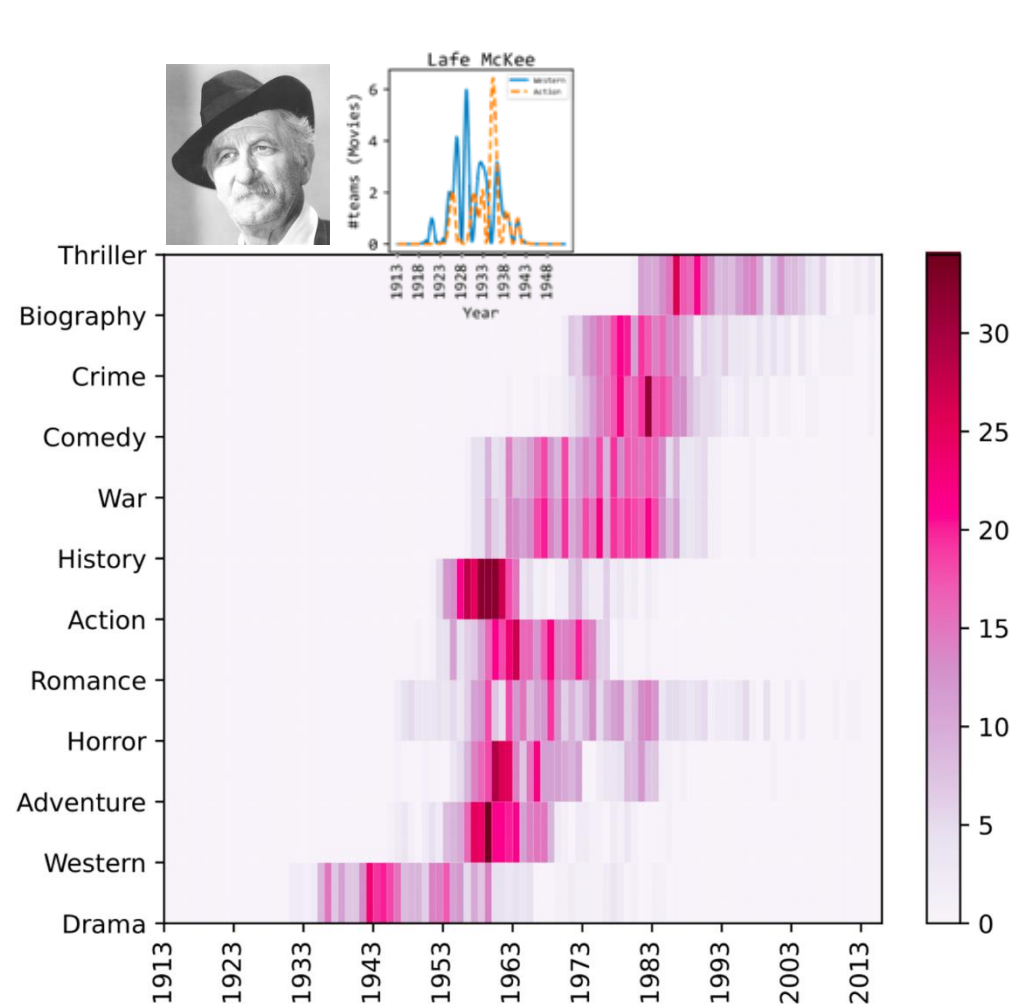


Figure 2: Training time vs. inference accuracy on dblp.v12.

Temporal Neural Team Recommendation



Effendy et al. WWW 2017.

Streaming Training Strategy

[Fani et. al., ECIR, 2024]

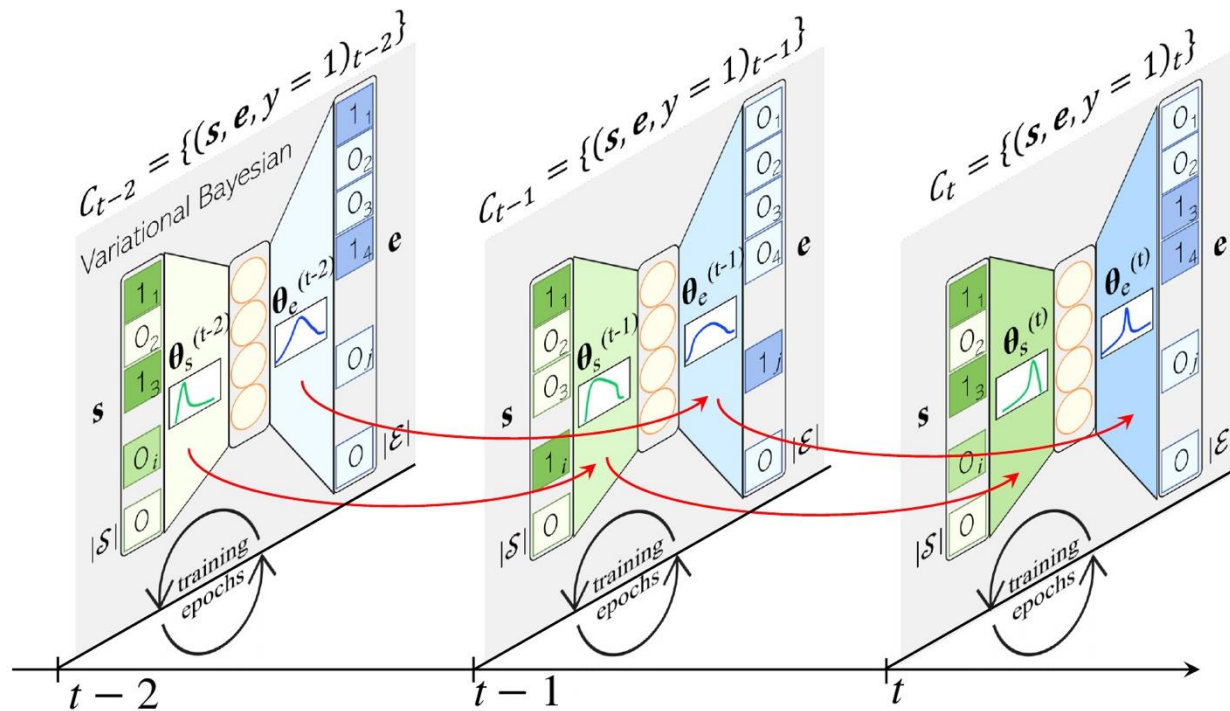
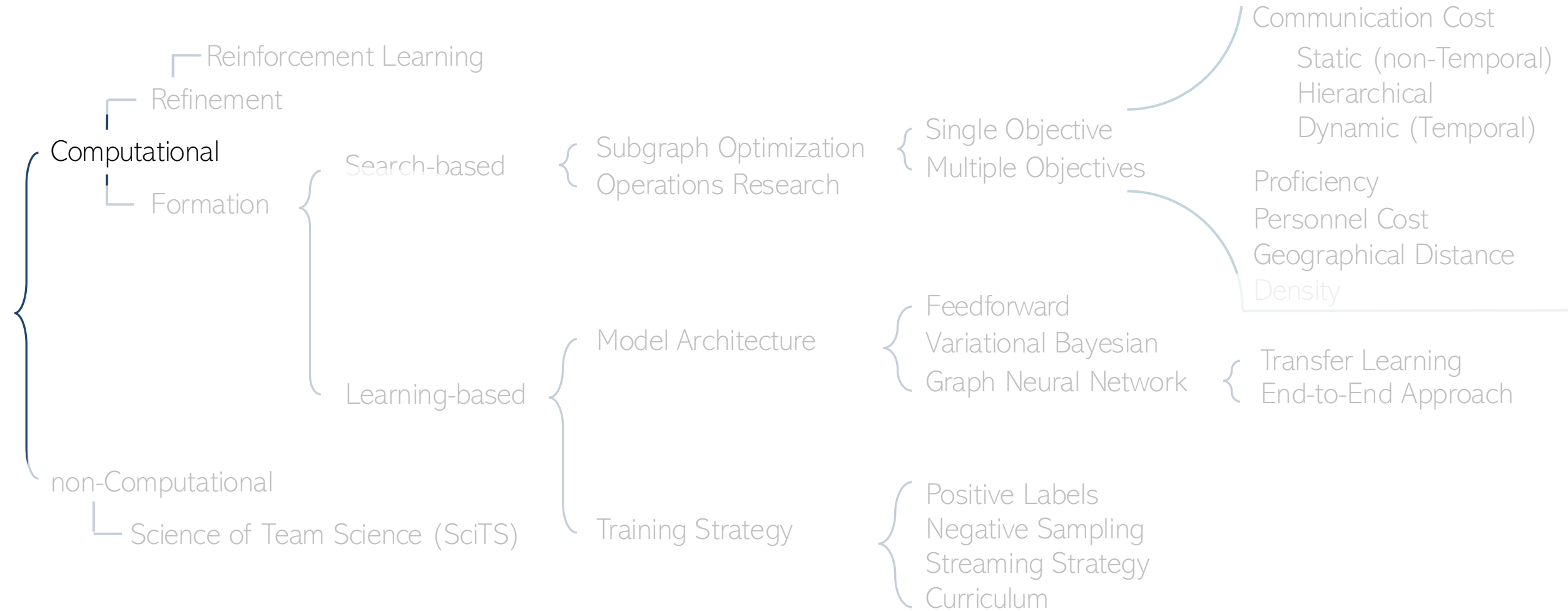


Table 2: Average performance of 5-fold neural models on the test set.

	%pr2	%pr5	%pr10	%rec2	%rec5	%rec10	%ndcg2	%ndcg5	%ndcg10	%map2	%map5	%map10	%aucroc
dblp													
bnn [36]	0.0570	0.0663	0.0710	0.0351	0.0993	0.2118	0.0538	0.0806	0.1330	0.0242	0.0411	0.0558	63.52
bnn_emb [35]	0.1124	0.1290	0.1251	0.0668	0.1909	0.3699	0.1083	0.1555	0.2397	0.0474	0.0792	0.1033	66.81
rrn [44]	0.0570	0.0391	0.0472	0.0380	0.0630	0.1552	0.0478	0.0523	0.0959	0.0217	0.0281	0.0446	50.73
tbn	0.1189	0.1413	0.1664	0.0706	0.2090	0.4984	0.1126	0.1689	0.3031	0.0484	0.0845	0.1223	73.08
tbn_emb	0.2996	0.2938	0.2811	0.1816	0.4433	0.8431	0.3048	0.3860	0.5721	0.1411	0.2095	0.2635	74.83
tbn_dt2v_emb	0.4299	0.3973	0.3612	0.2601	0.5963	1.0801	0.4284	0.5221	0.7465	0.1947	0.2864	0.3520	77.01
imdb													
bnn [36]	0.2128	0.5106	0.4255	0.1418	0.8511	1.3050	0.1646	0.5699	0.7848	0.0709	0.2600	0.3148	51.16
bnn_emb [35]	0.4255	0.5106	0.6383	0.2837	0.8511	1.9574	0.3292	0.5923	1.1358	0.1418	0.2813	0.4389	51.82
rrn [44]	0.0000	0.8511	0.8511	0.0000	1.4184	2.8369	0.0000	0.8163	1.4606	0.0000	0.3191	0.6265	52.22
tbn	0.8511	1.5319	1.4043	0.5319	2.4610	4.4965	0.7548	1.7381	2.6829	0.3369	0.8215	1.1674	63.46
tbn_emb	0.8511	1.1064	1.0638	0.5674	1.7518	1.3262	0.9474	1.4848	2.2007	0.4965	0.8138	1.0099	66.87
tbn_dt2v_emb	1.9149	1.1915	1.4468	1.2411	1.9504	4.5532	1.8667	1.8703	3.0303	0.9043	1.1099	1.4293	66.56
uspt													
bnn [36]	0.0657	0.0769	0.0910	0.0353	0.0976	0.2212	0.0655	0.0883	0.1481	0.0266	0.0433	0.0592	64.54
bnn_emb [35]	0.3663	0.4123	0.3748	0.1608	0.4509	0.8141	0.3652	0.4531	0.6094	0.1212	0.2027	0.2583	69.85
rrn [44]	0.0239	0.0383	0.0654	0.0140	0.0500	0.1370	0.0221	0.0408	0.0868	0.0096	0.0186	0.0340	51.60
tbn	0.1843	0.1841	0.2029	0.0933	0.2321	0.5158	0.1794	0.2152	0.3481	0.0681	0.1056	0.1429	75.44
tbn_emb	0.8272	0.7539	0.7042	0.3970	0.9021	1.6933	0.8457	0.9057	1.2657	0.3104	0.4533	0.5679	83.59
tbn_dt2v_emb	1.2268	1.0583	0.9324	0.6037	1.2928	2.2518	1.2322	1.2960	1.7348	0.4626	0.6659	0.8118	85.34
gith													
bnn [36]	3.0693	2.8515	2.6931	1.2164	2.8846	5.1174	3.1365	3.2893	4.2340	1.0104	1.5706	2.1633	56.18
bnn_emb [35]	7.3267	4.7129	3.3861	3.5441	5.1580	6.1885	6.4753	5.8418	6.2665	2.3424	3.0822	3.3837	62.65
rrn [44]	0.0000	0.1980	0.0990	0.0000	0.0619	0.0619	0.0000	0.1679	0.1090	0.0000	0.0206	0.0206	52.26
tbn	3.8614	2.8515	2.3564	1.8801	3.1525	4.5754	4.3319	3.9721	4.5031	1.8025	2.3978	2.8768	56.65
tbn_emb	4.9505	3.5248	3.1287	1.9434	3.0770	4.3718	5.0849	4.4715	4.9844	1.6957	2.1431	2.5949	62.20
tbn_dt2v_emb	5.7426	4.5941	3.8020	2.1874	3.8474	4.7855	5.6081	5.3287	5.6670	1.7131	2.4258	2.7858	64.89

Short timespan in the dataset

92 Computational Approach



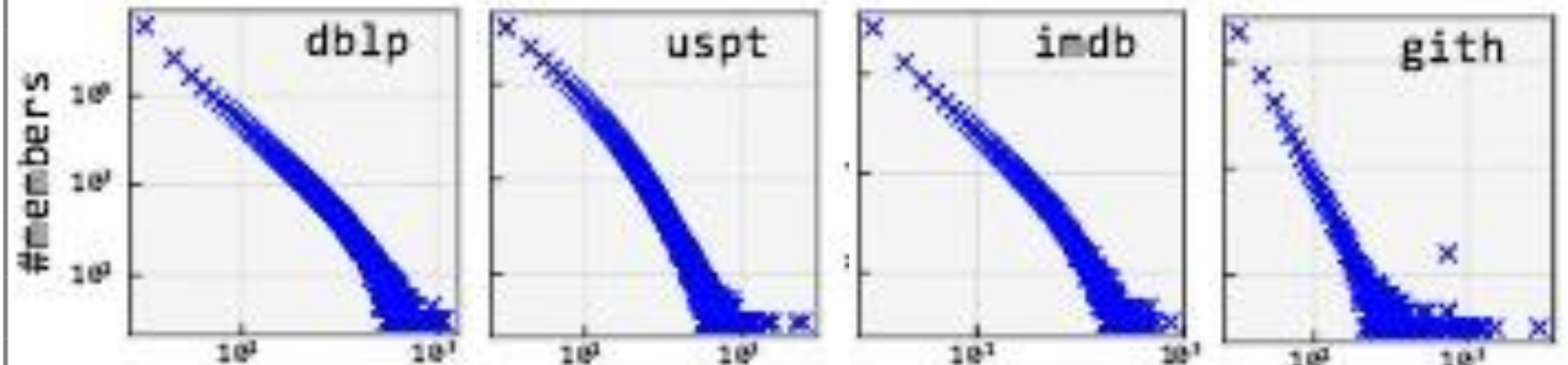
Team Allocation, Team Selection, Team Composition, Team Configuration, Team Recommendation, Team Formation

Experimental Setup → Dataset

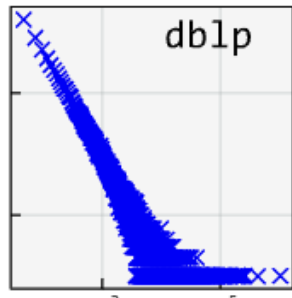
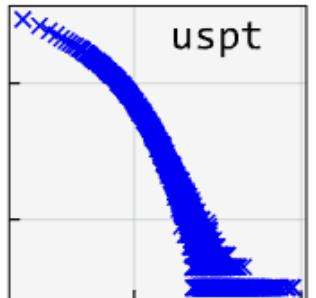
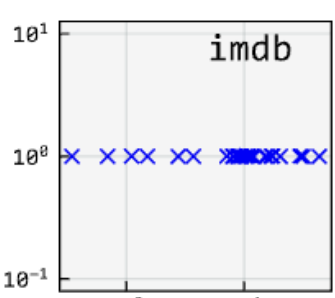
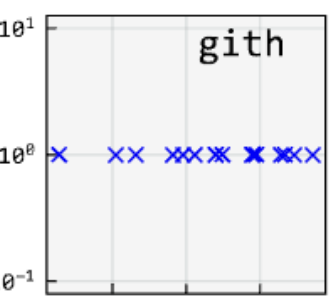
	dblp		uspt		imdb		gith	
	raw	filtered	raw	filtered	raw	filtered	raw	filtered
#teams	4,877,383	99,375	7,068,508	152,317	507,034	32,059	132,851	11,312
#unique experts	5,022,955	14,214	3,508,807	12,914	876,981	2,011	452,606	2,686
#unique skills	89,504	29,661	241,961	67,315	28	23	20	19
avg #expert per team	3.06	3.29	2.51	3.79	1.88	3.98	5.52	7.53
avg #skill per team	8.57	9.71	6.29	9.97	1.54	1.76	1.37	1.57
avg #team per expert	2.97	23.02	5.05	44.69	1.09	62.45	1.62	31.72
avg #skill per expert	16.73	96.72	19.49	102.53	1.59	10.85	2.03	5.18
#team w/ single expert	768,956	0	2,578,898	0	322,918	0	0	0
#team w/ single skill	5,569	56	939,955	8,110	315,503	15,180	69,131	6014
timespan (raw)	1979 – 2018		1976 – 2019		1914 – 2020		2008 – 2022	

Dataset

	dblp		uspt		imdb		gith	
	raw	filtered	raw	filtered	raw	filtered	raw	filtered
#teams	4,877,383	99,375	7,068,508	152,317	507,034	32,059	132,851	11,312
#unique experts	5,022,955	14,214	3,508,807	12,914	876,981	2,011	452,606	2,686
#unique skills	33,534	33,331	341,031	37,315	33	33	33	10
avg #expert per team								
avg #skill per team								
avg #team per expert								
avg #skill per expert								
#team w/ single expert								
#team w/ single skill								
timespan (raw)	1979 – 2018		1976 – 2019		1914 – 2020		2008 – 2022	



Dataset

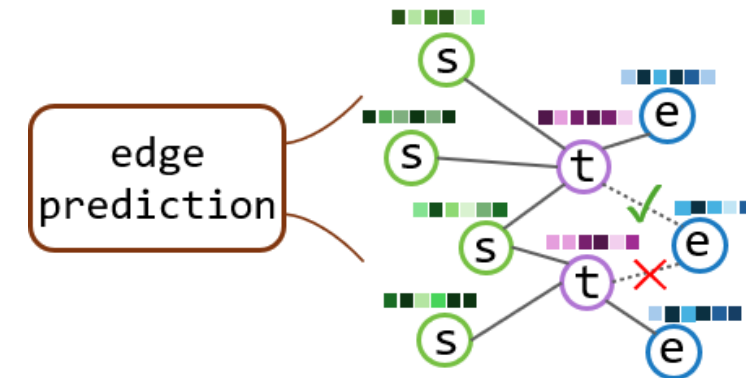
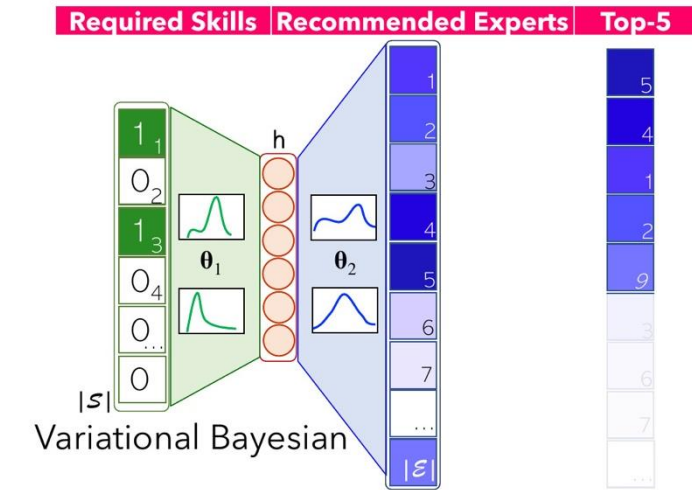
	dblp		uspt		imdb		gith	
	raw	filtered	raw	filtered	raw	filtered	raw	filtered
#teams	4,877,383	99,375	7,068,508	152,317	507,034	32,059	132,851	11,312
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#unique skills	89,504	29,661	241,961	67,315	28	23	20	19
avg #expert per team								
avg #skill per team								
avg #team per expert								
avg #skill per expert								
#team w/ single expert								
#team w/ single skill								
Timespan (raw)	1979 – 2018		1976 – 2019		1914 – 2020		2008 – 2022	

Dataset

	dblp		uspt		imdb		gith	
	raw	filtered	raw	filtered	raw	filtered	raw	filtered
#teams	4,877,383	99,375	7,068,508	152,317	507,034	32,059	132,851	11,312
#unique experts	5,022,955	14,214	3,508,807	12,914	876,981	2,011	452,606	2,686
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avg #expert per team	3.06	3.29	2.51	3.79	1.88	3.98	5.52	7.53
avg #skill per team	8.57	9.71	6.29	9.97	1.54	1.76	1.37	1.57
avg #team per expert	2.97	23.02	5.05	44.69	1.09	62.45	1.62	31.72
avg #skill per expert	16.73	96.72	19.49	102.53	1.59	10.85	2.03	5.18
#team w/ single expert	768,956	0	2,578,898	0	322,918	0	0	0
#team w/ single skill	5,569	56	939,955	8,110	315,503	15,180	69,131	6014
Timespan (raw)	1979 – 2018		1976 – 2019		1914 – 2020		2008 – 2022	

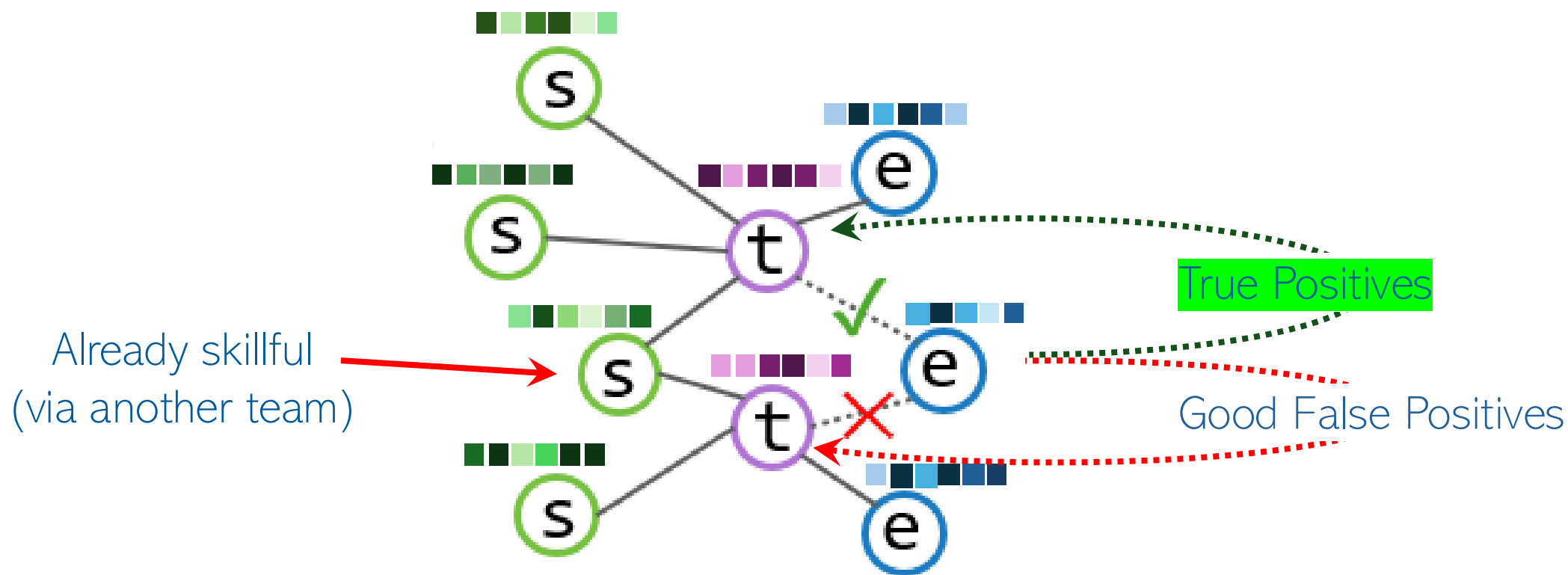
Experimental Setup → Metrics

Metrics		Sapienza <i>et al.</i> [2019]	Rad <i>et al.</i> [2020]	Rad <i>et al.</i> [2021b]	Dashti <i>et al.</i> [2022a]	Rad <i>et al.</i> [2022b]	Rad <i>et al.</i> [2022a]	Dashti <i>et al.</i> [2022b]	Rad <i>et al.</i> [2021a]
+Optimum Teams	Average Precision	✓	✓	✓	✓	✓	✓	✓	✓
	Reciprocal Rank	✓	✓		✓	✓	✓	✓	✓
	Discounted Cumulative Gain	✓	✓	✓	✓	✓	✓	✓	✓
	Recall	✓	✓	✓	✓	✓	✓	✓	✓
	Precision			✓			✓		
	Area Under Curve			✓			✓		
	Squared Error	✓							
	Skill Coverage		✓			✓			
Communication Cost						✓			



Metrics → Skill Coverage

How Bad is a False Positive Expert



Outline

- I) Introduction and Background
- II) Pioneering Techniques
- III) Learning-based Heuristics
- III) Team Refinement
- V) Challenges and New Perspectives
- VI) Applications

Hands-on: OpeNTF

Team Refinement

Refining a team

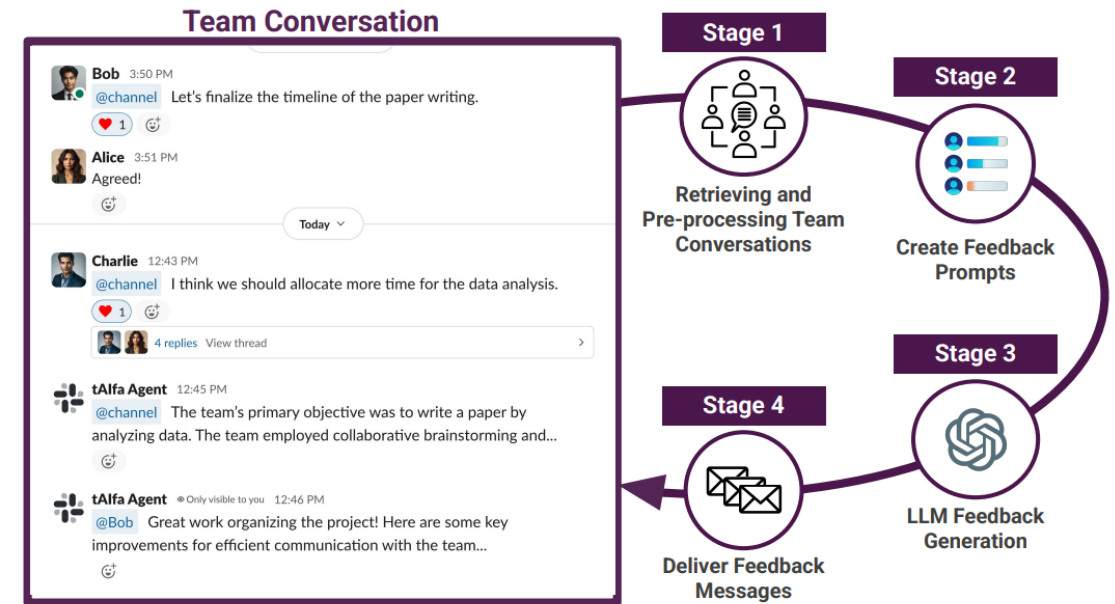
- Improving the effectiveness of initially formed team.
- Enhancing the collaboration among team members.

- Structural adjustment
- Team Expansion
- Membership substitution

Structural Adjustment

tAlfa, an AI agent that uses LLMs to provide personalized, automated feedback to teams and their members.

- Retrieve and structure team conversations.
- Evaluate team dynamics using rule-based metrics and LLM analysis.
- Use LLM to generate individual and team feedback messages.
- Deliver feedback privately and as group messages tagging all members.



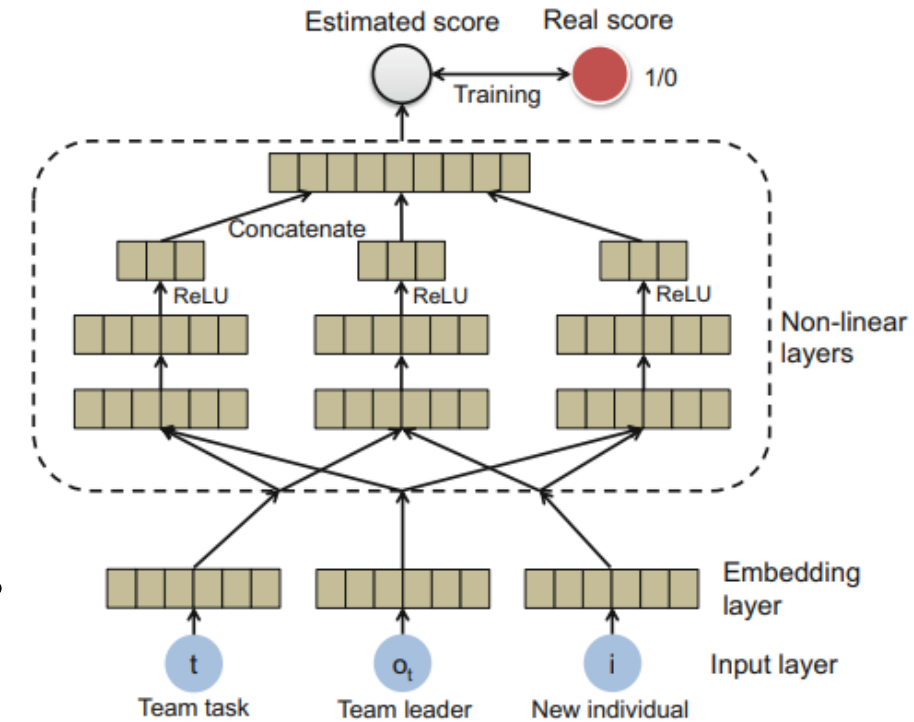
Evaluation Metrics:

- Average Conversation Duration
- Average Speaker Turn
- Average Word Count
- Self-Reported Measures

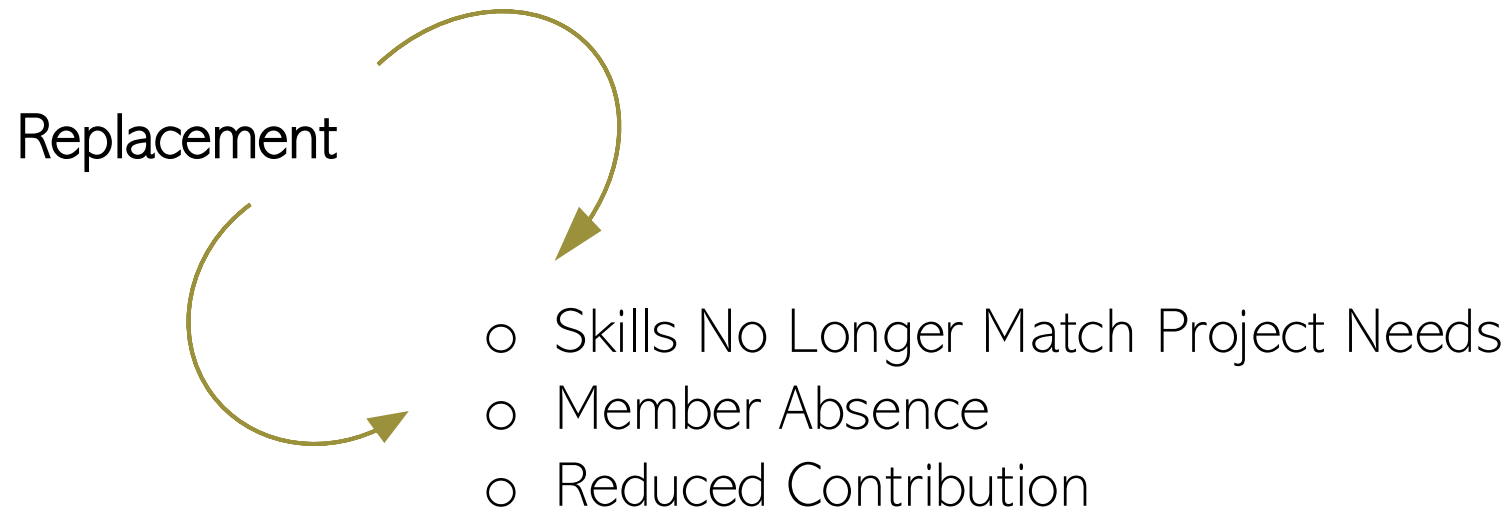
Team Expansion

TECE, predict a unified fitness score

- **Task–Candidate Interaction:** Learns how candidate skills align with task requirements using collaborative filtering.
- **Candidate–Leader Interaction:** Models social compatibility through interactions between the candidate and team leader.
- **Team State Interaction:** Captures current team dynamics via task–leader relationships to find complementary candidates.



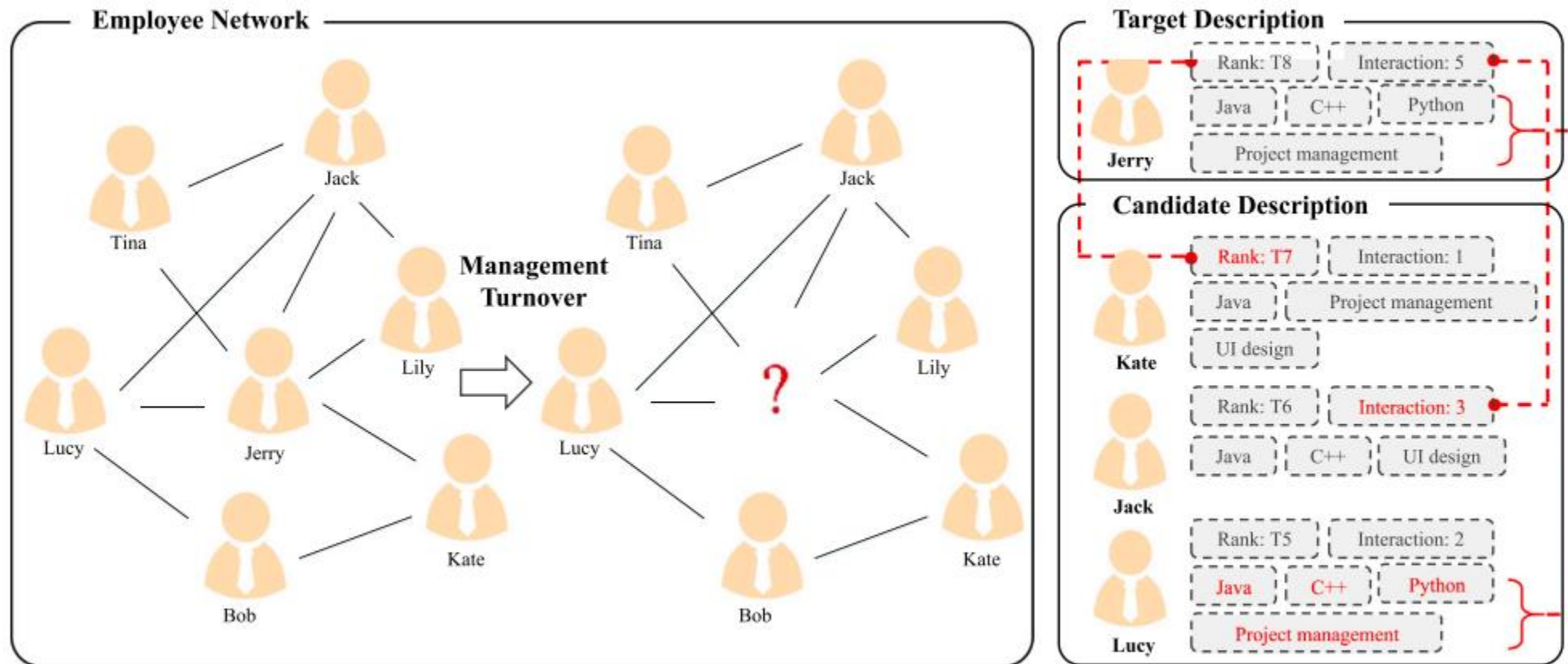
Membership Substitution



[Lv et. Al., Decision Support Systems, 2024]

[Zhao et. Al., Springer International Publishing AG, part of Springer Nature, 2018]

Membership Substitution



Outline

- I) Introduction and Background
- II) Pioneering Techniques
- III) Learning-based Heuristics
- IIII) Team Refinement
- V) Challenges and New Perspectives
- VI) Applications

Hands-on: OpeNTF

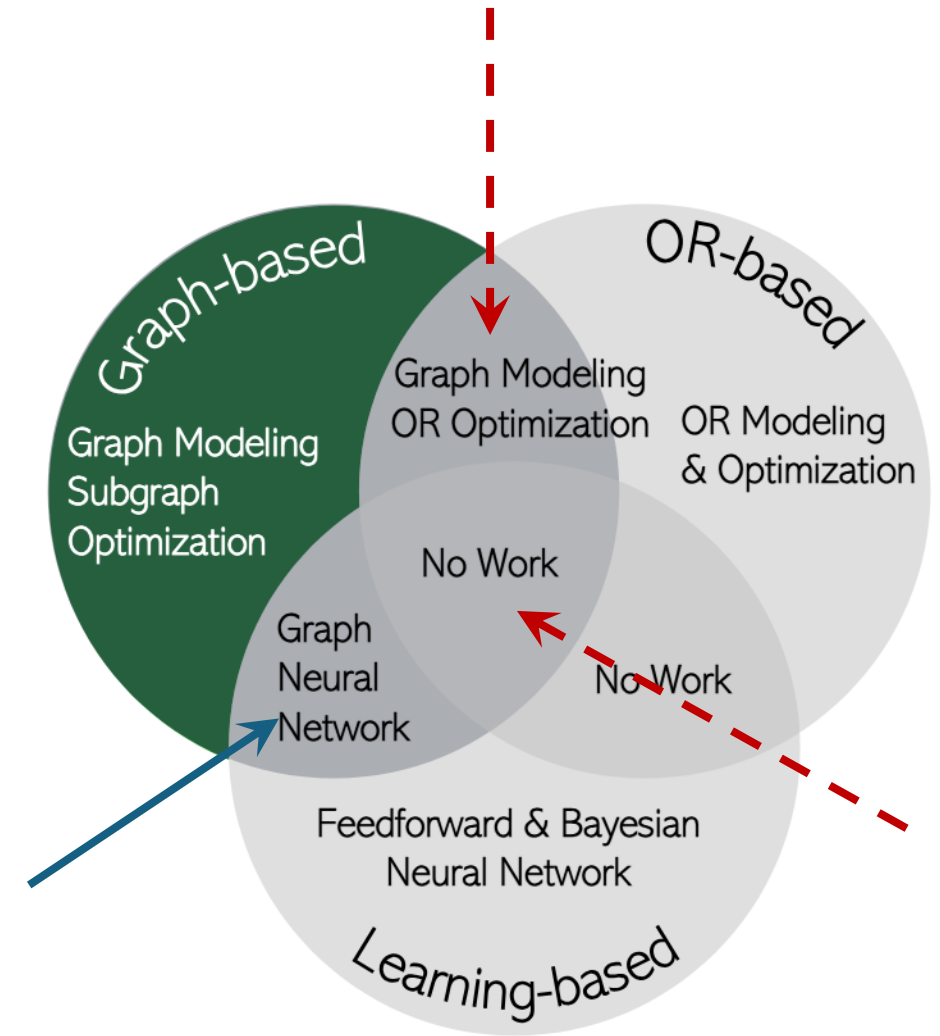
Immediate RQs

Temporal End-to-End Graph Neural Network

[Fani et. al., ECIR 2024] + [Ahmed et al. WISE 2024]

Spatial Team Recommendation

- Time Zones
- Regions



Immediate RQs

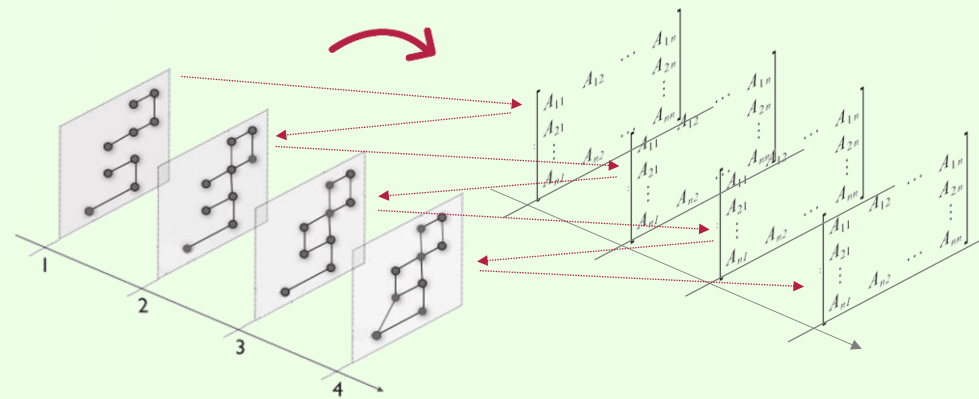
Temporal End-to-End Graph Neural Network

[Fani et. al., ECIR 2024] + [Ahmed et al. WISE 2024]

TEMPORAL LATENT SPACE MODELING

local Block Coordinate Gradient Descent (Zhu et al. TKDE 2016)

$$\arg \min \left[\sum_{t=1}^T \sum_{u,v \in \mathbb{U}} |w(u, v : t) - \mathbf{y}_{ut} \mathbf{y}_{vt}^\top|_F^2 \right. \\ \left. + \lambda \sum_{t=1}^T \sum_{u \in \mathbb{U}} (1 - \mathbf{y}_{ut} \mathbf{y}_{u(t-1)}^\top) \right] \\ \forall u \in \mathbb{U}; \mathbf{y}_{ut} \geq 0, \mathbf{y}_{ut} \mathbf{y}_{ut}^\top = 1$$

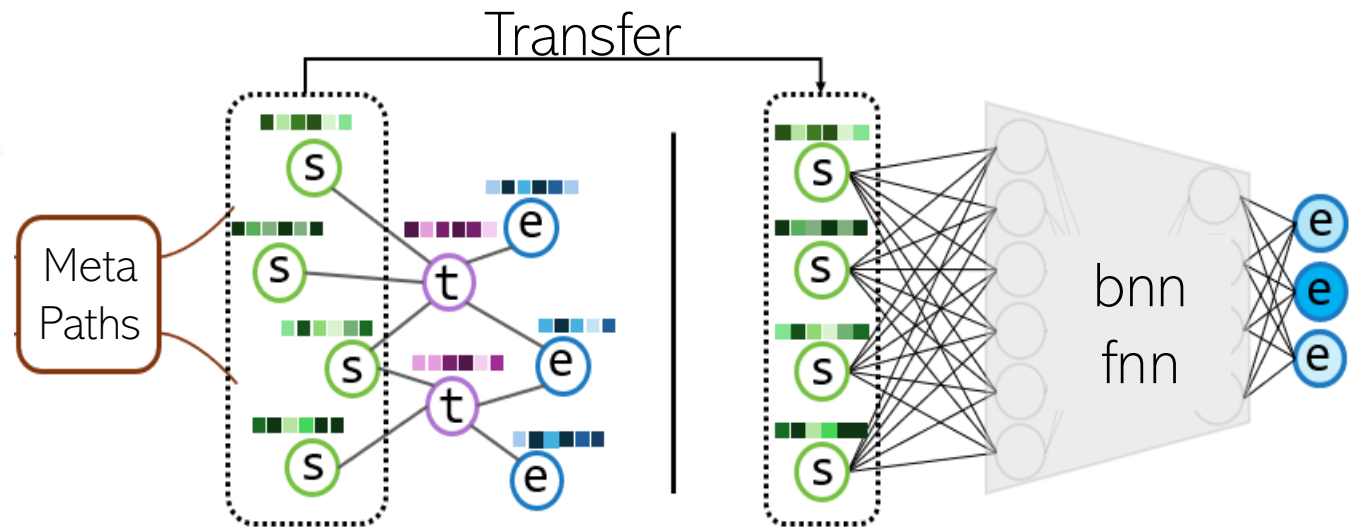
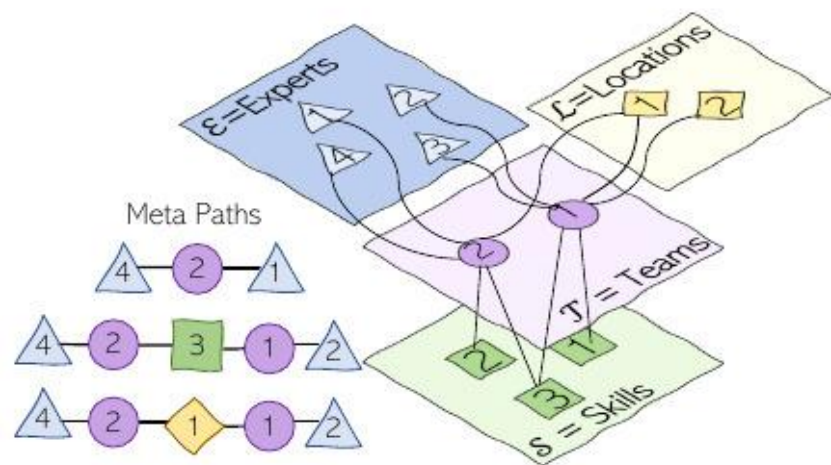


Fani et al. Temporal latent space modeling for community prediction. ECIR 2020

Immediate RQs

Geolocations used only for better skill vectors!

[Rad et. al., SIGIR 2021]



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Fair and Diverse Team Recommendation



Salima Mazari, female Afghan governor, Leading a Taliban resistance

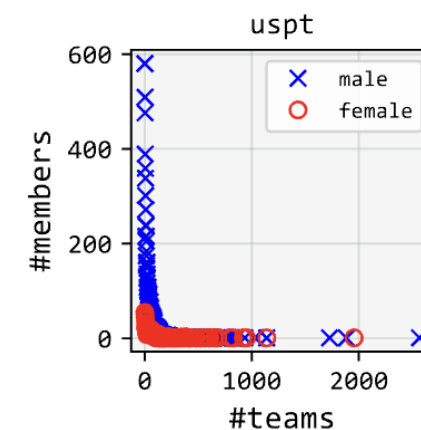
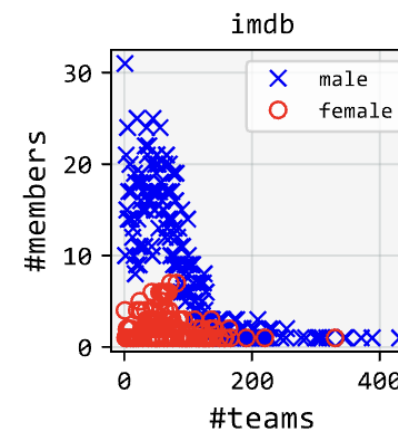
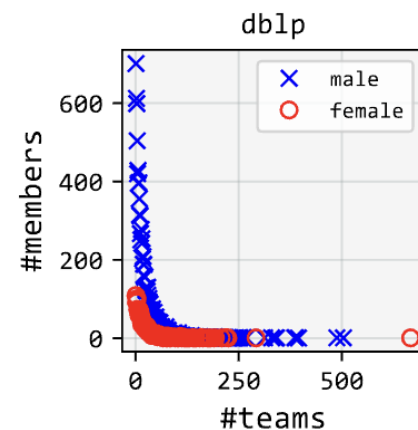
“Sometimes I'm in the office in Charkint, and other times I have to pick up a gun and join the battle”

<https://people.com/politics/meet-salima-mazari-afghan-governor-who-led-a-taliban-resistance/> - 2021

Fair and Diverse Team Recommendation

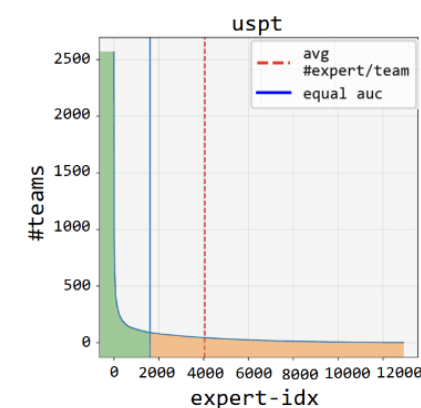
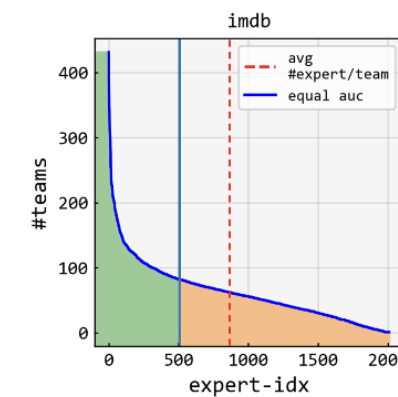
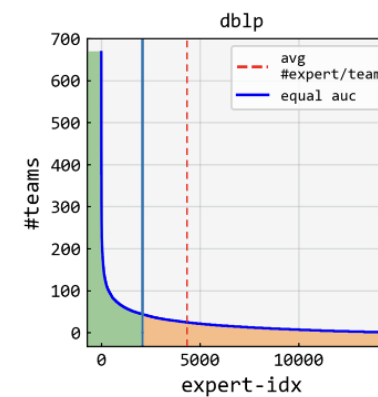
%female experts

dblp	imdb	uspt
14.20%	12.30%	13.80%

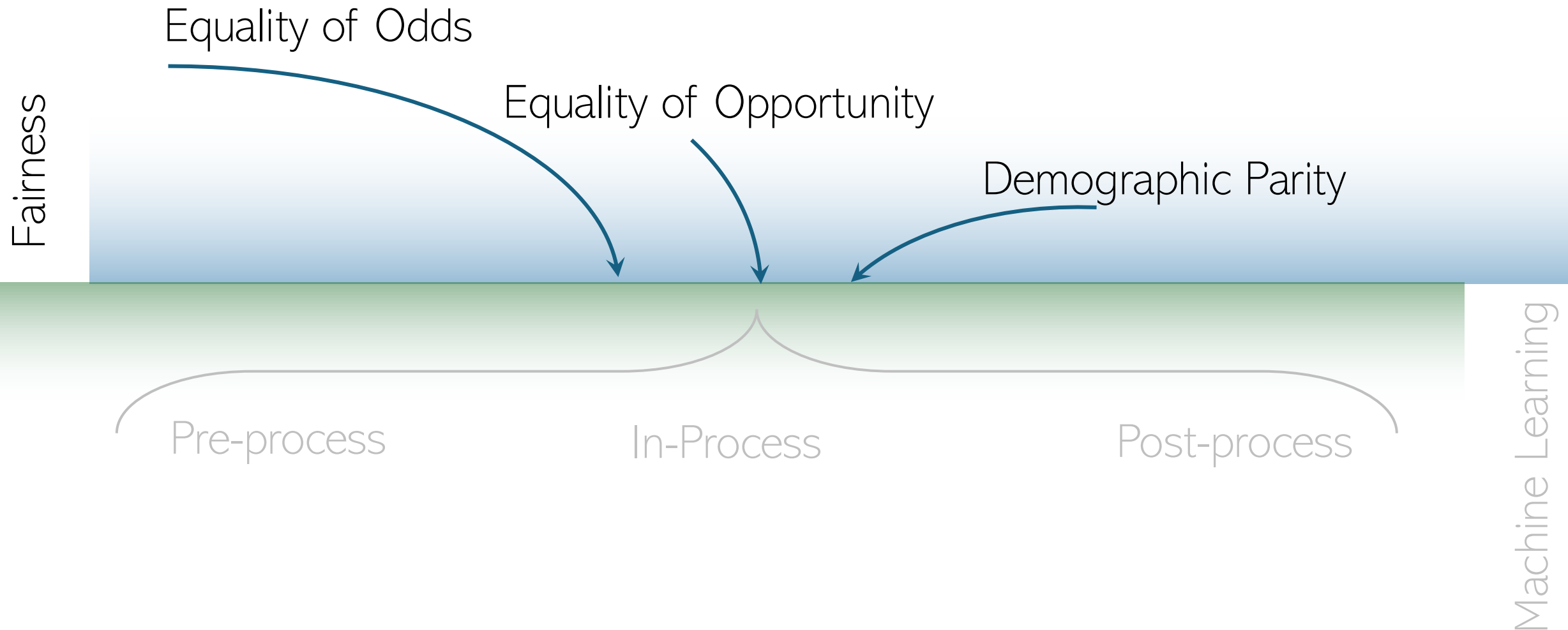


%popular experts (avg)

31.30% 42.60% 31.40%



Fair and Diverse Team Recommendation



Fair and Diverse Team Recommendation

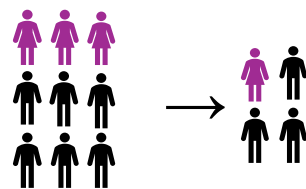
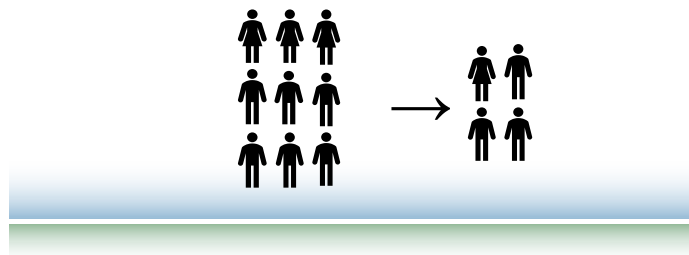
Demographic Parity: Membership of an expert is independent of **any** attribute

$$P(\text{recommended} \mid \text{male}) = P(\text{recommended} \mid \text{female})$$

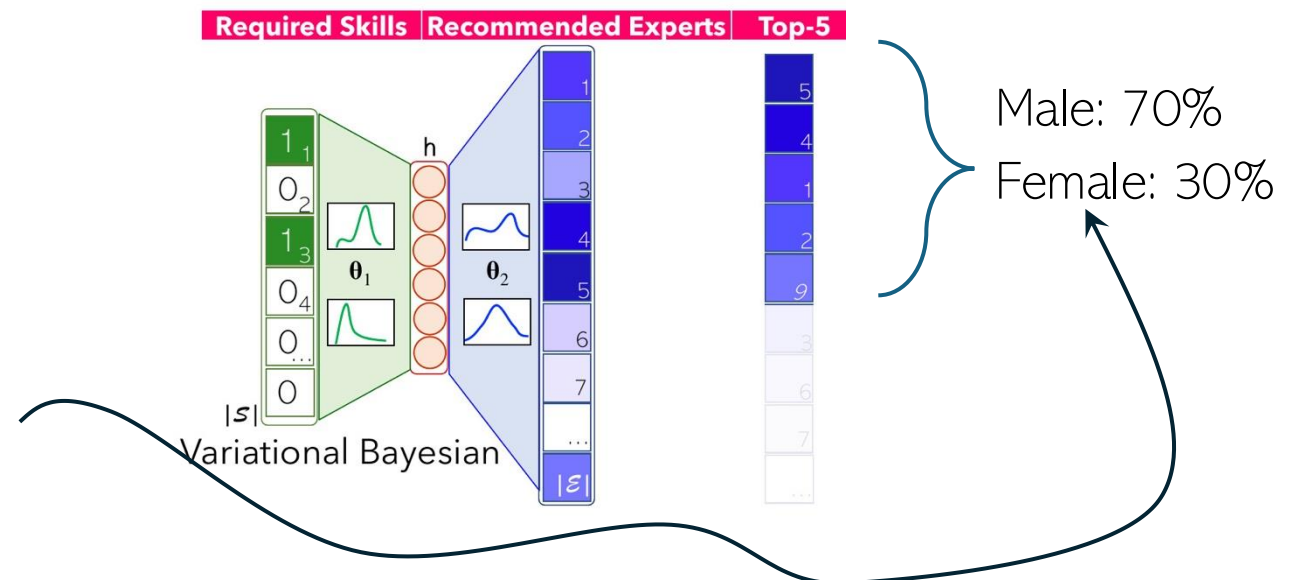
$$P(\text{recommended} \mid \text{popular}) = P(\text{recommended} \mid \text{non-popular})$$



$$P(\text{recommended} \mid \text{whoever you are}) = P(\text{recommended})$$



Male: 70%
Female: 30%



Fair and Diverse Team Recommendation

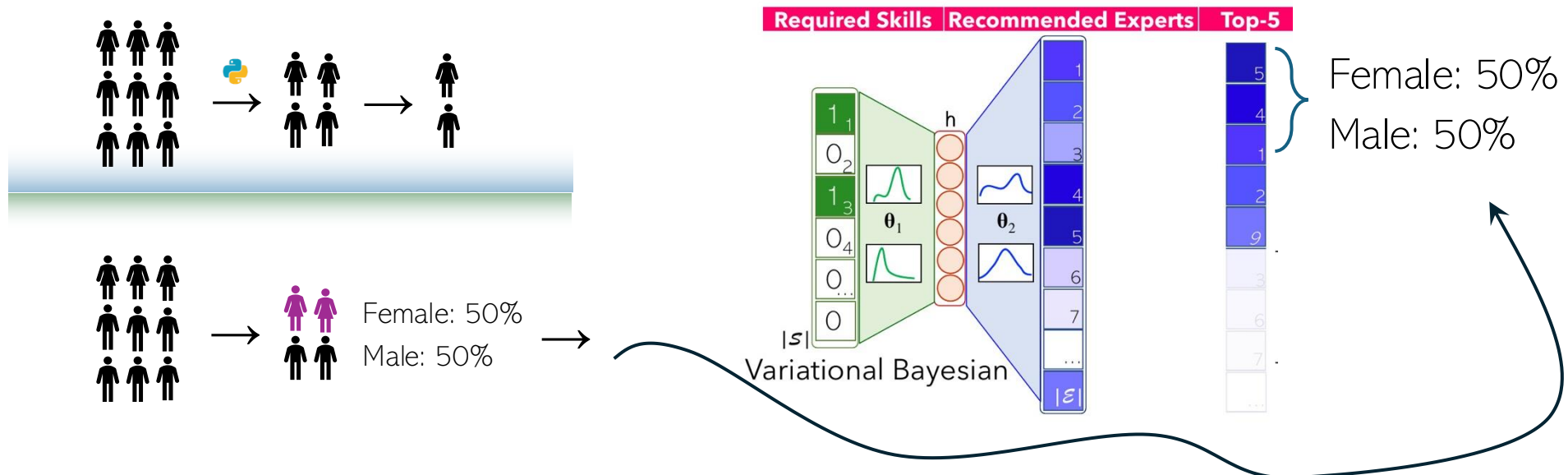
Equality of Opportunity: Membership of a **skilled** expert is independent of **any** attribute

$$P(\text{recommended} \mid \text{true male}) = P(\text{recommended} \mid \text{true female})$$

$$P(\text{recommended} \mid \text{true popular}) = P(\text{recommended} \mid \text{true non-popular})$$



$$P(\text{recommended} \mid \text{whoever you are as long as you have the skill}) = P(\text{recommended})$$

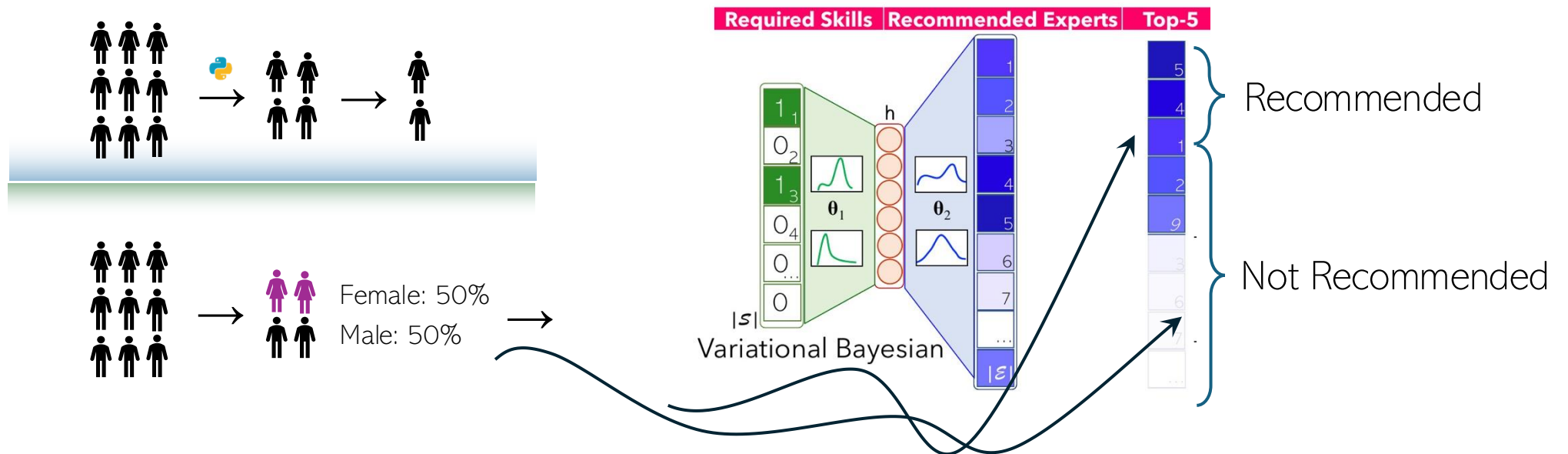


Fair and Diverse Team Recommendation

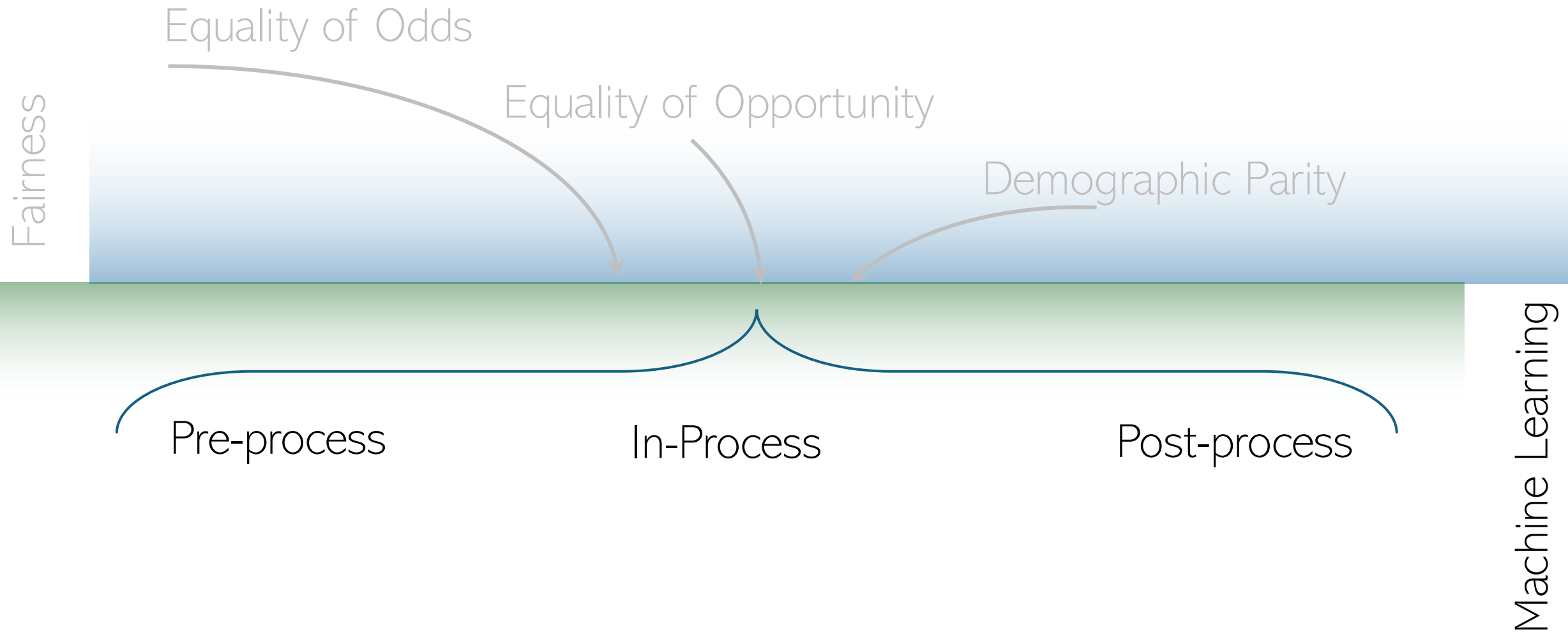
Equality of Odds

Membership of a **skilled** expert is independent of **any** attribute $\rightarrow$ True Positive Rate

Rejection of a **skilled** expert is independent of **any** attribute $\rightarrow$ False Positive Rate



Fair and Diverse Team Recommendation



Fair and Diverse Team Recommendation

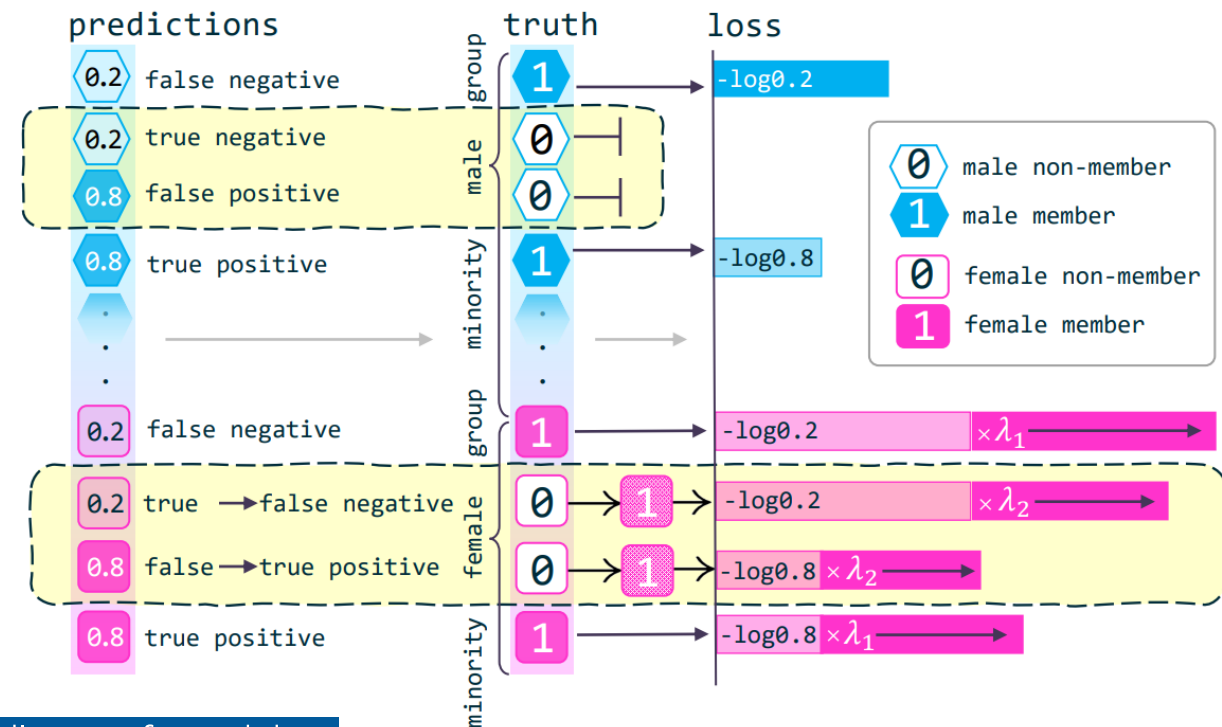
Debiasing Algorithms

- Pre-process: Re-sampling (over/under sampling) heuristics before model training
No work!
- In-process: Adjust the loss to balance efficacy and fairness
[Moasses et.al., BIAS-SIGIR, 2024]
[Barzegar et.al., WSDM 2025]
- Post-process:
[Geyik et.al., KDD, 2019] → [Loghmani et.al., BIAS-ECIR, 2022]

Fair and Diverse Team Recommendation

Debiasing Algorithms → In-Process

Moasses et.al., *vivafemme*: Mitigating Gender Bias in Neural Team Recommendation via Female-Advocate Loss Regularization. BIAS-SIGIR 2024



Increase the loss of female predictions regardless of truth!

Fair and Diverse Team Recommendation

Debiasing Algorithms → In-Process

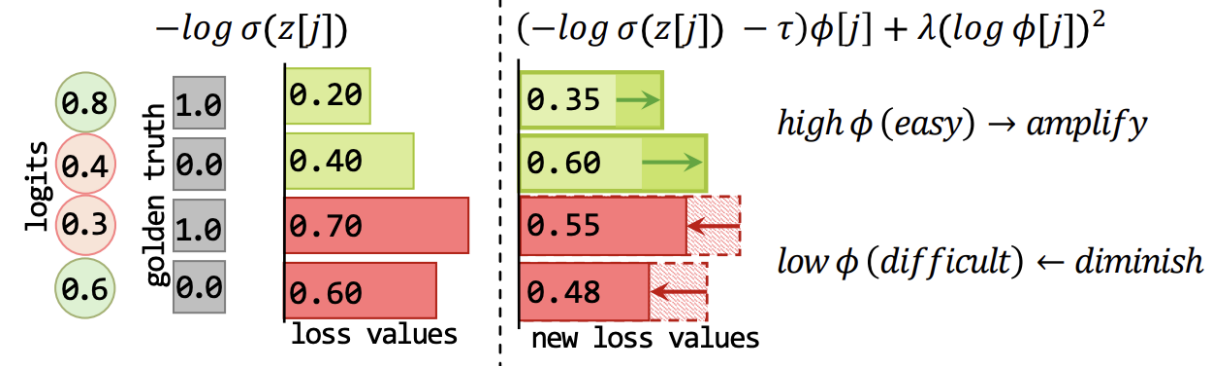
Barzegar et.al., Adaptive Loss-based Curricula for Neural Team Recommendation. WSDM 2025.

Easy Popular Experts

vs.

Difficult non-Popular Experts

dataset		neural model	our proposed curricula		
			static (sc)	parametric (pc)	non-parametric (npc)
dblp	non-variational		×	×	✓
	variational		✓	✓	✓
imdb	non-variational		×	×	✓
	variational		×	×	×



Saxena et al., Data Parameters: A New Family of Parameters for Learning a Differentiable Curriculum. NIPS 2019.

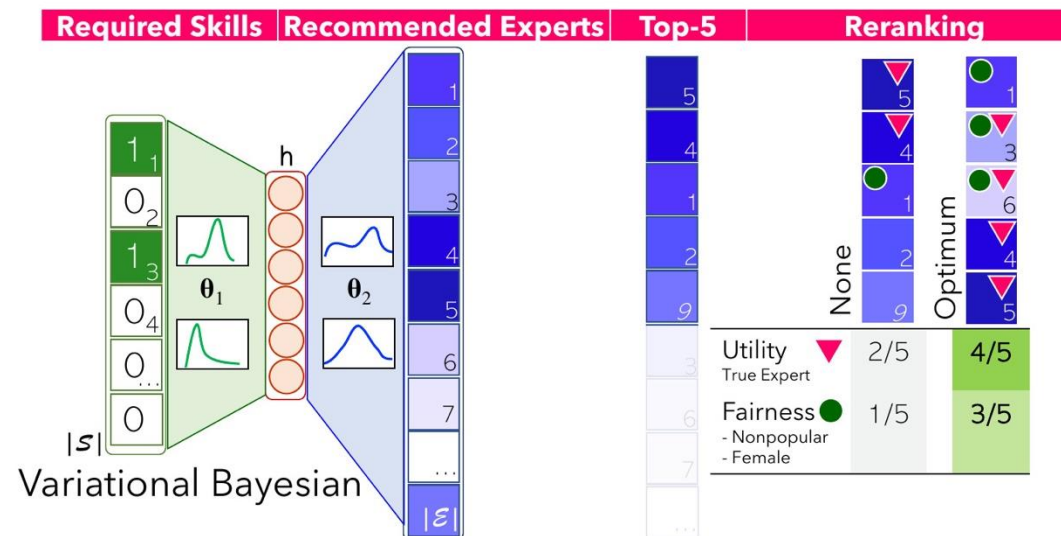
Castells et al., SuperLoss: A Generic Loss for Robust Curriculum Learning. NeurIPS 2020.

Fair and Diverse Team Recommendation

Debiasing Algorithms → Post-Process

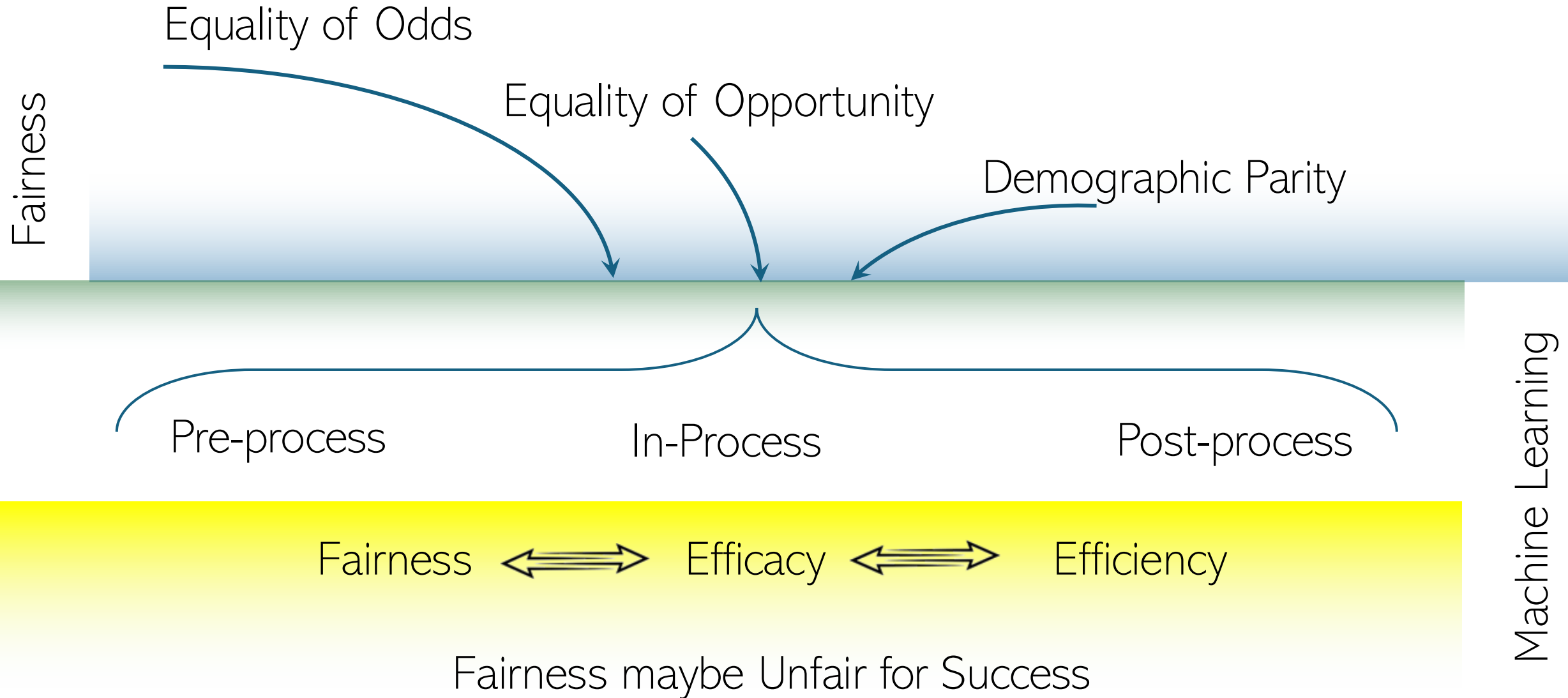
Loghmani et.al., **Bootless** Application of Greedy Re-ranking Algorithms in Fair Neural Team Formation, BIAS-ECIR 2022.

Geyik et al., Fairness-aware ranking in search recommendation systems with application to linkedin talent search. KDD 2019



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Fair and Diverse Team Recommendation

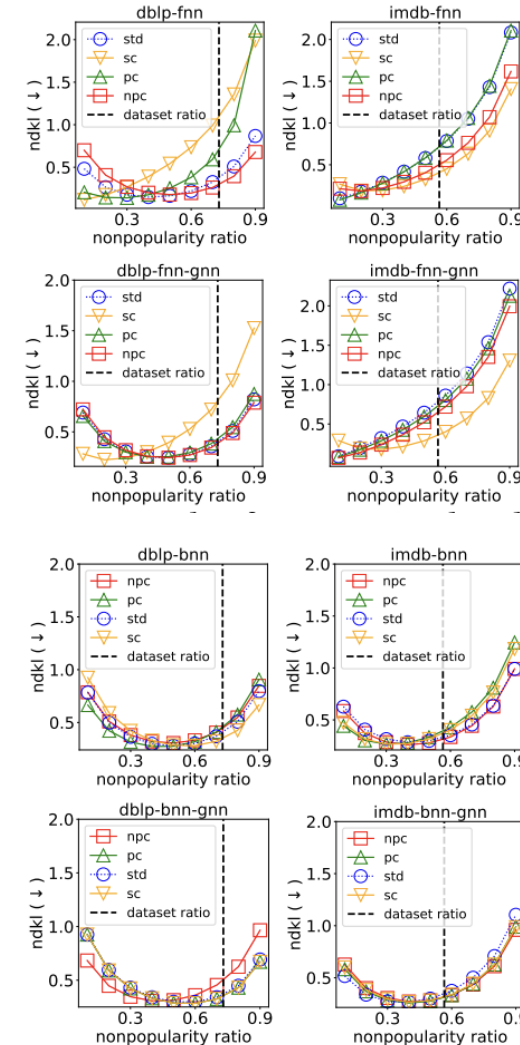


Fair and Diverse Team Recommendation

Barzegar et.al., Adaptive Loss-based Curricula for Neural Team Recommendation. WSDM 2025.

dblp	%precision		%recall		%ndcg		%map		%aucroc
	@2	@10	@2	@10	@2	@10	@2	@10	
random	0.0100	0.0200	0.0100	0.0600	0.0200	0.0400	0.0100	0.0200	49.9200
fnn-std [10]	0.2994	0.1799	0.1742	0.5300	0.2988	0.4024	0.1300	0.2049	68.3604
fnn-sc	0.0188	0.0164	0.0121	0.0509	0.0182	0.0338	0.0086	0.0154	50.1144
fnn-pc	0.2407	0.1751	0.1403	0.5134	0.2568	0.3884	0.1157	0.1981	61.4063
fnn-npc	0.5031$\pm$	0.3633$\pm$	0.2929$\pm$	1.0643$\pm$	0.5062$\pm$	0.7770$\pm$	0.2220$\pm$	0.3798$\pm$	71.4635
fnn-gnn-std	0.3975	0.2252	0.2318	0.6450	0.3967	0.5400	0.1745	0.2895	74.7247
fnn-gnn-sc	0.0188	0.0247	0.0110	0.0739	0.0197	0.0473	0.0089	0.0203	50.0975
fnn-gnn-pc	0.4721 $\pm$	0.2007	0.2712	0.5822	0.4795	0.5182	0.2094	0.2840	74.6674
fnn-gnn-npc	0.5182$\pm$	0.3792$\pm$	0.2991$\pm$	1.0912$\pm$	0.5269$\pm$	0.8005$\pm$	0.2277$\pm$	0.3898$\pm$	75.9575
bnn-std [27, 48]	0.0285	0.0272	0.0179	0.0856	0.0289	0.0586	0.0138	0.0288	50.0227
bnn-sc	0.0356	0.0266	0.0225	0.0837	0.0357	0.0595	0.0171	0.0299	49.8803
bnn-pc	0.0268	0.0300	0.0166	0.0954	0.0272	0.0630	0.0133	0.0306	50.0703
bnn-npc	0.0369	0.0223	0.0233	0.0698	0.0388	0.0534	0.0188	0.0278	50.0220
bnn-gnn-std [47]	0.0210	0.0263	0.0120	0.0811	0.0215	0.0509	0.0096	0.0217	50.0061
bnn-gnn-sc	0.0356	0.0241	0.0234	0.0776	0.0366	0.0576	0.0184	0.0306	49.9698
bnn-gnn-pc	0.0293	0.0247	0.0180	0.0773	0.0284	0.0526	0.0126	0.0254	50.0863
bnn-gnn-npc	0.0310	0.0309	0.0195	0.0960	0.0308	0.0624	0.0144	0.0278	49.9071

imdb	%precision		%recall		%ndcg		%map		%aucroc
	@2	@10	@2	@10	@2	@10	@2	@10	
random	0.1700	0.1800	0.0800	0.4500	0.1700	0.3100	0.0600	0.1200	49.8800
fnn-std [10]	0.9526	0.7750	0.4090	1.7625	0.9684	1.3405	0.3249	0.5839	58.9675
fnn-sc	0.2118	0.2093	0.1027	0.5234	0.2165	0.3692	0.0794	0.1534	50.2131
fnn-pc	0.9240	0.6878	0.3927	1.5619	0.9205	1.2018	0.3096	0.5329	54.8621
fnn-npc	0.9915$\pm$	0.8840$\pm$	0.4415$\pm$	2.0059$\pm$	1.0056$\pm$	1.5104$\pm$	0.3595$\pm$	0.6820$\pm$	63.0213
fnn-gnn-std	0.7319	0.6582	0.3376	1.5096	0.7449	1.1390	0.2645	0.5111	62.3177
fnn-gnn-sc	0.2326	0.2151	0.1154	0.5291	0.2429	0.3875	0.0915	0.1659	50.3146
fnn-gnn-pc	0.6645	0.6214	0.2991	1.3973	0.6645	1.0456	0.2277	0.4565	64.7053
fnn-gnn-npc	0.8150$\pm$	0.7408$\pm$	0.3555$\pm$	1.6556$\pm$	0.8162$\pm$	1.2426$\pm$	0.2686$\pm$	0.5317$\pm$	66.7753
bnn-std [27, 48]	0.2544	0.2481	0.1284	0.6062	0.2626	0.4375	0.1016	0.1859	49.7615
bnn-sc	0.1890	0.2450	0.0993	0.6145	0.1857	0.4070	0.0721	0.1630	50.0441
bnn-pc	0.1791	0.2149	0.0899	0.5177	0.1785	0.3553	0.0680	0.1431	50.1223
bnn-npc	0.2076	0.1983	0.0986	0.4871	0.2088	0.3488	0.0747	0.1462	49.9923
bnn-gnn-std [47]	0.1973	0.2030	0.1005	0.5029	0.2008	0.3557	0.0772	0.1495	50.0626
bnn-gnn-sc	0.1890	0.2471	0.0924	0.6003	0.1866	0.4062	0.0682	0.1595	50.0547
bnn-gnn-pc	0.1661	0.2424	0.0809	0.5944	0.1614	0.3922	0.0587	0.1508	50.0527
bnn-gnn-npc	0.1869	0.1900	0.0988	0.4896	0.1869	0.3393	0.0733	0.1424	49.9541



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Hands-on: OpeNTF



Active Engagement
Share Knowledge
Accountability
Time Management
Communication Skills
Conflict Resolution
Risk Management

Dead Poet Society, 1998, Peter Weir

Robin Williams: *"we don't read and write poetry because it's cute ..."*

Reviewer Assignment

 Reviewer.ly[Products](#)[About](#)[Resources](#)[Publications](#)[Contact Us](#)

Reviewerly: Modeling the Reviewer Assignment Task as an Information Retrieval Problem

Industry Talk

2024

The 33th ACM International Conference on Information and Knowledge Management (CIKM 2024) — Industry Day Talks

Negar Arabzdeh, Sajad Ebrahimi, Sara Salamat, Mahdi Bashari and Ebrahim Bagheri

[► Abstract](#)[Read paper](#)

Reviewer Assignment Problem: A Scoping Review

Research Paper

2023

Jelena, J.J., Jovanovic University of Belgrade, Serbia, & Ebrahim, E.B., Bagheri Toronto Metropolitan University, Canada

[► Abstract](#)[Read paper](#)

Palliative Care

Symptom Management: Helps relieve pain, fatigue, and other distressing symptoms

Emotional Support: Provides counseling and mental health support

Spiritual Care: Addresses existential concerns based on personal or religious beliefs

Family Support: Helps with decision-making

Coordination of Care: Ensures smooth communication between healthcare providers

Salins et al. Palliative and end-of-life care practices for critically ill patients and their families in a peri-intensive care setting: a protocol for an umbrella review. Palliative & supportive care 2024.

Patients
Families



Doctors
Nurses
Physiotherapists
Psychologists
Nutritionists
Volunteers

geographical distance, communication costs, time availability

Outline

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Hands-on: OpeNTF

https://fani-lab.github.io/OpeNTF

OpeNTF2: Fairness-aware Graph Neural Team Formation, SIGIR26

Neural machine learning methods for Team Formation/Recommendation problem.

machine-learning pytorch

data	(Adila#87)(skip-ci)(refactor gender inference codes)	2 months ago
docs	opentf2 models class diag(skip-ci)	last month
ipynb	Created using Colab (skip-ci)	last month
output	(#281) temporal (streaming) training refactors	last month
src	(#281) temporal (streaming) training refactors	last month
.gitignore	(#264)	8 months ago
.gitmodules	(#281)	2 months ago
Dockerfile	(#286)	9 months ago
computecanada.sh	computecanada run based on hydra exec	10 months ago
license.txt	no message	4 years ago
readme.md	Update readme.md (skip-ci)	last month
requirements.txt	dynamic import of a package in a single file from web (ski...	2 months ago

neural-team-formation

neural-team-recommendation

- Readme
- View license
- Activity
- Custom properties
- 27 stars
- 3 watching
- 17 forks
- Audit log
- Report repository

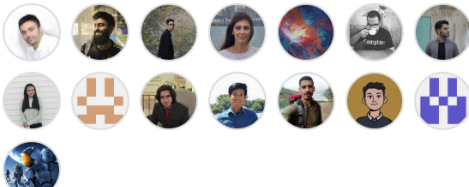
Releases 2

v0.2.0.0 Graph Neural Team F... Latest

on Nov 18, 2025

+ 1 release

Contributors 15

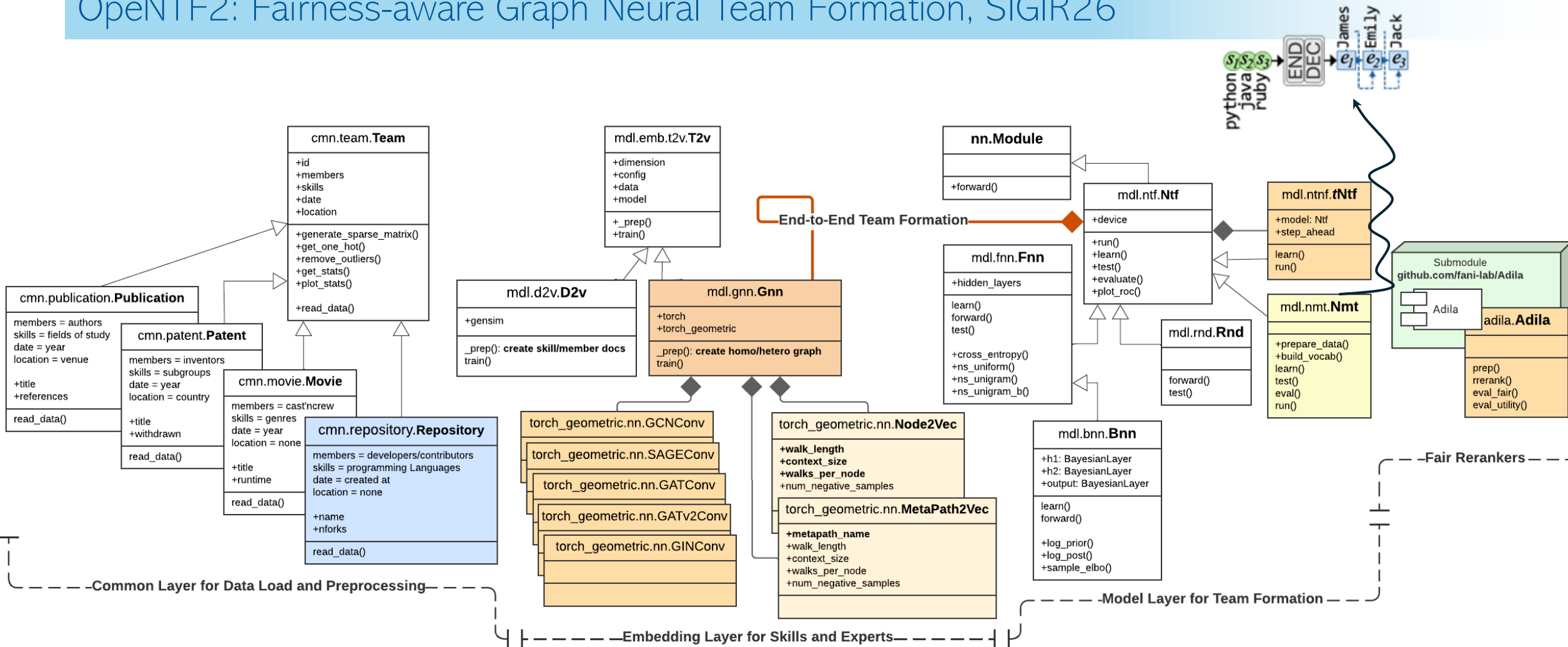


README License

OpeNTF : An Open-Source Neural Team Formation Benchmark Library

<https://fani-lab.github.io/OpeNTF>

OpeNTF2: Fairness-aware Graph Neural Team Formation, SIGIR26





Beyond Searching a Village: Learning to Recommend Diverse and Successful Collaborative Teams



fani-lab.github.io



recsys.acm.org/recsys26/

<https://fani-lab.github.io/OpenTF/tutorial/recsys26>

TIME AND LOCATION

ABSTRACT

TARGET AUDIENCE

OUTLINE

PIONEERING TECHNIQUES

NEURAL TEAM RECOMMENDATION

TIME AND LOCATION

Half-day, Afternoon, 1:30 PM - 5:30 PM (GMT/UTC -6), Monday, September 28, 2026

Elk/Deer Lake (4th floor), Marriott City Center, Downtown, Minneapolis, Minnesota, USA

Prev Tutorials at [UMAP24], [SIGIR-AP24], [WSDM25], [CIKM25] < [Full Outline] [Slides] [Recording]